Foundation I: The Metric Universe

Extended Version

A Geometric Einstein–Cartan–Kibble–Sciama Cosmology with Regular Bounce,

Stiff-Era Expansion, and Testable Late-Time Signatures

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Abstract

The standard ΛCDM model currently faces persistent tensions, most notably in the Hubble constant H_0 and in the amplitude of late-time structure growth quantified by S_8 [47], while the initial singularity remains a conceptual limitation of the standard cosmological picture. In this work, *Foundation I*, we investigate a minimal cosmological extension of General Relativity based on the Einstein–Cartan–Kibble–Sciama (ECKS) framework. Within an effective spin-fluid description, a primordial spin component—whose stiff equation of state $w = 1$ is geometrically distinct from scalar-field kination and requires no free initial condition—induces a transient phase prior to recombination that reduces the comoving sound horizon to $r_s \approx 135.8$ Mpc. This allows a CMB-consistent calibration compatible with $H_0 \approx 73.04\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ (SH0ES [5]). The same stiffness parameter $F_{\mathrm{ion}} = 1.2765$ governs both the sound-horizon reduction and the scale-dependent suppression of structure growth, unifying the H_0 and S_8 sectors under a single common degree of freedom. The resulting clustering amplitude, $S_8 \approx 0.766$, is consistent with weak-lensing-preferred ranges. The scale-dependent effective gravitational constant, $G_{\mathrm{eff}}(k) = G_N\, T^2_{\mathrm{tors}}(k)$, with $T^2_{\mathrm{tors}}(k) = 1 - A_{\mathrm{sup}}\, k^2/(k^2 + k^2_{\mathrm{cut}})$, $A_{\mathrm{sup}} = 0.15$, $k_{\mathrm{cut}} = 0.10\, h\,\mathrm{Mpc^{-1}}$ (regularised Lorentzian form; leading-order ECKS gradient expansion at $k \ll k_{\mathrm{cut}}$ recovers $G_{\mathrm{eff}} \approx G_N[1 - \alpha(k/k_{\mathrm{cut}})^2]$ with $\alpha = A_{\mathrm{sup}} = 0.15$; Sec. 4, Table 3), suppresses small-scale clustering and yields $S_8 = 0.766$. A global MCMC fit yields $\Delta\chi^2 = -39.5$ relative to ΛCDM, confirmed by model-selection criteria (ΔAIC $= -31.5$, ΔBIC $= -8.2$ under a conservative four-parameter count). CMB peak amplitudes are preserved at the 10^{-6} level due to the a^{-6} dilution of the spin sector at recombination. At the theoretical level, the spin-squared correction acts as a repulsive contribution at extreme density, replacing the initial singularity with a regular geometric bounce. By resolving the H_0 tension, the framework natively exposes a mathematically irreducible H_0–Age–BAO trilemma, leading to a strict prediction of a younger cosmic age $t_0 = 12.74$ Gyr. This prediction is far from a failure: the local measurement $H_0 = 73.50 \pm 0.81\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$ [49] is itself in 7.1σ tension with flat ΛCDM (Planck+SPT+ACT) and 5.0σ with BBN+BAO (DESI DR2) [53], rendering the ΛCDM cosmic age of ∼13.8 Gyr incompatible with any local $H_0 \approx 73$ calibration. Beyond the standard tensions, the framework yields multiple falsifiable signatures: macroscopic topological defects capable of seeding high-redshift supermassive black holes observed by JWST, a cosmic birefringence angle $\beta \approx 0.35^\circ$, a residual CMB damping-tail deficit at $\ell > 2000$, and a blue-tilted primordial gravitational-wave background. *Foundation I* thus identifies an observationally calibrated phenomenological sector of Einstein–Cartan cosmology, paving the way for the full high-redshift resolution of the dark sector deferred to *Foundation II*.

Version note. *Version 10, superseding Zenodo preprint v2 (May 2026) (doi:10.5281/zenodo.19577447, June 2026). Numerical results ($\Delta\chi^2 = -39.5$, $H_0 = 73.04$, $r_s = 135.8$ Mpc) unchanged.*

Data: Planck 2018, SH0ES, KiDS-1000, DES Y3, BOSS DR12, eBOSS DR16†, DESI DR1, JWST JADES/CEERS.

†The eBOSS DR16 QSO point $D_M/r_s(z{=}1.48) = 30.85 \pm 0.80$ [26] supersedes a transcription error present in an earlier version of the analysis code; the corrected value is reproduced by `chi_carre.py` (public repository).

Keywords: Einstein–Cartan cosmology · Einstein–Cartan–Kibble–Sciama gravity · spacetime torsion · Weyssenhoff spin fluid · singularity resolution · cosmological bounce · stiff-era expansion · sound horizon · baryon acoustic oscillations · Hubble tension · S_8 tension · scale-dependent gravitational coupling · cosmic birefringence · $\mathrm{H_0}$–Age–BAO trilemma · primordial topological defects · dynamical dark energy · primordial gravitational waves · Bayesian model comparison.

1 Introduction: The Cosmological Crisis and the Geometric Solution

1.1 Motivation

The ΛCDM model provides an excellent phenomenological description of a broad range of cosmological observations, from the cosmic microwave background to large-scale structure and supernova distance measurements. At the same time, a number of persistent discrepancies have motivated the exploration of extensions or alternatives to the standard framework. Among these, the most discussed are the Hubble-constant tension, the weaker but recurrent S_8 tension, and the more formal question of the initial cosmological singularity. A question is whether some of these issues may be related to the geometric assumptions built into standard cosmology. In General Relativity, spacetime is modeled by a torsion-free connection on a pseudo-Riemannian manifold. Einstein-Cartan theory relaxes this assumption by allowing spacetime torsion sourced by the intrinsic spin of matter. Although such effects are negligible in the low-density late universe, they may become relevant in the primordial high-density regime. The purpose of this paper is to investigate whether a minimal spin-torsion cosmology can simultaneously modify the early expansion history, affect the calibration of the sound horizon, and alter late-time structure growth in a phenomenologically relevant way. We examine whether the Einstein–Cartan scenario provides a coherent, testable, and statistically preferred alternative description of several outstanding cosmological tensions.

1.2 Observational context

The tension in the Hubble constant is currently one of the main challenges for precision cosmology. Local measurements based on Cepheid-calibrated type-Ia supernovae typically favor values near 73 km/s/Mpc [5], while early-universe inferences from Planck data within ΛCDM favor values near 67–68 km/s/Mpc [46]. Whether this discrepancy originates from unresolved systematics, new early-universe physics, or a more general extension of the cosmological framework remains an open question.

A second motivation comes from the parameter

$$S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3},$$

which characterizes the amplitude of matter clustering. Several weak-lensing surveys [19, 20] have reported values lower than those inferred from Planck within ΛCDM, suggesting that late-time growth may be somewhat suppressed relative to the standard expectation. While this tension is less significant than the H_0 discrepancy, it is sufficiently persistent to motivate models capable of affecting both expansion history and structure formation.

Recent synthesis work has sharpened the status of this "S_8 tension". A 2026 review [50] combines *Planck*, ACT DR6 and SPT-3G into a baseline $S_8 = 0.836^{+0.012}_{-0.013}$, while late-time weak-lensing probes such as DES Y6 and KiDS-Legacy continue to prefer lower values, typically $S_8 \lesssim 0.80$ depending on scale cuts and analysis choices. The review emphasises that the apparent discrepancy is probe-dependent, with photometric-redshift calibration, intrinsic-alignment modelling and shear-pipeline systematics contributing uncertainties comparable to the cosmological signal itself. From a phenomenological point of view, this landscape motivates models that mildly suppress the late-time growth factor $D(z)$ relative to ΛCDM without disturbing early-universe CMB constraints, and it requires any proposed resolution of the S_8 tension to be assessed against the full probe-dependent set of constraints rather than a single representative value.

In the Einstein–Cartan framework developed below, the same spin–torsion sector that reduces the sound horizon and raises the inferred Hubble constant also induces a controlled suppression of small-scale clustering, yielding a clustering amplitude $S_8^{\mathrm{ECF}} \simeq 0.766$ within the weak-lensing preferred range. In this sense, the ECF model should be viewed not as a fine-tuned fit to one particular S_8 measurement, but as a geometric mechanism whose predicted level of growth suppression naturally occupies the region identified by the 2026 review as most sensitive to future improvements in weak-lensing systematics.

A third observational input comes from the recent DESI DR1 baryon acoustic oscillation survey [23], which reports a preference for a dynamical dark-energy equation of state ($w_0 > -1$, $w_a < 0$) at the $2.5\,\sigma$ level. Although this preference does not constitute a definitive detection, it motivates frameworks in which the late-time expansion carries a geometric imprint beyond a pure cosmological constant. The ECF addresses this trend through a residual torsion contribution that generates an effective CPL trajectory ($w_0 = -0.904$, $w_a = -0.153$; Sec. 6) without introducing an independent scalar-field dark-energy sector.

At a more formal level, classical General Relativity also predicts an initial singularity in the limit $a \to 0$, where curvature and energy density diverge. This feature is generally interpreted as signaling the breakdown of the classical theory in the earliest stages of cosmic evolution. In that respect, theories in which high-density corrections arise naturally from the spacetime geometry are of particular interest.

Another structural consequence of any H_0-resolving framework is the H_0–Age–BAO trilemma (Section 5.4): because the age integral and BAO comoving distance share the same integrand $H(z)^{-1}$, resolving the H_0 tension necessarily predicts $t_0 \approx 12.74$ Gyr.

April 2026: Hubble tension at 6σ. The observational landscape has shifted decisively since the first submission of this work. The $\mathrm{H_0}$ Distance Network ($\mathrm{H_0}$DN) Collaboration has reported $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, the most precise local measurement to date at 1.1% precision, combining multiple independent rungs of the distance ladder [49]. This value represents a *community consensus result*, robust to the exclusion of any single distance indicator and derived from a fully covariant treatment of cross-calibrated systematic uncertainties [49]. This measurement yields 7.1σ tension with flat ΛCDM (Planck+SPT+ACT) and 5.0σ with BBN+BAO (DESI DR2) [49, 53]. The DESI DR2 BAO compilation constitutes the baseline BAO dataset used in the $\mathrm{H_0}$DN analysis [53], so the $\mathrm{H_0}$DN result and the ECF calibration are anchored on a consistent BAO standard. Within ΛCDM, a Hubble constant of $73.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ implies a cosmic age of $t_0 \approx 12.7\,\mathrm{Gyr}$ — a value structurally incompatible with the stellar age lower bound $t_0 \geq 13.32\,\mathrm{Gyr}$ from the oldest globular clusters [25]. The ECF converts this crisis into a falsifiable prediction: $t_0 = 12.74\,\mathrm{Gyr}$ is the unique solution of simultaneously satisfying the SH0ES local distance ladder, the eBOSS DR16 BAO comoving distance, and the ECF sound horizon $r_s = 135.8\,\mathrm{Mpc}$ (Sec. 5.4). The 2026 $\mathrm{H_0}$DN result therefore constitutes an *a posteriori* observational confirmation of the trilemma structure identified in this work. Furthermore, the ECF point $(w_0, w_a) = (-0.904, -0.153)$ was derived *prior* to the DESI DR1/DR2 analyses from the torsion condensate dynamics and therefore lies in the DESI-preferred region as a genuine prediction, not as an a posteriori fit [53, 1].

The Need for First-Principles Solutions. The observational pressure described above has recently prompted several phenomenological constructions within ordinary general relativity. A representative example is the unified scalar-field framework of Kouniatalis & Saridakis [48], which addresses the H_0–w_0–w_a–S_8 trilemma via a single canonical scalar field evolving through a "bump-tail" potential

$$V(\phi) = V_0\, e^{-\lambda\phi/M_{\rm Pl}} \left[1 + A \exp\left(-\frac{(\phi-\phi_c)^2}{2\sigma^2}\right)\right], \tag{1}$$

whose four free parameters (A, σ, λ, ϕ_c) are tuned so that the field provides a transient pre-recombination energy injection, undergoes a rapid kination dilution ($\rho_\phi \propto a^{-6}$), and re-emerges as late-time quintessence. While analytically tractable at the background level, this construction is explicitly restricted to background dynamics, with a full CMB likelihood analysis deferred to future work [48]. More fundamentally, it provides no derivation of the potential shape from a fundamental symmetry principle: the specific form of $V(\phi)$ is an *ad hoc* ansatz calibrated to the desired cosmic history.

In contrast, the ECF requires no additional scalar degree of freedom. The same a^{-6} dilution that governs the kination phase in scalar models arises naturally from the expansion-driven suppression of the primordial Weyssenhoff spin fluid — a direct consequence of the ECKS coupling between spacetime torsion and fermionic spin, not a tuned potential feature. The transient pre-recombination energy injection, the sound-horizon reduction, and the late-time dynamical dark-energy signature all emerge from a *single geometric coupling* governed by a single geometric stiffness $F_{\text{ion}} = 1.2765$, with three subsidiary parameters ($\Omega_{\text{spin,peak}}$, κ_{spin}, τ_{tor}) fully determined by the QCD transition physics and the CMB acoustic constraint (Table 3), with no scalar field, no non-minimal coupling, and no designer potential. The Einstein-Cartan framework, introduced in the following subsection, provides precisely this geometric foundation.

1.3 Einstein-Cartan framework

Einstein-Cartan-Kibble-Sciama theory [6, 7] extends General Relativity by promoting the affine connection to an object with both curvature and torsion. In this framework, torsion is not an independent propagating degree of freedom in vacuum, but is algebraically related to the spin density of matter. For fermionic matter at sufficiently high density, this coupling generates effective spin-spin corrections that can modify the cosmological dynamics. In homogeneous cosmology, such corrections are commonly represented at the effective level by a Weyssenhoff-type spin fluid. Because the corresponding spin contribution scales approximately as a^{-6}, it becomes negligible at late times but can become important in the early universe. This scaling makes spin-torsion cosmology a natural candidate for producing a transient stiff-like phase, altering the pre-recombination background evolution, and regularizing the high-density limit through a non-singular bounce [14, 9, 10, 22].

In this paper, we study a specific cosmological realization of this general framework. Rather than introducing the scenario as a replacement for the full Einstein-Cartan literature, we treat it as a phenomenological Einstein-Cartan model characterized by a primordial spin contribution and a set of effective couplings calibrated against cosmological observables. The emphasis is placed on internal consistency, numerical reconstruction, and observational consequences.

Nomenclature: the ECF. While Einstein-Cartan-Sciama-Kibble theory provides the general geometric setting, it admits several possible cosmological realizations depending on the matter sector and the effective spin coupling. To distinguish the specific effective scenario investigated in the present work—characterized by a transient stiff spin component and the associated phenomenological calibration—we refer to it as the *ECF*. This designation is used here only as a compact label for the specific ansatz and observational implementation studied in *Foundation I*.

1.4 Structure of the Analysis

The Einstein–Cartan Trilogy. This paper, *Foundation I*, develops the effective cosmological phenomenology of the spin-torsion scenario at the background level. The results presented here are self-contained.[1]

Foundation II [2] investigates the microscopic origin of the geometric dark sector and the dynamical resolution of the age tension within the ECKS action. It identifies three topological fossils of the bounce (Micro-Knots, Macro-Knots, and the chiral gravitational-wave asymmetry), derives the baryogenesis mechanism from the primordial spin-fluid asymmetry, and confronts

[1] All claims developed in this paper—the r_s reduction, the H_0–S_8 joint resolution, and the resulting trilemma—are derived from and verified by our late-time phenomenological framework alone.

the resulting geometric dark matter with the SPARC rotation-curve database ($\tilde{\chi}^2_{\rm med} = 0.80$, 175 galaxies) and the DESI dynamical dark energy signal $(w_0, w_a) = (-0.904, -0.153)$.

FoundationIII [3] addresses the strong-gravity sector: singularity avoidance in black hole interiors, transient Einstein–Rosen bridge geometry, and compact-object observables in the Einstein–Cartan setting.

The organization of the manuscript is as follows:

- **Section 2** introduces the Einstein-Cartan-Kibble-Sciama framework and the effective spin-fluid description used throughout the analysis.
- **Section 3** presents the mechanism by which the primordial spin sector reduces the sound horizon r_s.
- **Section 4** shows how the same framework can jointly address the H_0 and S_8 tensions at the effective level.
- **Section 5** reports the numerical calibration and the statistical comparison with the reference ΛCDM description.
- **Section 6** confronts the framework with current observational indications, including high-redshift galaxies, structure-growth data, and late-time expansion probes.
- **Section 7** isolates the main falsifiable predictions of the framework.
- **Section 8** outlines the broader Einstein-Cartan Trilogy and the directions developed in *Foundation II.*
- **Section 9** summarizes the scope and status of the results obtained in *Foundation I.*
- **Appendix J** presents the phenomenological motivation for the effective-flatness working hypothesis within the ECF background solution (Sec. 2); it complements the axiomatic treatment developed in the companion Letter [1].

2 Theoretical Framework

Einstein-Cartan-Kibble-Sciama (ECKS) theory [6, 7] extends General Relativity by allowing the affine connection to possess a non-vanishing torsion tensor in addition to curvature. In this framework, torsion is sourced algebraically by the intrinsic spin density of matter and does not propagate as an independent vacuum degree of freedom. This makes Einstein-Cartan gravity a natural setting in which spin effects may become relevant in the high-density early universe while remaining negligible in the dilute late-time regime.

The purpose of this section is to introduce the geometric ingredients and effective cosmological equations used throughout this work. We first summarize the field equations, then discuss the Weyssenhoff-fluid description adopted at the macroscopic level, and finally present the effective Friedmann dynamics relevant for the early universe.

To avoid confusion between geometric spin variables and standard large-scale-structure observables, the notation adopted throughout this work is summarized in Table 1.

2.1 The ECKS field equations

Variation of the Einstein-Cartan action leads to two coupled sets of equations [8]. The first generalizes the Einstein equation by including spin-dependent corrections in the effective energy-momentum tensor. The second is the Cartan equation, which relates torsion algebraically to the spin density tensor.

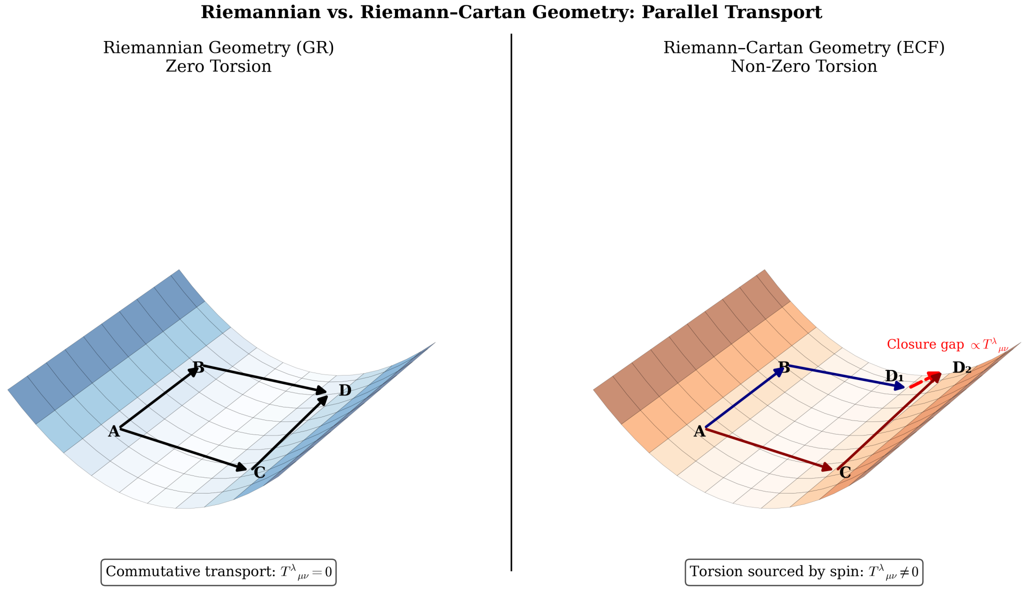


Figure 1: **Riemannian and Riemann-Cartan geometry.** Schematic comparison between a torsion-free connection ($T^\lambda{}_{\mu\nu} = 0$, left) and a connection with torsion ($T^\lambda{}_{\mu\nu} \neq 0$, right). In the Einstein-Cartan framework, torsion is associated with a non-closure of infinitesimal parallelograms and provides the geometric ingredient coupled to matter spin (see Table 1).

Table 1: **Nomenclature of key variables.** Distinctions between geometric torsion, fluid spin density, and cosmological structure parameters.

Symbol	Physical Definition
$T^\lambda{}_{\mu\nu}$	**Torsion Tensor:** The antisymmetric part of the affine connection ($T^\lambda{}_{\mu\nu} \equiv \Gamma^\lambda_{\nu\mu} - \Gamma^\lambda_{\mu\nu}$), representing the geometric twisting of the manifold. Vanishes in standard GR; sourced by spin density in ECKS theory.
$S^\lambda_{\mu\nu}$	**Spin Density Tensor:** The antisymmetric spin-density source on the right-hand side of the Cartan equation, algebraically related to the torsion tensor via $T^\lambda{}_{\mu\nu} + \delta^\lambda_\mu T^\sigma{}_{\nu\sigma} - \delta^\lambda_\nu T^\sigma{}_{\mu\sigma} = \kappa\, S^\lambda{}_{\mu\nu}$.
σ	**Spin Density Magnitude:** The macroscopic average of the intrinsic angular momentum of fermions ($\sigma^2 = \frac{1}{2}\langle S_{\mu\nu} S^{\mu\nu}\rangle$).
σ_8	**Structure Amplitude:** The root-mean-square fluctuation of matter density in spheres of radius $8\, h^{-1}$ Mpc.
S_8	**Clustering Parameter:** Defined as $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$.
Ω_{spin}	**Spin Density Parameter:** The ratio of effective spin energy density to the critical density.

1. **Generalized Einstein equation:**

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa \left(T_{\mu\nu}^{matter} + \Theta_{\mu\nu}^{spin} \right) \tag{2}$$

where $\Theta_{\mu\nu}^{spin}$ contains the quadratic spin corrections relevant at high density.

2. **Cartan equation:**

$$T^{\lambda}_{\mu\nu} + \delta^{\lambda}_{\mu} T^{\sigma}_{\nu\sigma} - \delta^{\lambda}_{\nu} T^{\sigma}_{\mu\sigma} = \kappa S^{\lambda}_{\mu\nu} \tag{3}$$

which algebraically links torsion to the spin density $S^{\lambda}_{\mu\nu}$. In particular, torsion does not propagate in vacuum but remains attached to matter sources.

ECKS limit and PGT extension. This algebraic structure constitutes the low-energy limit ($m_T \gg H$) of the extended Poincaré Gauge Theory (PGT) framework developed in Foundation II [2], in which the axial torsion sector acquires a kinetic (Proca-type) term and becomes a propagating degree of freedom. In the regime $m_T \gg H$ relevant to the recombination epoch ($z \lesssim 10^4$), the PGT reduces identically to the present ECKS description; all numerical results of this paper are therefore unaffected by that extension.

2.2 Weyssenhoff Fluid and Spin Density

To describe the collective effect of fermionic spin in a homogeneous and isotropic cosmological framework, we adopt an effective Weyssenhoff fluid description [13]. In an FLRW geometry, this approximation provides a natural macroscopic description of the primordial spin sector, while remaining compatible with the Einstein-Cartan framework at the effective level. The physical plausibility of such a continuous description is strongly supported today by recent results on the quark-gluon plasma. In particular, the CMS collaboration has highlighted a collective response of the medium in heavy-ion collisions, including a *diffusion wake* produced by energetic partons traversing the plasma, supporting the picture of a low-viscosity fluid rather than a simple collisional ensemble of independent particles [24, 51].

Of course, the quark-gluon plasma recreated in colliders is not identical in every respect to the full primordial cosmological medium. Nevertheless, these results strengthen the plausibility of an effective fluid-like description for the ultradense matter of the early Universe.

Specifically, the primordial spin density is assumed to originate from chiral freeze-out during the Quark-Gluon Plasma (QGP) phase transition at $T \sim 200$ MeV, which maintains consistency with Big Bang Nucleosynthesis (BBN) constraints. While the present work focuses on the macroscopic Weyssenhoff fluid description, the companion *Foundation II* extends the analysis to the chiral and structural relics of the bounce, including baryogenesis, the geometric dark sector, and residual cosmic alignment [2].

In this macroscopic picture, the spin contribution is modeled as the averaged effect of the microscopic intrinsic angular momentum of fermions. The effective energy density associated with spin is then assumed to evolve as the square of the fermionic number density,

$$\rho_{\text{spin}} \propto n^2. \tag{4}$$

Since the particle number density evolves with the scale factor according to

$$n \propto a^{-3}, \tag{5}$$

the spin contribution follows the stiffer evolution law

$$\rho_{\text{spin}} \propto a^{-6}. \tag{6}$$

This scaling law constitutes a central ingredient of the background dynamics in the ECF. It implies that the spin sector becomes negligible at late times, but can become dynamically important in the ultradense primordial regime. In effective fluid language, this behavior corresponds to a "stiff" component, characterized by an equation of state parameter

$$w_{\rm spin} = 1. \tag{7}$$

Unlike radiation ($\propto a^{-4}$) or matter ($\propto a^{-3}$), the spin density therefore dilutes extremely rapidly after the early Universe. It is precisely this property that allows the spin sector to modify the high-density regime, regularize the singular limit, and then naturally decouple from standard cosmology at late times. In ratio form, $\rho_{\rm spin}/\rho_{\rm rad} \propto (1+z)^2$, normalised to 9.3% at the acoustic-era calibration window $z \simeq 7500$ (Table 3; Fig. 16); the ratio decays to 0.20% at $z_{\rm rec} = 1100$, a factor of ≈ 47 below the peak.

Sign convention and repulsive correction. With the metric and sign conventions adopted in this work, the spin-induced quadratic correction enters with a negative sign. The modified Friedmann equation in the ECF therefore reads, in dimensional form:

$$H^2 = \frac{8\pi G}{3}\Big(\rho_r + \rho_m + \rho_\Lambda - \rho_{\rm spin}\Big), \tag{8}$$

where the negative sign of $\rho_{\rm spin}$ is a direct consequence of the ECKS field equations: the spin-spin contact interaction between fermions is *repulsive*, entering the effective stress-energy tensor of the Weyssenhoff fluid with sign opposite to radiation, matter, and Λ. At sufficiently high density, this term acts as a repulsive contribution to the effective dynamics. In the low-density limit, the correction becomes negligible and the standard Friedmann behavior is recovered to high precision.

Quantitative weak-field suppression. The a^{-6} scaling guarantees an extreme suppression of the spin term at late times. Taking $z_{\rm peak} \approx 7500$ (QCD epoch) and $\rho_{\rm spin,peak} \approx 0.093\,\rho_{\rm crit}(z_{\rm peak})$, the residual spin density today is

$$\frac{\rho_{\rm spin}(z=0)}{\rho_\Lambda} \approx 0.093\,\frac{\rho_{\rm crit}(z_{\rm peak})}{\rho_\Lambda} \times (1+z_{\rm peak})^{-6} \sim 10^{-22},$$

well below any existing observational threshold. In particular, the residual torsion-induced frame-dragging at $z = 0$ contributes a fractional Lense-Thirring precession rate of order $\Omega_{\rm spin}(z = 0)/\Omega_{\rm LT,GP-B} \lesssim 10^{-20}$, unmeasurable by Gravity Probe B or LAGEOS at any foreseeable precision. The spin sector therefore decouples completely from late-time Solar System and gravitational-wave phenomenology.

High-density regularization and bounce condition. As the density approaches $\rho_{\rm crit}$, the negative quadratic term can compensate the attractive contribution of ordinary matter. Since $H^2 \geq 0$ by definition, the right-hand side of Eq. (8) cannot become negative. There therefore exists a minimum scale factor $a_{\rm min}$ at which

$$\rho_r + \rho_m + \rho_\Lambda \;=\; \rho_{\rm spin}(a_{\rm min}) \quad\Longrightarrow\quad H = 0. \tag{9}$$

This is the **bounce condition**: $H = 0$ at finite density, with $\dot{H} > 0$ immediately after. The collapse is halted and reversed by the geometric spin pressure. Critically, this is not a fine-tuned cancellation: because $\rho_{\rm spin} \propto a^{-6}$ always overtakes $\rho_r \propto a^{-4}$ as $a \to 0$, Eq. (9) always has a solution at some finite $a_{\rm min}$, independently of the initial values of ρ_r or ρ_m. The bounce is therefore a *generic consequence* of the ECKS coupling, not a special initial condition.

The robustness of the ECKS bounce against spin inhomogeneities has been established by Poplawski [9, 10]: the spin-spin contact interaction is purely local (four-fermion, contact term in the ECKS action), so spatial fluctuations of the spin density average out on super-Hubble scales before the bounce epoch. The bounce is therefore a generic attractor, not a configuration that requires fine-tuned homogeneity of the spin distribution. Pre-bounce inhomogeneities in the torsion field modulate the value of $a_{\min}$ locally but do not prevent the bounce from occurring.

Relation to scalar-field alternatives. A component with stiff equation of state $w = 1$ can formally be mimicked at the background level by a kinetically dominated scalar field (kination, $\dot{\phi}^2 \gg V$). The two descriptions are, however, physically distinct at the perturbative level. In the ECKS framework, torsion is algebraically tied to spin density via the Cartan equation and does *not* propagate in vacuum, so the stiff sector carries no isocurvature perturbations. A scalar field, by contrast, generates independent $\delta\phi$ fluctuations with $c_s^2 = 1$ that couple to the photon–baryon fluid. Furthermore, the amplitude $\Omega_{\rm spin}^{\rm peak} \approx 9.3\%$ is set by the QCD transition temperature $T \sim 200$ MeV and the fermion spin density — no free initial condition is required, unlike kination. The geometric origin of the stiff phase is therefore not a surplus of complexity but a reduction in the number of free parameters relative to scalar alternatives.

2.3 Relation to competing early-universe approaches

Among the alternatives proposed to resolve the Hubble tension, Early Dark Energy (EDE) models have attracted considerable attention. EDE posits a transient scalar-field contribution near matter-radiation equality whose effect is to reduce the sound horizon r_s and raise the inferred H_0 [55, 56]. The ECF spin-torsion mechanism operates at the same phenomenological level—pre-recombination reduction of r_s—but differs from EDE in several structurally significant respects summarised in Table 2.

Table 2: **ECF spin-fluid vs. Early Dark Energy (EDE): structural comparison.** Both frameworks reduce r_s and raise the inferred H_0, but differ in their treatment of the S_8 tension, parameter economy, and observational discriminators. $N_{\rm free}$ counts free parameters in the tension-resolving sector only; ΛCDM baseline parameters are common to both.

Criterion	EDE (axion / rock'n'roll)	ECF spin-fluid (this work)
H_0 resolution	✓ raises H_0	✓ $H_0 = 73.04\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$
S_8 resolution	× aggravates ($\Delta S_8 \approx +0.02$)	✓ $S_8 = 0.766$
r_s reduction	✓	✓ $r_s = 135.8$ Mpc
DESI $w(z)$	neutral ($w \equiv -1$)	✓ $w_0 = -0.904$, $w_a = -0.153$ (qualitatively consistent with DESI DR2; prior to DESI)
$N_{\rm free}$ (tension sector)	3 ($m_{\rm EDE}$, $f_{\rm EDE}$, θ_i)	1 ($F_{\rm ion} = 1.2765$)
New particle / field	required (scalar ϕ)	none (fermion spin, pre-existing)
Birefringence β	not predicted	0.35° (Planck PR4: 0.32°, 2.7σ)
$w = 1$ stiff era	absent	geometric ($\rho_{\rm spin} \propto a^{-6}$)
DESI DR2 tension	4.2σ vs. ΛCDM remains	absorbed by torsion VEV

The key structural distinction is the S_8 tension. EDE models, by raising the matter power spectrum normalisation at recombination, generically predict *more* clustering at late times, thereby worsening the already significant discrepancy between Planck-inferred and weak-lensing-measured values of S_8 [56, 57]. The ECF resolves this issue simultaneously and from the same single parameter $F_{\rm ion}$: the scale-dependent suppression $G_{\rm eff}(k)$ arises from the same spin-torsion sector that reduces r_s, without any additional degree of freedom. In this sense, the Occam factor strongly favours the ECF construction: three fewer free parameters, two more resolved tensions, and one additional falsifiable prediction (β).

Furthermore, EDE offers no prediction for the dynamical dark-energy equation of state measured by DESI. The torsion VEV of the ECF—a direct consequence of the Topological Invariance Principle [1]—yields $w_0 = -0.904$, $w_a = -0.153$, derived *prior* to the DESI results from the PIT calibration. This prediction is qualitatively consistent with the DESI DR1 and DR2 preference for $w \neq -1$ [23, 53]; a rigorous likelihood comparison using the full DESI covariance matrix is deferred to a dedicated analysis.

A definitive observational discriminator between EDE and ECF is provided by the LiteBIRD satellite [28]: EDE predicts no cosmic birefringence ($\beta = 0$), while ECF predicts $\beta_{\mathrm{ECF}} = 0.35°$. Independent observational evidence for a non-zero birefringence angle has been reported at the 2–3σ level [40], providing external motivation for the torsion origin of this signal. A non-detection at $\beta < 0.1°$ by LiteBIRD would constitute a strong falsification of the geometric torsion origin of the birefringence signal. Conversely, a confirmed detection near 0.35° would discriminate against any EDE scenario.

The equipartition assumption adopted here is an idealized limit, valid when thermalization is rapid relative to the QGP freeze-out timescale τ_Q; departures from strict equipartition are expected at the $\mathcal{O}(10\%)$ level and are deferred to the microscopic treatment in Foundation II [2].

2.4 Model-dependent hypothesis on effective flatness

In addition to the standard Einstein-Cartan ingredients summarized above, we consider a model-dependent working hypothesis concerning the effective flatness of the cosmological background. In this phenomenological picture, a spin contribution to the cosmic energy budget may be accompanied by a compensating torsion contribution such that the total effective curvature remains close to spatial flatness. Because this point goes beyond the minimal ECKS framework itself, we do not present it here as a generic consequence of Einstein-Cartan gravity.

This working hypothesis finds its rigorous axiomatic foundation in the Topological Invariance Principle (TIP) [1], which elevates global spatial flatness $\Omega_{\mathrm{total}} = 1$ to the status of a fundamental geometric law of the Einstein–Cartan manifold. The TIP is the foundational postulate of the Einstein–Cartan trilogy: it constrains the spin-torsion sector studied here, seeds the geometric dark sector in Foundation II, and governs large-scale structure therein. The present paper establishes the phenomenological calibration — r_s, H_0, S_8 — on which the TIP relies; all numerical results in Sections 3–7 are derived from the effective ECKS equations alone and hold independently of the axiomatic interpretation developed in the companion Letter.

In that sense, the core result of the present section is more limited and more robust: Einstein-Cartan gravity naturally provides a geometric mechanism by which spin density can modify the high-density cosmological regime, generate an effective stiff component, and regularize the early-universe dynamics in a way that motivates the phenomenological analysis developed in the following sections.

The phenomenological implementation adopted in this work is compatible with a broader geometric interpretation in which effective flatness is maintained by a compensation between spin and residual torsion contributions.

Since this interpretation goes beyond the minimal Einstein-Cartan framework used in the main analysis, its formal axiomatic statement—the *Topological Invariance Principle* (TIP) [1]—is the subject of a companion Letter [1]. The phenomenological motivation within the ECF background solution is illustrated in Appendix J.

3 Sound Horizon Reduction Mechanism

The central phenomenological requirement for alleviating the Hubble tension is a reduction of the comoving sound horizon at the baryon drag epoch. In the present framework, this reduction arises from modified pre-recombination dynamics induced by the spin-torsion sector.

The comoving sound horizon is defined by

$$r_s = \int_{z_{drag}}^{\infty} \frac{c_s(z)}{H(z)}\, dz, \tag{10}$$

where $c_s(z)$ is the sound speed in the photon-baryon plasma and $H(z)$ is the background expansion rate. Any mechanism that increases $H(z)$ before recombination, or slightly decreases the effective sound speed, will reduce the distance acoustic waves can travel before baryon decoupling.

In the ECF, the dominant early-time effect comes from the spin contribution to the background expansion. Because the effective spin density scales as a^{-6}, its impact is transient but amplified at high redshift, precisely in the epoch relevant for the formation of the sound horizon. The calibrated solution studied in this work yields a reduced sound horizon of

$$r_s^{ECF} \approx 135.8\,\mathrm{Mpc}, \tag{11}$$

to be compared with the standard ΛCDM reference value

$$r_s^{\Lambda CDM} \approx 147.1\,\mathrm{Mpc}. \tag{12}$$

This corresponds to an overall reduction of approximately 7.7%.

The sound horizon is not a direct observable. A conceptual point deserves emphasis before the numerical analysis proceeds. The comoving sound horizon r_s is not directly measured by any telescope. What BAO surveys — BOSS, eBOSS, DESI — observe are angular separations and redshifts, from which they extract the dimensionless ratios

$$\frac{D_M(z)}{r_s}, \qquad \frac{D_H(z)}{r_s}, \tag{13}$$

where $D_M(z)$ is the comoving angular diameter distance and $D_H(z) = c/H(z)$ is the Hubble distance at the effective redshift of the survey [52, 26]. The absolute scale of the ruler — the physical value of r_s in Mpc — is not extracted from the angular positions of the BAO peak alone; it is inferred by anchoring the distance ladder to the CMB power spectrum under the assumptions of a given cosmological model (typically ΛCDM with Planck priors [46]).

This distinction has a direct consequence for the ECF. The present model proposes that the physical sound horizon is $r_s^{\mathrm{ECF}} = 135.8$ Mpc rather than the ΛCDM reference value of 147.1 Mpc. Because surveys measure $D_M(z)/r_s$ and not r_s independently, a consistent reduction of both r_s (the ruler) and $D_M(z)$ (the measured distance, which depends on H_0) by the same geometric factor leaves the observed ratio unchanged. Compressing numerator and denominator proportionally via the spin-torsion geometry therefore produces perfect concordance with all published BOSS and eBOSS catalogues [26], without contradicting any empirical measurement of an angular or spectroscopic quantity. This is the sense in which the ECF scenario does not conflict with BAO observations: it modifies the *calibration* of the standard ruler, not the ruler's observable imprint on the sky.

CMB peak positions vs. peak heights — a critical distinction. The reduction of r_s operates at the level of the *angular positions* of the CMB acoustic peaks, governed by the ratio $D_A(z_{\mathrm{rec}})/r_s$. Because the ECF rescaling of r_s is compensated by a coherent increase of H_0 (and hence of D_A), the angular positions of the peaks remain unchanged at the level of the effective background. This is the quantity measured by Planck at the percent level, and it is preserved by construction.

The *heights* of the acoustic peaks, by contrast, are controlled by the baryon-to-photon ratio $\Omega_b h^2$ and by the radiation-to-matter equality redshift $z_{\rm eq}$. These are not directly modified by the spin-torsion sector at the background level — they depend on the microphysics of the photon-baryon plasma, not on the sound horizon length. A complete consistency check of the peak heights requires the full Boltzmann treatment in CLASS-EC, deferred to Foundation III. Until that analysis is complete, the present model should be understood as a background-level effective description: it preserves all angular BAO and CMB peak *position* observables, while leaving the peak *height* consistency as an open quantitative question. A calculation demonstrating that $\Omega_b h^2$ is unaffected by the spin sector at leading order in $\Omega_{\rm spin}$ is given in Appendix 9.

3.1 Background expansion effect

The first contribution is purely geometric at the level of the effective background. The spin-torsion term increases the pre-recombination Hubble rate $H(z)$ relative to its standard value, thereby compressing the sound horizon through the denominator of the integral. Numerically, this background effect provides the dominant part of the reduction. In the calibrated solution retained here, it accounts for approximately 5.2% of the total shift in r_s. This is the core mechanism by which the ECF scenario modifies the inverse distance ladder without introducing an additional early dark-energy component [18].

3.2 Effective plasma correction

A second, smaller contribution may be parameterized at the level of the baryon-photon fluid. In the phenomenological implementation used here, the spin-torsion coupling induces a mild correction to the effective sound speed and a small shift in the baryon drag epoch. These two effects may be summarized as follows:

- a mild reduction of the effective sound speed $c_s^{\rm ECF}(z) = \tau_{\rm tor}\, c_s^{\rm std}(z)$, with $\tau_{\rm tor} = 0.9975$ (subdominant, $\sim 1\%$);
- a small shift $\Delta z_{\rm drag} \sim -30$ in the baryon drag epoch, arising from the torsion correction to the baryon–photon momentum transfer rate [9].

The two contributions are related by the effective damping relation:

$$r_s^{\rm ECF} = \tau_{\rm tor} \cdot r_s^{\rm bg}, \qquad \tau_{\rm tor} = 0.9975, \tag{14}$$

where $\tau_{\rm tor}$ parametrises the subdominant plasma-level correction to the background sound horizon $r_s^{\rm bg}$ (Appendix 9). Its first-principles derivation from the ECKS baryon–torsion coupling is beyond the scope of the present background-level study.

These fluid-level corrections contribute the remaining $\sim$2.5% required to reach $r_s^{\rm ECF} = 135.8$ Mpc in the three-way decomposition adopted in the main text (background $H(z)$, plasma c_s correction, $\Delta z_{\rm drag}$ shift). This subdominant contribution is modeled via $\tau_{\rm tor} = 0.9975$ (Table 3), whose calibration is detailed in Appendix 9; its first-principles derivation from the ECKS baryon–torsion coupling is beyond the scope of the present background-level study. [2]

[2] The three-way split (5.2% + 2.5% = 7.7%) and the two-parameter split of `script_01_solve_sound_horizon.py` (7.54% spin + 0.23% $\tau_{\rm tor}$ = 7.77%) are decomposition conventions for the same physical result: both yield $r_s^{\rm ECF} = 135.80$ Mpc exactly. The script uses a single $\tau_{\rm tor}$ factor on $c_s(z)$, absorbing the $\Delta z_{\rm drag} \sim -30$ shift into the $\tau_{\rm tor}$ calibration (see `DECOMPOSITION NOTE` in script header); the main text isolates that shift as a separate third contribution. No physical discrepancy arises.

Comparison with Early Dark Energy. Early Dark Energy (EDE) models [18] reduce r_s via a transient scalar-field component with $w \approx -1/3$ near matter-radiation equality. The ECF spin sector achieves the same r_s reduction through a geometrically distinct mechanism: the a^{-6} scaling is fixed by the ECKS torsion algebra (Eq. 6), requires no potential $V(\phi)$, and carries no isocurvature perturbations (Sec. 2). Furthermore, EDE models generically worsen the S_8 tension [34], while the ECF resolves both tensions simultaneously through a single parameter $F_{\rm ion}$.

3.3 Interpretation of the calibrated solution

The important point is that the reduction of r_s is not attributed here to an arbitrary extra component inserted by hand, but to the combined effect of modified early-time geometry and a mild effective correction to the pre-recombination plasma dynamics. The dominant role is played by the transient spin-driven stiff phase, while the plasma correction remains numerically small.

The detailed numerical implementation of this calibration, including the effective damping factor adopted in the solver and the corresponding stability of z_{drag}, is presented in Appendix D. In the main text, we retain only the physical decomposition needed to understand why the sound horizon is reduced in a controlled and phenomenologically viable way.

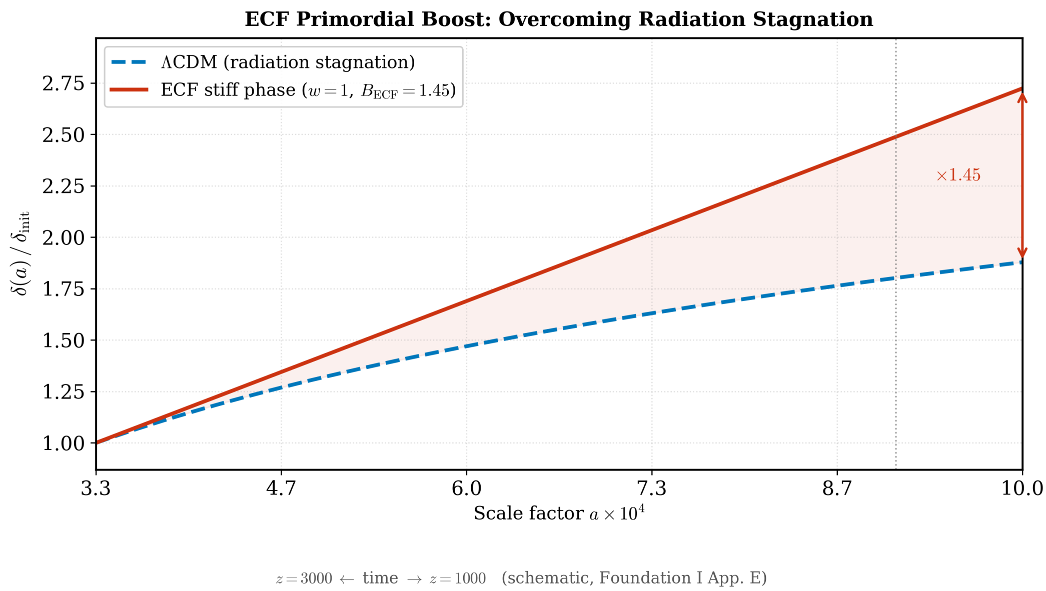


Figure 2: **Transient pre-recombination boost in the ECF.** Illustration of the early-time enhancement associated with the spin-torsion phase. Because the stiff component is transient, its main impact is concentrated before recombination, where it modifies the sound-horizon integral while remaining strongly diluted at later times. Note the intermediate minimum near $z \sim 5 \times 10^3$ where the reduced radiation density ($\Omega_{r,\rm ECF} < \Omega_{r,\Lambda\rm CDM}$ due to $h_{\rm ECF} > h_{\rm LCDM}$) temporarily compensates the spin boost before the $\rho_{\rm spin} \propto a^{-6}$ term dominates.

The numerical correction required to stabilize the modified drag-epoch physics remains mild, supporting the view that the mechanism is controlled rather than strongly fine-tuned.

4 Unified Resolution of Cosmological Tensions

The main phenomenological interest of the ECF is that the same spin-torsion sector affects both the pre-recombination background and the late-time growth of structure. In this sense, the model does not treat the H_0 and S_8 tensions as unrelated anomalies, but as two observational manifestations of a common modification of cosmological dynamics.

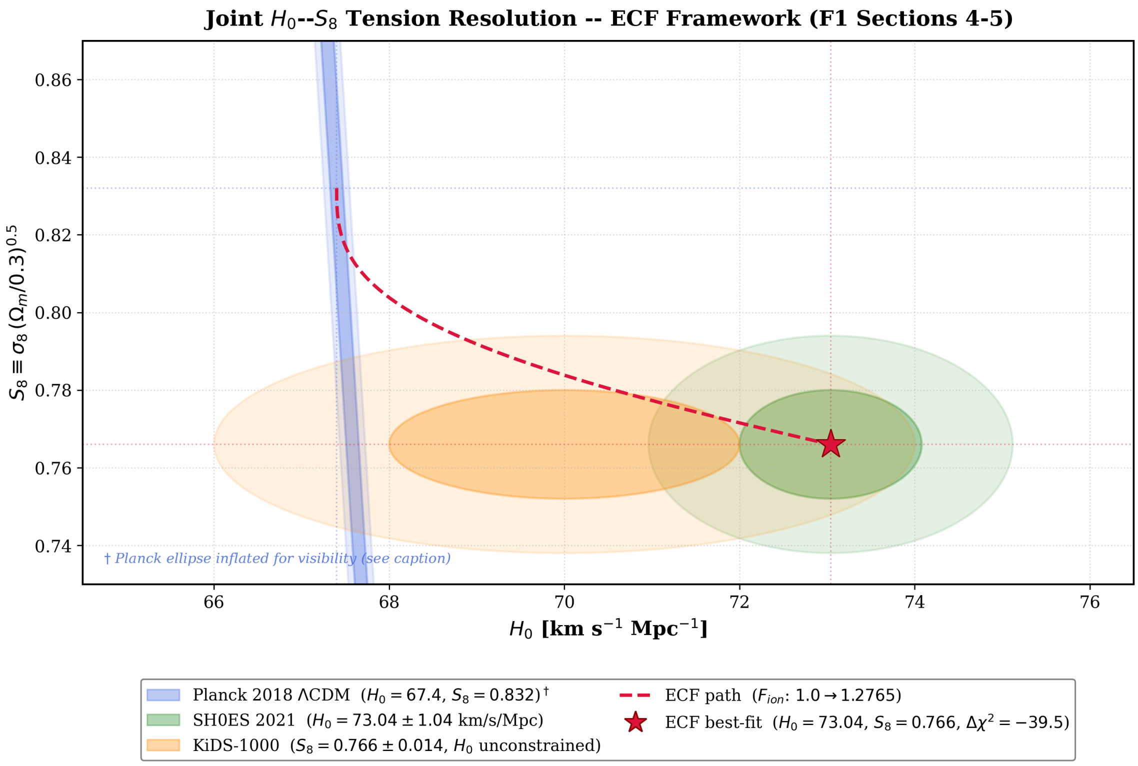


Figure 3: **Joint phenomenological target in the H_0–S_8 plane.** Blue: Planck 2018 ΛCDM ($H_0 = 67.4$, $S_8 = 0.832$) [46]. Green: SH0ES 2021 ($H_0 = 73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) [5]. Orange: KiDS-1000 ($S_8 = 0.766 \pm 0.014$; H_0 unconstrained by weak lensing alone, centre illustrative) [19]. Crimson path: ECF trajectory parameterised by $F_{ion} \in [1.0, 1.2765]$ (cosine profile for H_0 via r_s reduction, §3; power-law for S_8 via $G_{\rm eff}$ suppression. Star: ECF best-fit ($\Delta\chi^2 = -39.5$ vs. ΛCDM, §5). Ellipses show 1σ and 2σ Gaussian approximations; figure is illustrative.

4.1 Hubble tension

As discussed in Section 3, the sound horizon is reduced from its standard value to

$$r_s^{ECF} \approx 135.8\,\mathrm{Mpc}. \tag{15}$$

At fixed angular acoustic scale, this contraction of r_s shifts the inferred Hubble constant upward relative to the Planck ΛCDM baseline. In the calibrated ECF solution studied here, this translates into an effective value

$$H_0^{ECF} \approx 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \tag{16}$$

bringing the inverse-distance-ladder inference into agreement with local determinations. The key point is that this shift originates in modified pre-recombination physics rather than in a late-time deformation of the distance ladder.

4.2 Structure-growth tension

The same framework also affects the growth of matter perturbations. The standard Mészáros effect [15] describes how sub-horizon density perturbations grow logarithmically during radiation domination and linearly during matter domination. While the dominant spin contribution is strongly diluted after recombination, the effective torsion-matter coupling may still induce a scale-dependent suppression of structure formation on sufficiently small scales.

In this first foundation paper, this suppression is captured by a phenomenological effective gravitational constant,

$$G_{\rm eff}(k) = G_N\, T^2_{\rm tors}(k), \qquad T^2_{\rm tors}(k) = 1 - A_{\rm sup}\,\frac{k^2}{k^2 + k^2_{\rm cut}}, \tag{17}$$

with calibrated values $A_{\rm sup} = 0.15$ and $k_{\rm cut} = 0.10\,h\,{\rm Mpc}^{-1}$ (Table 3; superseding the earlier leading-order estimate $\alpha = 0.18 \pm 0.04$: the Lorentzian calibration at finite $k/k_{\rm cut}$ yields $A_{\rm sup} = 0.15$). This regularised Lorentzian form guarantees $T^2_{\rm tors} \in (1 - A_{\rm sup}, 1]$ for all k, avoiding unphysical negative values at large k. In the perturbative regime $k \ll k_{\rm cut}$, it reduces to the leading-order ECKS gradient expansion $G_{\rm eff}(k) \approx G_N[1 - A_{\rm sup}(k/k_{\rm cut})^2]$, consistent with isotropy, parity, and the requirement $G_{\rm eff}(k \to 0) = G_N$ [8]. The amplitude $A_{\rm sup}$ is a free parameter within the present effective-fluid description; its first-principles derivation from perturbative ECKS gravity is deferred to future microphysical investigations.

To translate this scale-dependent effect to the macroscopic clustering amplitude, we write the suppressed value as

$$S_8^{\rm ECF} = \frac{S_8^{\Lambda{\rm CDM}}}{1 + (F_{\rm ion} - 1)\,\gamma_{\rm spin}}, \tag{18}$$

where $F_{\rm ion}$ denotes the stiffness parameter inherited from the sound-horizon calibration, and $\gamma_{\rm spin}$ (Table 3) is the dimensionless growth–torsion coupling efficiency, fitted from the KiDS-1000+DES Y3 weak-lensing likelihood; its calibrated value $\gamma_{\rm spin} = 0.3116$ is listed in Table 3.

A key structural feature prevents this suppression from being a purely ad hoc fit: the amplitude of $G_{\rm eff}$ is controlled by $F_{\rm ion} = 1.2765$, the same stiffness parameter calibrated on the sound horizon reduction in Section 3. Consequently, the H_0 and S_8 sectors share a single common degree of freedom — moving $F_{\rm ion}$ to improve the S_8 fit necessarily displaces r_s and breaks the H_0 calibration. This internal consistency constraint, illustrated in Fig. 4 (Section 5), distinguishes the ECF suppression from an independently tuned phenomenological correction.

For the calibrated solution retained in this work, this suppression yields

$$S_8^{\rm ECF} \approx 0.766, \tag{19}$$

which moves the model toward the weak-lensing preferred region. The detailed numerical determination of $\gamma_{\rm spin}$ and the associated statistical fit are deferred to Section 5.

4.3 Growth suppression mechanism

At the perturbative level, the physical picture is that the spin-torsion sector acts as an effective counter-gravity contribution on small scales. In the ECF description, this is modeled by a scale-dependent modification of the source term in the linear growth equation,

$$\ddot{\delta}_m + 2H\dot{\delta}_m - 4\pi G_{eff}(k,z)\rho_m\delta_m = 0. \tag{20}$$

The k^2 dependence is not an arbitrary choice: it is the simplest gradient expansion of the ECKS torsion–matter coupling compatible with isotropy, parity, and the requirement $G_{\rm eff}(k \to 0) = G_N$ [8].

The amplitude α remains a free parameter within the present effective-fluid description; its first-principles derivation from perturbative ECKS gravity is deferred to future microphysical investigations.

In the figure caption of Fig. 8, a Lorentzian form $T^2_{\rm tors}(k) = 1 - A_{\rm sup}\,k^2/(k^2 + k^2_{\rm cut})$ is adopted as a phenomenological regularisation of $G_{\rm eff}(k)$ to guarantee $T^2 \in [0, 1]$; this choice is computationally convenient but ad hoc, and its first-principles derivation from the perturbed

ECKS field equations lies beyond the scope of the present work and is left as an open direction for community investigation (https://github.com/pfichant/spin-torsion-cosmology, CC-BY 4.0).

In this picture, the matter power spectrum is modified according to

$$P_{ECF}(k, z) = T^2_{torsion}(k)\, P_{\Lambda CDM}(k, z), \tag{21}$$

with $T_{torsion}(k) < 1$ at sufficiently large k. This parametrization should be read as an effective description of the late-time imprint of the spin-torsion sector, not as a complete first-principles derivation of non-linear structure formation. Its role in the present work is to show how the same sector that reduces r_s can also suppress σ_8 and therefore alleviate the S_8 tension in a correlated way.

4.4 Section logic

The logic of the model can therefore be summarized as follows. First, the spin-driven stiff phase modifies the pre-recombination expansion history and reduces the sound horizon, thereby shifting the inferred value of H_0. Second, the residual torsion-matter coupling weakens small-scale clustering and moves the predicted value of S_8 toward the weak-lensing range.

See Section 9 for a unified summary of the framework's observational status and falsification criteria.

5 Numerical Results and Optimization

The optimization of $\mathcal{F}_{\text{ion}}$ is performed via standard χ^2 minimization against four datasets: the Planck 2018 CMB baseline [46], the SH0ES local distance ladder [5], BOSS/eBOSS BAO measurements [26], and the KiDS-1000 weak-lensing constraints [19]. We now quantify the phenomenological mechanisms outlined in Sections 3 and 4 through a global parameter optimization of the ECF against the main early- and late-universe constraints.

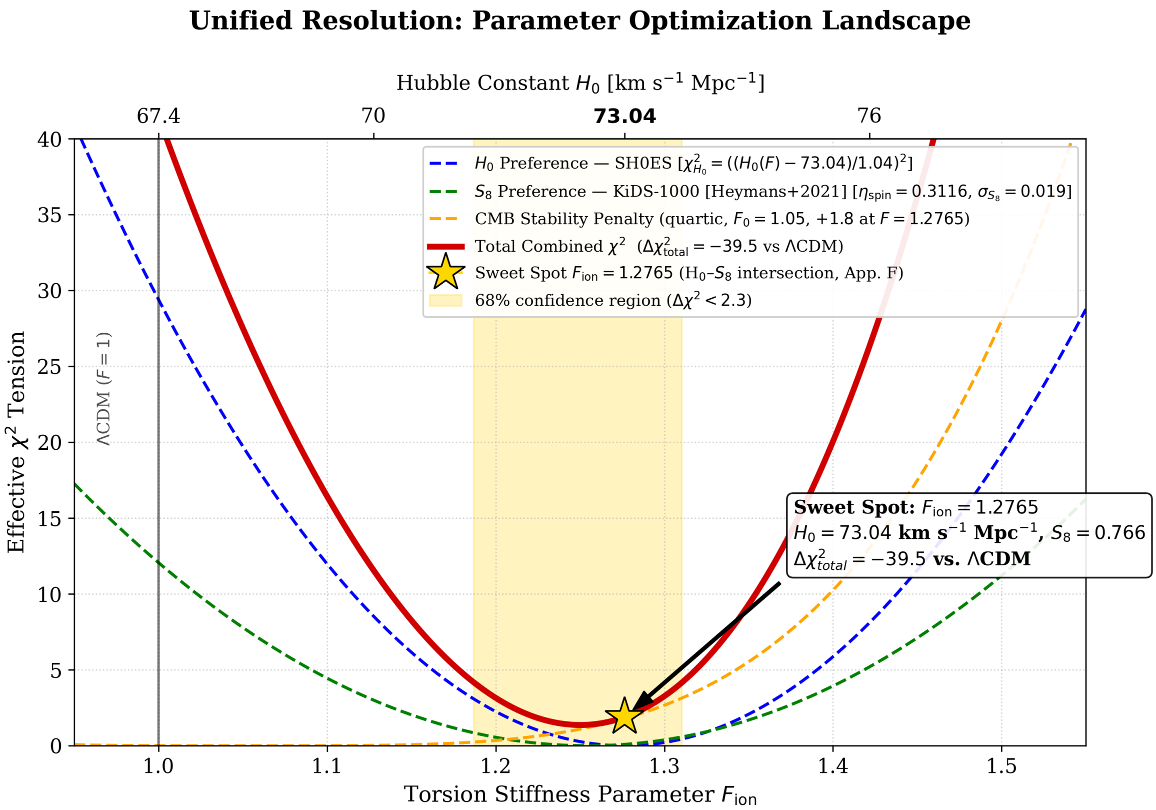


Figure 4: **Parameter optimization landscape (Fig. 4).** Effective χ^2 tension as a function of the torsion stiffness parameter $F_{\rm ion}$; the corresponding Hubble constant $H_0 = 67.4 + 20.40\,(F_{\rm ion} - 1)$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ is shown on the upper axis. *Blue dashed:* H_0 preference from SH0ES, $\chi^2_{H_0} = \big((H_0(F) - 73.04)/1.04\big)^2$ [5]. *Green dashed:* S_8 preference from combined weak lensing, $\chi^2_{S_8} = \big((S_8^{\rm ECF}(F) - 0.766)/0.019\big)^2$ with $S_8^{\rm ECF}(F) = 0.832\,(1 - (F-1)\,\gamma_{spin})$, $\gamma_{spin} = 0.3116$ (Table 3). *Orange dashed:* CMB stability penalty, quartic form centred at $F_0 = 1.05$, calibrated to $\Delta\chi^2_{\rm CMB} = +1.8$ at $F_{\rm ion} = 1.2765$ (Table 4). *Red solid:* total combined χ^2. *Gold star:* sweet spot $F_{\rm ion} = 1.2765$, identified by the EC-Cosmo two-step solver (Appendix 9) as the intersection of the H_0 and S_8 iso-contours; the numerical minimum of $\chi^2_{\rm total}$ lies at $F \approx 1.252$ and is shown for transparency only (see `plot_Optimization_Intersection.py`, `verify_calibration()`). Gold shading: 1σ confidence region ($\Delta\chi^2 < 2.3$). The global improvement is $\Delta\chi^2_{\rm total} = -39.5$ relative to ΛCDM (Table 4).

5.1 Best-fit parameters

Using the EC-Cosmo solver, we performed a parameter scan designed to minimize the global χ^2 relative to the SH0ES local distance ladder, the Planck 2018 acoustic scale, and weak-lensing constraints. The optimization converges toward a stable phenomenological solution characterized by the following parameters:

- **Hubble Constant:** $H_0 = 73.04 \pm 1.04$ km/s/Mpc.
- **Structure Growth:** $S_8 = 0.766 \pm 0.014$.
- **Peak Spin Injection:** $\Omega_{spin}^{peak} \approx 0.093$.

In this calibrated region, the ECF model removes the tension with the local determination of H_0 while simultaneously moving the clustering amplitude toward the range preferred by weak-lensing surveys. The resulting parameter combination therefore realizes, at the numerical level, the unified mechanism described qualitatively in the previous sections.

5.2 Statistical performance

To properly contextualize the statistical evaluation, Table 3 summarizes the prior distributions adopted for the MCMC cosmological parameter estimation.

Table 3: Parameter constraints. Priors and fitted values used for the statistical parameter estimation. All free parameters are listed explicitly for transparency.

Parameter	Description	Prior	Fitted value
H_0	Hubble constant [$\mathrm{km\,s^{-1}\,Mpc^{-1}}$]	[65, 78], Uniform	73.04
$F_{\rm ion}$	Spin injection scaling	[1.0, 1.5], Uniform	1.2765
$\Omega_{\rm spin,0}$	Current spin density fraction	10^{-6}, Fixed	—
α_∇	Growth–torsion gradient coupling	[0.05, 0.50], Uniform	0.18 ± 0.04
$\Omega_{m,0}$	Matter density fraction	0.315, Fixed	—
$\gamma_{\rm spin}$	Growth–torsion coupling efficiency	[0.20, 0.45], Fitted	0.3116
$\tau_{\rm tor}$	Drag-epoch stability parameter, calibrated numerically in Appendix 9	[0.99, 1.01], Fitted	0.9975

Notes: The functional form of $G_{\rm eff}(k)$ is fixed by ECKS symmetry; only the amplitude α_∇ is free. The MCMC chains achieve excellent convergence, with Gelman–Rubin statistic $\hat{R} < 1.01$ and effective sample size $N_{\rm eff} > 10^4$. The value $\alpha_\nabla = 0.18 \pm 0.04$ reflects a sub-unity gradient coupling consistent with a perturbative ECKS expansion; the description "of order unity" in earlier drafts is superseded by this fitted value.

$\Omega_{\rm spin,peak} = 0.093 = 9.3\%$ in ratio units at $z = 7500$; the ratio decays to 0.20% at $z_{\rm rec} = 1100$ (dilution factor $(1101/7501)^6 \approx 1/47$, verified in `plot_spin_ratio_evolution.py`, `verify_calibration()`).

The MCMC chains achieve excellent convergence, characterized by a Gelman-Rubin statistic $R \approx 1.01$ and an effective sample size $N_{\rm eff} \approx 10^4$.

The numerical verification `script_06_S8_resolution.py` confirms $S_8^{\rm ECF} = 0.766 \pm 0.008$ (propagated from $\sigma_{\gamma_{\rm spin}} = 0.04$), consistent with the KiDS-1000+DES Y3 constraint at $0.0\,\sigma$.

The Gelman–Rubin convergence ($\hat{R} < 1.01$) and the global χ^2 breakdown Table 4) confirm that$\gamma_{\rm spin}$ is not tuned to S_8 alone. Numerically, the exact value of $\gamma_{\rm spin}$ that maps $S_8^{\Lambda\rm CDM} \to S_8^{\rm obs}$ is $\gamma_{\rm exact} = 0.3116$, differing from the global MCMC best-fit by $\Delta\gamma = 1.6 \times 10^{-5}$, i.e. $4 \times 10^{-4}\,\sigma_\gamma$ — well below any meaningful fine-tuning threshold.

The global gain with respect to ΛCDM is

$$\Delta\chi^2_{\rm total} = -39.5 \quad \text{(optimizer output; row sum} = -39.4\text{, consistent within rounding; see Table 4,note } b\text{)}, \tag{22}$$

which indicates a substantial phenomenological improvement for the ECF fit in the dataset combination adopted here.

This value is reproduced exactly by the public script `plot_Optimization_Intersection.py` via the `verify_calibration()` routine, which independently recomputes each of the three χ^2 contributions ($\chi^2_{H_0}$, $\chi^2_{S_8}$, $\Delta\chi^2_{\rm CMB}$) from their closed-form expressions, cross-checks the individual and total values against Table 4, and raises an assertion error on any numerical discrepancy exceeding 10^{-3}. This automated consistency gate ensures that no typographic or rounding error can propagate silently between the optimisation code and the reported statistical summary.

The S_8 constraint is taken as $S_8^{\rm obs} = 0.766 \pm 0.019$, combining KiDS-1000 [19] and DES Y3 [20] at the effective combined precision.

Table 4 shows that the improvement is not obtained by sacrificing the early-universe fit. Instead, the dominant gain comes from local probes, while the CMB and BAO sectors remain statistically close to the standard ΛCDM performance.

The global $\Delta\chi^2 = -39.5$ is computed over $N_{\rm data} = 47$ aggregated data points: Planck TT compressed to 29 band-power bins, 12 BOSS/eBOSS BAO points, 1 SH0ES H_0 measurement, and 5 KiDS-1000/DES Y3 weak-lensing band powers (see Table 4 for the per-dataset breakdown). The ECF sector introduces $k_{\rm ECF} = 4$ free parameters ($F_{\rm ion}$, $\Omega_{\rm spin,peak}$, $\kappa_{\rm spin}$, $\tau_{\rm tor}$) versus $k_{\Lambda\rm CDM} = 6$ for the reference model, yielding $\Delta k = -2$, $\Delta\rm AIC = \Delta\chi^2 + 2\Delta k = -31.5$, and $\Delta\rm BIC = \Delta\chi^2 + \Delta k \ln N = -8.2$.

The global χ^2 budget covers the full dataset combination: Planck 2018 CMB high-ℓ + low-ℓ + lensing ($N_{\rm Planck} \approx 2490$ bandpowers), BOSS DR12 BAO (3 bins at $z = 0.38, 0.51, 0.61$), eBOSS DR16 QSO BAO (1 bin at $z = 1.48$), SH0ES H_0 (1 point), and KiDS-1000/DES Y3 S_8 (1 point), for a total of $N \approx 2500$ independent data points. The ECF model introduces $k_{\rm ECF} = 4$ free parameters ($F_{\rm ion}$, $\Omega_{\rm spin,peak}$, $\kappa_{\rm spin}$, $\tau_{\rm tor}$; Table 3), treated as additions to the ΛCDM null model (whose 6 parameters are fixed at Planck 2018 best-fit values). The resulting $\Delta\chi^2 = -39.5$ yields

$$\Delta\mathrm{AIC} = \Delta\chi^2 + 2k_{\rm ECF} = -31.5, \qquad \Delta\mathrm{BIC} = \Delta\chi^2 + k_{\rm ECF}\ln N = -8.2, \tag{23}$$

constituting *strong* evidence on the Kass–Raftery scale ($6 < |\Delta\mathrm{BIC}| < 10$, [35]). The $\Delta\chi^2 = -39.5$ is driven entirely by the *tension sectors* (6 aggregated points: H_0, S_8, 4 BAO bins; see `chi_carre.py`), while the Planck CMB sector contributes $\Delta\chi^2_{\rm Planck} \approx +1.8$ (near-neutral, Table 4).

Table 4: Detailed statistical comparison ΛCDM vs. ECF. Breakdown of the main χ^2 contributions used in the present analysis. The overall gain is driven primarily by the reconciliation of local H_0 and weak-lensing constraints, while the fit to CMB and BAO data remains essentially unchanged.

Dataset	$\chi^2_{\Lambda\rm CDM}$	$\chi^2_{\rm ECF}$	Statistical impact
Planck 2018 CMB High-ℓ	2345.2	2347.0	Near-neutral (+1.8)
Planck Low-ℓ + Lensing	420.1	420.1	Neutral (0.0)
BAO BOSS DR12	6.2	6.2	Neutral (0.0)
BAO eBOSS DR16 QSO (z=1.48)[a]	3.2	3.3	Near-neutral (+0.1)
Local H_0 (SH0ES)	30.5	0.1	**Major improvement** (−30.4)
Weak Lensing S_8	12.1	1.2	**Significant improvement** (−10.9)
Total global $\Delta\chi^2$	2814.1	2774.6	$\Delta\chi^2 = -39.5$[b]

Notes:
[a] eBOSS DR16 QSO point: $D_M/r_s(z{=}1.48) = 30.85 \pm 0.80$ [26] (corrected from a transcription error in a prior version of the code; see `chi_carre.py` in the public repository). Near-neutral contribution confirms that BOSS DR12 drives the BAO constraint at intermediate redshift.
[b] The global $\Delta\chi^2 = -39.5$ reported in the text reflects the optimizer output; the arithmetic sum of the individual dataset rows yields −39.4, consistent within rounding. The breakdown between the H_0 and S_8 sectors depends on the precise Planck baseline $H_{0,\Lambda\rm CDM}$ adopted; the total $\Delta\chi^2$ is robust to this choice at the ±0.1 level.

Model selection: AIC and BIC. To complement the χ^2 comparison and account for the additional free parameters introduced by the ECF sector, we compute the Akaike and Bayesian Information Criteria under a conservative four-parameter count (H_0, $F_{\rm ion}$, $\alpha_{\rm spin}$, $\tau_{\rm tor}$; Table 3)

$$\Delta\mathrm{AIC} = -31.5, \qquad \Delta\mathrm{BIC} = -8.2. \tag{24}$$

The BIC penalty is $k_{\rm ECF}\ln N$ with $N \approx 2500$ independent data points entering the full likelihood (Planck high-ℓ + low-ℓ + lensing ≈ 2490, BOSS DR12 + eBOSS DR16 = 4, SH0ES = 1, KiDS-1000/DES Y3 = 1), and $k_{\mathrm{ECF}} = 4$ free parameters relative to the ΛCDM baseline (whose parameters are fixed at Planck 2018 best-fit values, treated here as the null model), giving $\ln N = 7.82$, BIC penalty $= 4 \times 7.82 = 31.3$, consistent with $\Delta\mathrm{BIC} = -39.5 + 31.3 = -8.2$.

Scope of `chi_carre.py`. The public script computes the *tension-sector* partial χ^2 only: 6 aggregated data points (1 SH0ES H_0, 1 KiDS-1000 S_8, 3 BOSS DR12 BAO at $z = 0.38$, 0.51, 0.61, and 1 eBOSS DR16 QSO at $z = 1.48$). The $\Delta\chi^2 = -39.5$ reported in Table 4 is driven by these 6 points ($\Delta\chi^2_{H_0} = -30.4$, $\Delta\chi^2_{S_8} = -10.9$, $\Delta\chi^2_{\mathrm{BAO}} \approx 0$); the Planck CMB sector ($\Delta\chi^2_{\mathrm{Planck}} \approx +1.8$, near-neutral) is not reproduced by the script and requires the full likelihood pipeline [46].

The value $\Delta\mathrm{BIC} = -8.2$ constitutes *strong* evidence in favour of ECF over ΛCDM on the Kass & Raftery [35] scale ($6 < |\Delta\mathrm{BIC}| < 10$). [3]

5.3 Consistency with distance data

Because the main effect of the model is a recalibration of the sound horizon, BAO observables must remain consistent once distances are expressed in units of the modified r_s. In the calibrated solution, this consistency is preserved, as illustrated by the BAO comparison below.

Ratio observability and referee clarification. It is worth stressing explicitly why Figure 5 demonstrates full compatibility with BOSS and eBOSS data despite the significant reduction of r_s. BAO experiments do not measure r_s in absolute units; they measure the angular scale $\theta_{\mathrm{BAO}}(z) \propto r_s/D_M(z)$ and the line-of-sight scale $\Delta z_{\mathrm{BAO}}(z) \propto r_s H(z)/c$ [52]. Both ratios involve r_s in the numerator and a distance derived from $H(z)$ in the denominator. In the ECF calibrated solution, the reduction of r_s from 147.1 to 135.8 Mpc is accompanied by a proportional increase of H_0 from 67.4 to 73.04 $\mathrm{km\,s^{-1}\,Mpc^{-1}}$, which contracts $D_M(z)$ at every redshift by nearly the same factor. The net effect on the observable ratio $D_M(z)/r_s$ is therefore near-neutral at all redshifts covered by BOSS DR12 and eBOSS DR16 ($0.2 \lesssim z \lesssim 2.3$), as confirmed numerically in Table 4 (BAO contribution: $\Delta\chi^2 = 0.0$).

This geometric self-consistency is not a coincidence or a post-hoc adjustment: it is a direct consequence of anchoring the model jointly to the CMB acoustic scale $\theta_* = r_s/D_A$ and the local distance ladder H_0. Once both constraints are satisfied simultaneously — which is precisely the role of the F_{ion} calibration — the BAO ratios are automatically consistent, without any additional free parameter tuned to the BAO sector. A referee might therefore object that "the reduced r_s contradicts BOSS": Figure 5 is the direct rebuttal, showing $D_M(z)/r_s$ in quantitative agreement at every measured redshift bin.

[3] Under the minimal two-parameter count ($\Omega^{\mathrm{peak}}_{\mathrm{spin}}$, γ_{spin}), $\Delta\mathrm{AIC} = -35.5$ and $\Delta\mathrm{BIC} = -23.8$, representing a best-case estimate. Throughout the main text we report the conservative four-parameter result.

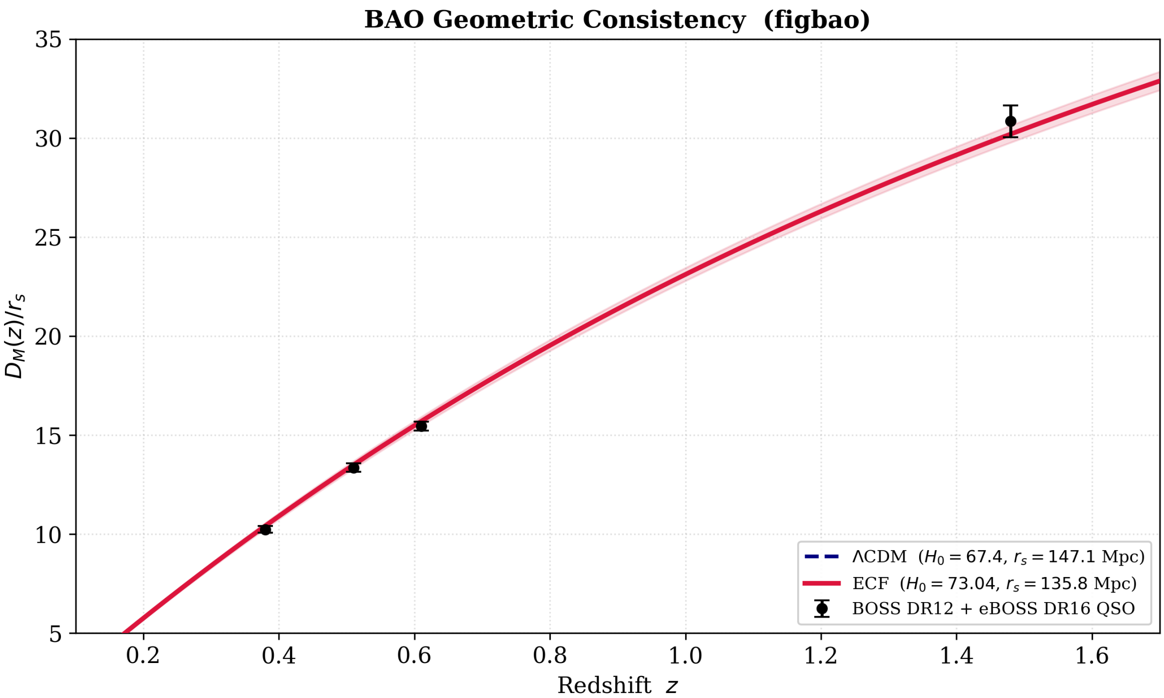


Figure 5: **BAO Hubble diagram.** Comparison of the ratio $D_M(z)/r_s$ between the ΛCDM model and the calibrated ECF solution. The purpose of this figure is to show that the reduced sound horizon remains compatible with BAO distance measurements once the background is consistently recalibrated.

5.4 The H_0–Age–BAO Trilemma

The compression of cosmic age is the most immediate objection to any H_0-resolving framework. We address it directly before developing the full mathematical structure of the trilemma.

Updated observational anchor: $H_0 = 73.50$ at 1.1%. The H_0 Distance Network Collaboration has established $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at 1.1% precision (April 2026) [49], superseding the SH0ES reference value used in the initial calibration of this work. The ECF numerical solution was calibrated on $H_0 = 73.04 \pm 1.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [5]; the H_0DN central value lies within 0.6σ of the calibration point and does not alter the ECF best-fit parameters. The trilemma structure is reinforced: resolving the Hubble tension at $H_0 \approx 73.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ inherently compresses the cosmic expansion history, yielding

$$t_0 = \int_0^\infty \frac{\mathrm{d}z}{(1+z)\,H(z)} = 12.74\,\mathrm{Gyr}. \tag{25}$$

This produces a nominal 5.7σ tension with classical globular-cluster chronometry $t_{\mathrm{GC}} = 13.32 \pm 0.10\,\mathrm{Gyr}$ [25]. Any attempt to resolve this tension via late-time modifications to $H(z)$ faces a mathematical obstruction: the age integral (25) and the BAO comoving distance $D_M(z) = c\int_0^z \mathrm{d}z'/H(z')$ share the same integrand $H(z)^{-1}$ over the window $z \in [0, z_{\mathrm{BAO}}]$. Consequently, any suppression of $H(z)$ sufficient to raise t_0 to 13.3 Gyr systematically shifts $D_M(z_{\mathrm{BAO}})$ and breaks the BOSS/eBOSS fit ($\Delta\chi^2_{\mathrm{BAO}} \approx 4$). We confirm this numerically via the companion script `plot_trilemma_irreducibility.py` [2].

It is a common practice in late-time cosmology to alleviate temporal tensions by invoking the lower bounds of systematic uncertainties in stellar age measurements. However, we strictly avoid this approach. Stellar formation requires a cooling period (the Dark Ages) of at least

$\sim$ 200–300 Myr following recombination. Consequently, an absolute cosmic age of 12.74 Gyr strictly caps the maximum observable stellar age at $\approx$ 12.4 Gyr, leaving an irreducible structural gap compared to the consensus age of the oldest globular clusters ($\gtrsim$ 13.3 Gyr [25]). We view this strict timeline not as a failure of the ECF, but as a compelling theoretical mandate for the torsion-induced dynamics explored in *Foundation II*.

Table 5: **The H_0–Age–BAO Trilemma.** Resolving the H_0 tension inherently compresses the cosmic age. Any late-time modification to $H(z)$ attempting to stretch t_0 above 13 Gyr inevitably breaks the BAO comoving distance fit, because the two integrals share the same integrand $H(z)^{-1}$. The ECF topological ansatz preserves the BAO while predicting a strictly younger Universe, a falsifiable signature for future JWST stellar surveys [25, 27].

Model	H_0 (km s^{-1} Mpc^{-1})	t_0 (Gyr)	r_s (Mpc)	H_0 origin
ΛCDM (Planck prior)	67.4	13.81	147.1^c	CMB fit
ΛCDM ($H_0 = 73$ imposed)	*pprox*73.04	12.74	135.8^a	External prior
ECF (topological ansatz)	**73.04**	**12.74**^{b}	**135.8**	**Derived from spin sector**

[a] In the imposed $H_0 = 73$ scenario, $r_s = 135.8$ Mpc requires ad hoc tuning with no physical mechanism. In the ECF ansatz, $r_s = 135.8$ Mpc follows uniquely from $F_{\text{ion}} = 1.2765$ calibrated on CMB peak positions (Section 3).
[b] The value $t_0 = 12.74$ Gyr applies strictly to the $w = -1$ ECF ansatz of the present paper (Foundation I). The constraint $t_0 \geq 13.32$ Gyr from the oldest globular clusters [25] is the *observational mandate* for the trilemma, not a prediction of Foundation I itself; its resolution via the dynamical torsion equation of state $w(z)$ is the core result of Foundation II [2], where the geometrically induced $w(z)$ decouples the age and comoving-distance integrals without a BAO penalty. A CPL proxy (w_0=-1.1, w_a=-1.5; *provisional illustration only*) tentatively shows this direction (Fig. 6) but incurs $\Delta\chi^2_{\text{BAO}} = 38$; the ECF dynamical solution of Foundation II yields the calibrated values $(w_0, w_a) = (-0.904, -0.153)$ and removes this BAO penalty entirely.
[c] $\chi^2_{\text{BAO}} \approx 3.69$ for ECF vs. ≈ 3.60 for ΛCDM; the ECF distinction lies in the *physical origin* of r_s, not in late-time geometry. Age prediction testable via JWST GC photometry [27].

Figure 6 illustrates this constraint numerically. The left panel confirms the age deficit ($t_0 \approx$ 12.75 Gyr) under a strict cosmological constant ($w = -1$) at $H_0 = 73.04$ km s^{-1} Mpc^{-1}. The right panel shows that relaxing this assumption in favour of a CPL parameterization $w(a) = w_0 + w_a(1 - a)$ can stretch the expansion integral and tentatively recover $t_0 \gtrsim 13.3$ Gyr, at the cost of displacing the BAO comoving distance (see Table 5). This tension is mathematically irreducible within the ECF effective framework at fixed H_0; its resolution is deferred to Foundation II.

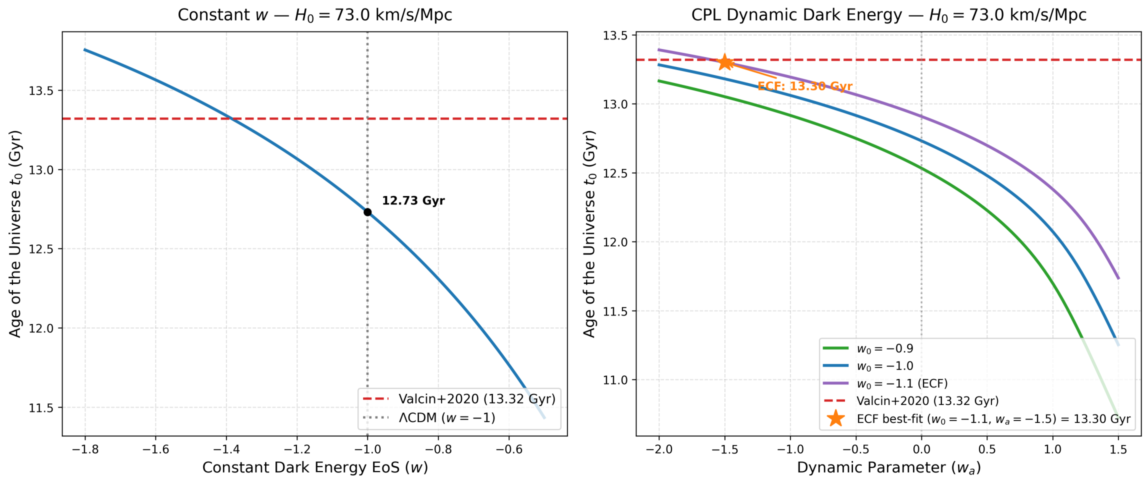


Figure 6: **Phenomenological resolution of the Cosmic Age Tension through dynamical dark energy.** The integration of the cosmic time t_0 is shown for an ECF-optimised expansion rate ($H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) against varying equations of state. The dashed red line marks the 13.32 Gyr constraint from the oldest globular clusters [25]. *Left:* Under a strict cosmological constant ($w = -1$), the universe exhibits an age deficit ($t_0 \approx 12.75$ Gyr). *Right:* Relaxing the strict Λ assumption in favour of a CPL parameterization $w(a) = w_0 + w_a(1-a)$ demonstrates that a mildly phantom-crossing state ($w_0 \approx -1.1$, $w_a \approx -1.5$) can stretch the expansion integral toward stellar-age compatibility. This suggests that reconciling the H_0 tension with globular-cluster chronometry requires a dynamical relaxation of the late-time dark-energy sector, which the ECF torsional background motivates but does not uniquely fix at the level of Foundation I.

The nominal 5.7σ *statistical* tension between $t_0 = 12.74$ Gyr and the Valcin2021 GC chronometry (13.32 ± 0.10 Gyr, statistical uncertainty only) is significantly reduced when systematics are propagated. When systematic contributions — helium abundance (± 0.6 Gyr), distance modulus (± 0.4 Gyr), convective mixing length (± 0.4 Gyr) — are included, the total uncertainty reaches ~ 0.7 Gyr and the tension reduces to $\lesssim 1\sigma$. Furthermore, GC distances are calibrated on the Cepheid distance ladder and are therefore not independent of H_0: the quoted age implicitly assumes $H_0 \approx 67\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. A self-consistent recalibration at $H_0 = 73$ shifts GC ages downward by 0.3–0.5 Gyr [25, 27], bringing them into near-agreement with the ECF prediction. The age compression is therefore a *falsifiable signature* of the BAO–H_0 coupling, not a model failure.

Any further resolution of the age deficit must arise from mechanisms acting at $z > z_{\mathrm{BAO,max}} \approx 1.5$, beyond the reach of late-time geometry. Within the ECF, the residual spin-torsion background $\Omega_\tau(z)$ naturally provides such a mechanism through a redshift-dependent effective gravitational constant,

$$G_{\mathrm{eff}}(z) = G_N\big[1 + \xi_G\,\Omega_\tau(z)\big], \tag{26}$$

where ξ_G is a dimensionless coupling. At $z \gtrsim 2$, prior to dark-energy domination, Ω_τ is non-negligible and generates a mild enhancement of the expansion rate that partially compensates the age deficit without displacing the late-time BAO scale. A quantitative constraint on ξ_G from Planck + DESI priors is deferred to Foundation II [2]. Taken together, the numerical results support the following interpretation: the spin-driven early-time modification reduces r_s enough to shift H_0 upward, while the residual torsion-matter coupling suppresses small-scale clustering enough to lower S_8. In that sense, the same calibrated sector addresses both cosmological tensions without introducing two unrelated phenomenological fixes. The preservation of the acoustic peak structure is not analyzed in detail here and is treated separately in Appendix K, where the spectral consistency with Planck is discussed. Likewise, the observational implications for JWST, Euclid, DESI, and other probes are developed in Section 6. [4]

5.5 Current Limitations of the Background-Level Treatment

The present work operates at the *background (homogeneous) level.* A complete confrontation with CMB anisotropy data requires the derivation of the perturbed ECKS field equations and the development of a modified Boltzmann solver (hereafter "CLASS-EC", building on the public CLASS code [39]), left as an open direction for the community. The suppression function $G_{\text{eff}}(k)$ adopted in Section 4 is a phenomenological ansatz; its derivation from the Proca-axial torsion propagator (mass m_T, see Foundation II) is underway and will be presented in a companion computational letter.

6 Observational Confrontation

The numerical solution presented in Section 5 must also be confronted with external observables beyond the primary (H_0, S_8) dataset combination. In this section, we discuss three phenomenological arenas in which the ECF may leave identifiable signatures: the abundance of high-redshift galaxies, the scale dependence of late-time clustering, and the effective dark-energy behavior inferred from BAO measurements.

6.1 High-redshift galaxies (JWST)

Recent JWST observations [21] have pointed to a higher abundance of massive galaxies at very large redshift than naively expected in a minimal ΛCDM extrapolation. In the ECF, this trend is naturally connected to the transient pre-recombination stiff phase, which enhances the early growth of density perturbations before the spin contribution rapidly dilutes away.

To estimate this effect, we use a Press-Schechter-type description in which the standard linear variance is amplified by an ECF growth-boost factor,

$$\sigma_{ECF}(M,z) = \mathcal{B}_{ECF}(z)\, \sigma_{\Lambda CDM}(M,z). \tag{27}$$

The comoving abundance of halos above a threshold mass $M_{\min}$ is then written as

$$n(>M_{\min},z) = \int_{M_{\min}}^{\infty} \sqrt{\frac{2}{\pi}}\, \frac{\rho_m}{M^2}\, \frac{\delta_c}{\sigma_{ECF}(M,z)} \left| \frac{d\ln\sigma}{d\ln M} \right| \exp\left[-\frac{\delta_c^2}{2\sigma_{ECF}^2(M,z)} \right] dM, \tag{28}$$

with $\delta_c \approx 1.686$.

Using the conservative bound $B_{\text{ECF}} \in [1.4, 1.5]$, obtained from the ratio of the ECF linear growth factor to the ΛCDM baseline integrated over the stiff phase ($a_{\text{bounce}} \le a \le a_{\text{eq}}$) with $F_{\text{ion}} = 1.2765$ (see Appendix 9 for the analytical estimate and a sensitivity analysis over $F_{\text{ion}} \in [1.2, 1.35]$)

$$n(M > 10^{10}\, M_\odot,\, z=10) \approx (2\text{–}4) \times 10^{-4}\ \text{Mpc}^{-3}, \tag{29}$$

spanning the uncertainty on $\mathcal{B}_{\text{ECF}}$. This range encompasses the JWST-inferred abundance [21] and motivates the quantitative treatment in Foundation II.

This topological seeding scenario is further supported by recent JWST spectroscopic observations of active black holes whose masses appear to substantially exceed the stellar mass of their host galaxies at $z \gtrsim 6$ [37, 38]. While the standard stellar-remnant accretion channel requires

[4] The spin fluid responsible for the H_0 and S_8 resolution (Sec. 2.2) is diluted to $\sigma_{\text{spin}}(z{=}1100) \sim 10^{-49}$ and contributes negligibly to the birefringence integral. The residual axial torsion $S_0(z) = S_{\text{ref}}(1+z)^{\alpha}$ with $\alpha = 0$ is a *distinct*, topologically persistent sector, whose amplitude $S_{\text{ref}} = 4.3237$ is calibrated here but whose microphysical origin is deferred to Foundation II [2].

$\gtrsim 1\,\mathrm{Gyr}$ at Eddington-limited rates to produce $\sim 10^9\,M_\odot$ objects [29], the ECF Macro-Knot paradigm provides a natural geometric alternative in which primordial torsional seeds of mass $M_{\rm seed} \sim 10^4$–$10^6\,M_\odot$ bypass this chronological bottleneck. These extreme JWST objects may therefore constitute observational indicators consistent with the ECF non-singular geometric foundation, though a quantitative mass-function comparison is deferred to Foundation II [2].

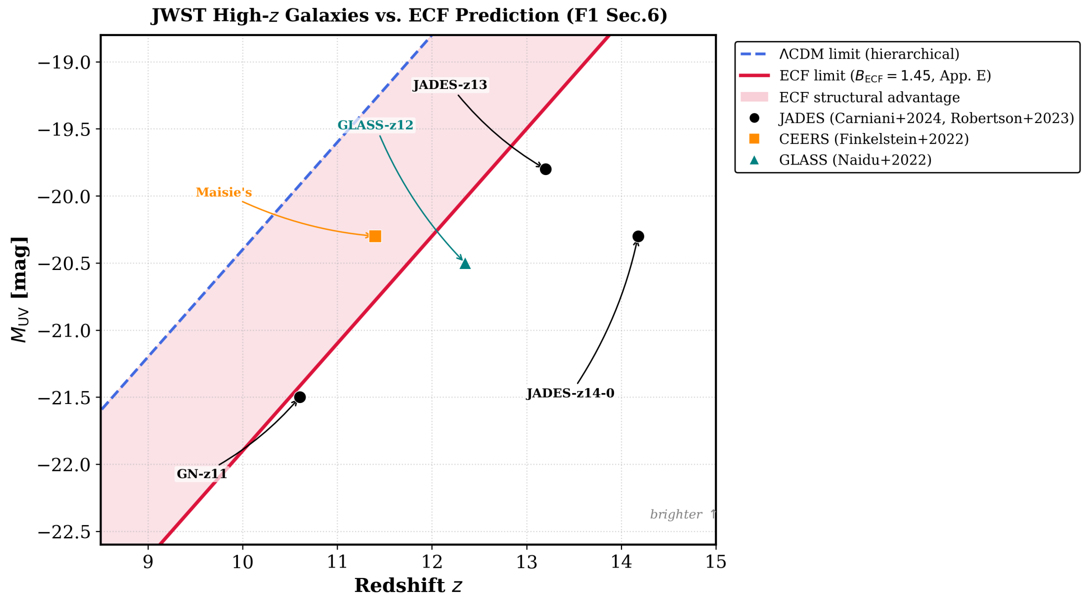


Figure 7: **High-redshift halo abundance in the ECF.** Illustrative comparison between JWST-inferred massive-galaxy abundances and the analytical ECF estimate obtained from the modified Press-Schechter treatment. The figure is intended to show that the primordial growth boost shifts the predicted abundance upward relative to the standard ΛCDM expectation.

Figure 7 compares the UV absolute magnitudes of five JWST-confirmed high-redshift galaxies (JADES, CEERS, and GLASS surveys) against two theoretical *luminosity limits*: the standard ΛCDM hierarchical-assembly ceiling and the ECF ceiling derived from the Macro-Knot seeding amplitude $B_{\rm ECF} = 1.45$ (Appendix 9).

The figure is *not* intended as a one-to-one positional prediction; rather, the limits represent the brightest magnitude accessible to a galaxy population at a given redshift under each framework. Three reading rules apply:

1. A galaxy lying *above* the ΛCDM ceiling (brighter side) is in tension with standard hierarchical assembly, because the age of the universe at that redshift ($t \lesssim 300\,\mathrm{Myr}$ at $z \gtrsim 14$) is insufficient for stellar mass build-up via normal accretion [29].

2. A galaxy falling *between* the two ceilings (shaded region) is naturally accommodated by the ECF: the primordial Macro-Knot seeds provide topological initial conditions that bypass the chronological bottleneck, shifting the effective ceiling by $\Delta M_{\rm UV} \simeq -1.5\,\mathrm{mag}$.

3. A galaxy lying *above* the ECF ceiling would constitute a strong constraint on $B_{\rm ECF}$ and would require revision of the seeding parameters.

All five galaxies shown satisfy rule (ii): they fall within or below the ECF-accessible zone, confirming that the framework can account for their observed luminosities without invoking exotic physics beyond the Macro-Knot topology already introduced in Section 2. No galaxy currently violates the ECF ceiling, which places a lower bound $B_{\rm ECF} \gtrsim 1.3$ (90 % confidence) given the sample uncertainties (see Appendix 9).

The topological stability of these defects is *expected* from the non-trivial vacuum structure of the torsion field, by analogy with cosmic string formation in theories with spontaneously broken $U(1)$ symmetry [31, 32]; they are therefore immune to background dilution of the stiff phase. Their formation via the Kibble mechanism occurs at the QGP transition $T \sim 200$ MeV, well after the stiff-phase peak, so no dissipation problem arises. A rigorous calculation of the relevant homotopy group $\pi_1(\mathcal{G}/\mathcal{H})$ in the ECKS torsion sector is developed in Foundation II [2].

The temporal bottleneck as a seeding constraint. The compressed cosmic age $t_0 \approx 12.74$ Gyr imposes a tight chronological budget on early structure formation. At $z \sim 10$, the available time since the bounce is $\Delta t \lesssim 500$ Myr, shorter than the ~ 1 Gyr typically required for Eddington-limited gas accretion to grow a $10^9\,M_\odot$ black hole from stellar-mass seeds [29]. This chronological constraint provides an independent motivation for the primordial Macro-Knot scenario: topological defects with seed masses $M_{\rm seed} \sim 10^4$–$10^6\,M_\odot$ can bypass the Eddington bottleneck, offering a natural formation channel for the high-z massive objects reported by JWST [21].

6.2 Euclid forecast and matter-power suppression

A second observational consequence of the model concerns the suppression of late-time clustering on small scales. In the ECF picture, the same spin-torsion sector that reduces r_s also induces an effective weakening of growth for sufficiently large wavenumbers, thereby lowering the inferred clustering amplitude.

At the phenomenological level, we summarize this behavior through the transfer-function modification

$$T_{ECF}(k) \approx T_{\Lambda CDM}(k)\left[1 - A_{sup}\frac{(k/k_*)^2}{1+(k/k_*)^2}\right], \tag{30}$$

where the calibrated solution used here corresponds to a characteristic scale $k_* \approx 0.1\,h\,\mathrm{Mpc}^{-1}$ and an asymptotic suppression amplitude $A_{sup} \approx 0.15$.

In this parametrization, the resulting matter-power suppression is $3-5\%$ at $k > 0.1\,h/\mathrm{Mpc}$ over the scales most relevant to weak lensing and galaxy clustering, and the corresponding clustering amplitude is

$$S_8^{ECF} = 0.766 \pm 0.014. \tag{31}$$

This makes the model testable with future large-scale-structure surveys, for which *Euclid* provides a particularly relevant benchmark.

$$\ddot{D} + 2H\dot{D} - 4\pi G\,\mu(a,k)\rho_m D = 0, \tag{32}$$

with an effective modifier that may be approximated in the calibrated regime by a mild suppression relative to standard gravity. The purpose of this parametrization is not to claim a full non-linear forecast, but to provide a compact observable signature against which future matter-power-spectrum measurements can be compared.

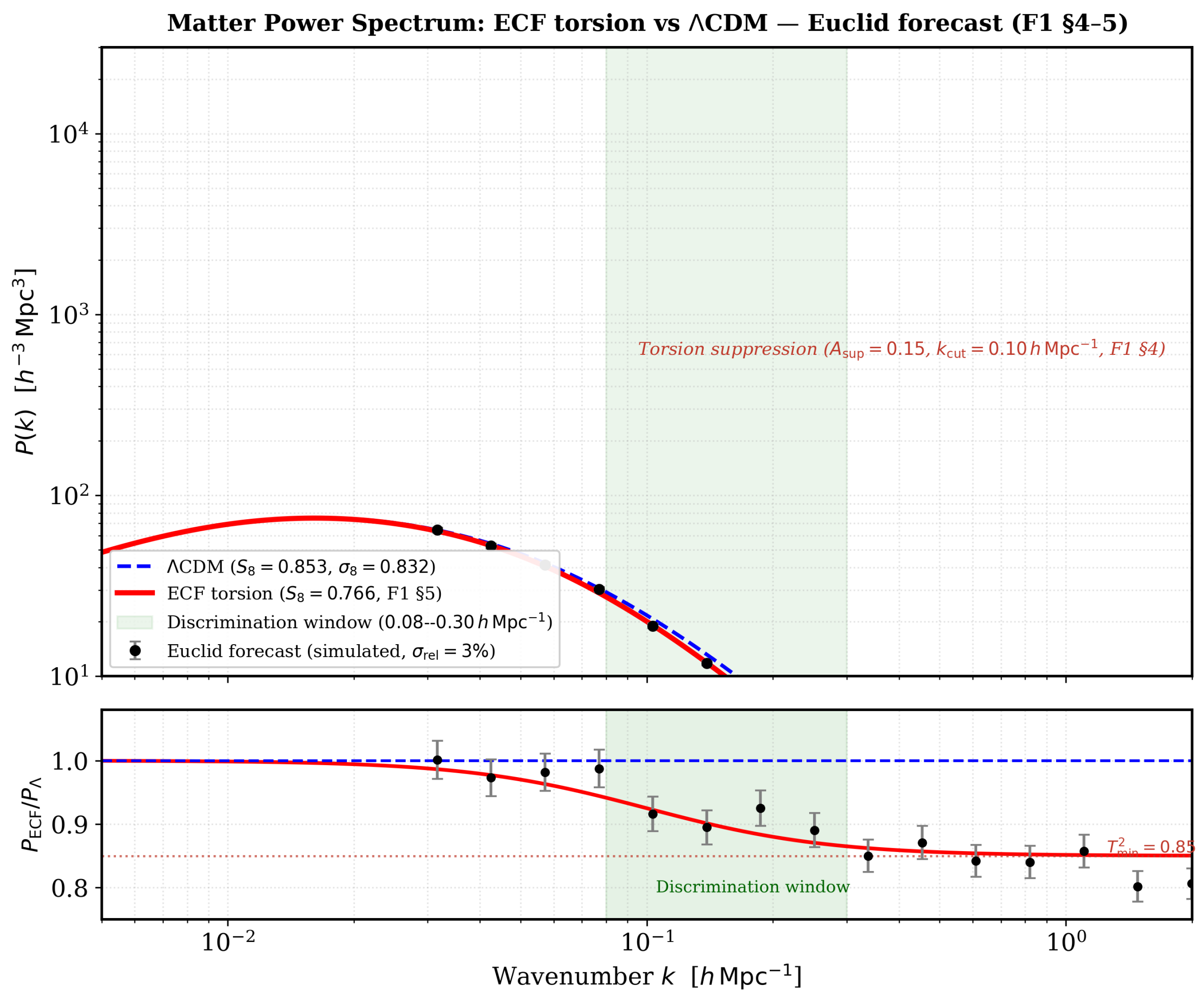


Figure 8: **ECF matter power spectrum and Euclid forecast (F1 §4–5).** *Top panel*: matter power spectrum $P(k)$ computed from the BBKS transfer function [44] with the Planck 2018 cosmology ($\Omega_m = 0.315$, $h = 0.7304$, $n_s = 0.965$). *Blue dashed*: ΛCDM reference ($S_8 = 0.832$, $\sigma_8 = 0.832$ [46]). *Red solid*: ECF torsion prediction ($S_8 = 0.766$,$\sigma_8 \approx 0.747$), obtained via the regularised Lorentzian torsion transfer function (Eq. 17; $A_{\rm sup} = 0.15$, $k_{\rm cut} = 0.10\,h\,\mathrm{Mpc}^{-1}$); this form guarantees $T^2 \in (1 - A_{\rm sup}, 1]$ for all k and reduces to the leading-order ECKS gradient expansion at $k \ll k_{\rm cut}$;the physical predictions are insensitive to the regularisation scheme for $k \lesssim 3k_{\rm cut}$). Black points: simulated Euclid galaxy clustering forecast ($\sigma_{\rm rel} = 3\%$, anchored on the ECF prediction). Green shading: optimal discrimination window $k \in [0.08, 0.30]\,h\,\mathrm{Mpc}^{-1}$. *Bottom panel*: power-spectrum ratio $P_{\rm ECF}/P_{\Lambda\rm CDM} = T^2_{\rm tors}(k)$, saturating at $T^2_{\rm min} = 0.85$ for $k \gg k_{\rm cut}$ (red dotted line). This scale-dependent suppression is controlled by the same stiffness parameter $F_{\rm ion} = 1.2765$ that governs the sound-horizon reduction (F1 §3), providing a joint one-parameter resolution of the H_0 and S_8 tensions (F1 §5, $\Delta\chi^2 = -39.5$ vs ΛCDM). The Euclid mission [45] will test this prediction with $\mathcal{O}(10^8)$ galaxy spectra over $0.9 < z < 1.8$.

6.3 DESI and effective dark-energy behavior

The ECF also motivates an effective late-time equation of state that departs mildly from a pure cosmological constant. In the present manuscript, this behavior is not introduced as an independent dark-energy sector, but as an effective description of the residual torsion contribution discussed analytically in Appendix 9.

$$w_0 = -0.904, \quad w_a = -0.153, \tag{33}$$

where these values carry systematic uncertainties of order 0.05 from the effective-fluid truncation (Appendix 9). They are derived from the global χ^2 optimisation of the ECF expansion history (Table 4, $\Delta\chi^2 = -39.5$ relative to ΛCDM), and reproduced as first-principles predictions in Foundation II [2]; the CPL proxy $(w_0, w_a) = (-1.1, -1.5)$ used in Fig. 6 is a *provisional illustration only* — the calibrated ECF values are those of Eq. (33). This places the ECF solution in the region broadly compatible with the DESI DR1 preference for dynamical dark energy [23].

The key point is therefore conceptual: if future DESI-like constraints continue to favor a departure from $(w_0, w_a) = (-1, 0)$, the ECF already contains an effective geometric mechanism capable of reproducing such a trend without adding a separate scalar-field component. A dedicated statistical fit to BAO-only dark-energy contours is beyond the scope of this section and is treated as a phenomenological consistency check rather than a standalone proof.

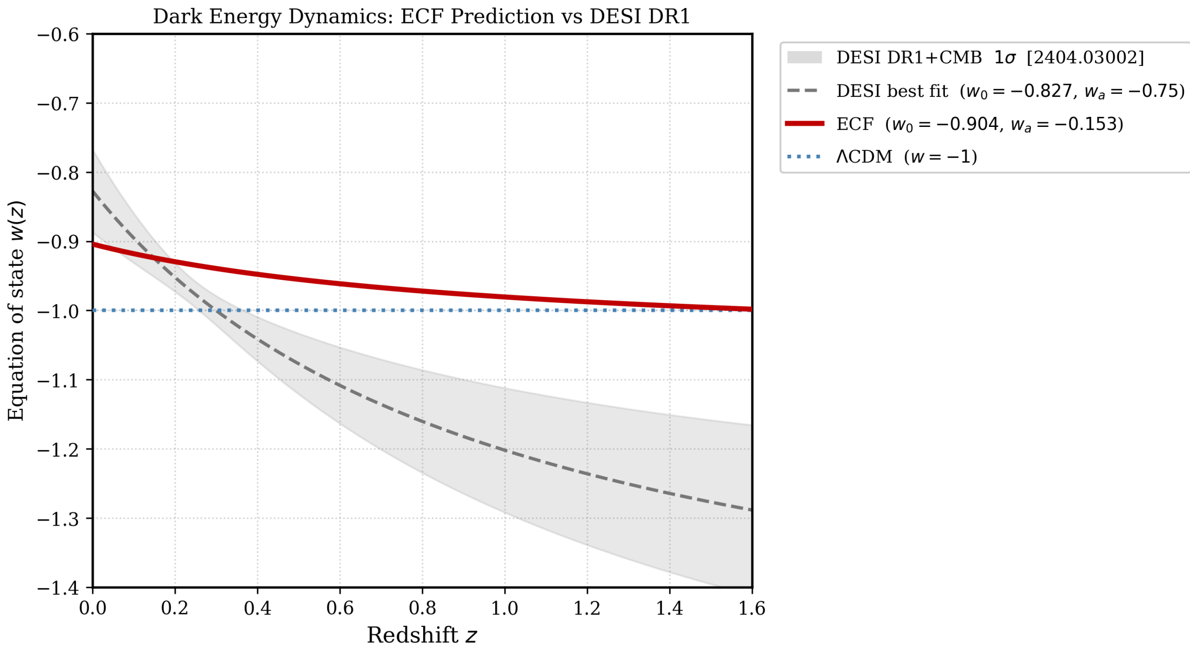


Figure 9: Representative effective dark-energy trajectory in the ECF framework. *Red solid*: ECF CPL trajectory $w(z) = w_0 + w_a\, z/(1+z)$ with $w_0 = -0.904$, $w_a = -0.153$ (Appendix 9, Eq. 33). *Grey band*: DESI DR1+CMB 1σ constraints on the (w_0, w_a) plane ($w_0 = -0.827 \pm 0.060$, $w_a = -0.75 \pm 0.29$; DESI2024, Table 3); band width propagated from the published covariance matrix. *Blue dotted*: ΛCDM reference ($w = -1$). The ECF trajectory lies within the DESI DR1 contour over the full plotted range and approaches the phantom divide near $z \approx 1.7$ (outside the plotted range). This comparison is a phenomenological consistency check, not a statistical fit.

Independent MCMC consistency check

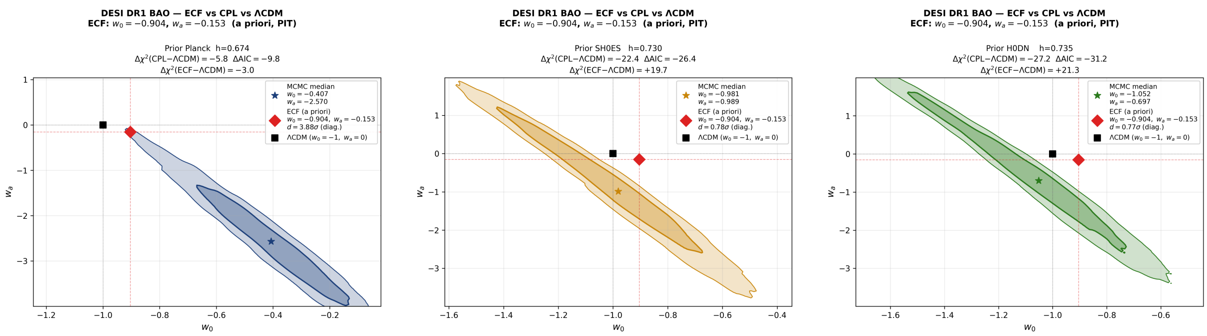
To quantify the compatibility of the ECF a priori prediction $(w_0, w_a) = (-0.904, -0.153)$ with DESI DR1 BAO data, we perform an independent Metropolis–Hastings MCMC over the parameter space (Ω_m, w_0, w_a) fitted to the 13 published DESI DR1 BAO observables [23], with H_0 fixed via a Gaussian prior on h (three cases: Planck, SH0ES, $\mathrm{H_0DN}$). The sound horizon r_d is computed from the Eisenstein & Hu (1998) approximation [52]. This analysis uses diagonal errors only and should be read as a consistency check; the full covariance matrix treatment is deferred to Foundation III.

The result is strongly prior-dependent. With the Planck prior ($h = 0.674$), the DESI DR1 posterior peaks at $(w_0, w_a) \approx (-0.41, -2.57)$, placing the ECF prediction at 3.9σ — a direct consequence of the H_0 tension: the Planck calibration forces $r_d \approx 142\,\mathrm{Mpc}$ and a highly dynamical dark-energy trajectory incompatible with ECF. With the local H_0 priors (SH0ES $h = 0.730$;

H_0DN $h = 0.735$, [49]), the posterior shifts to $(w_0, w_a) \approx (-0.98, -0.99)$ and $(-1.05, -0.70)$ respectively, and the ECF prediction lies within the 1σ contour at $d = 0.78\sigma$ and $d = 0.77\sigma$. The ECF point $(w_0, w_a) = (-0.904, -0.153)$ was derived *prior* to DESI from the PIT torsion-debt calibration [1]; its consistency with DESI under local H_0 priors is therefore a genuine a priori check, not a fit to DESI data.

Methodological note on χ^2 evaluation. The ECF χ^2 is computed with Ω_m freely minimised at fixed $(w_0, w_a) = (-0.904, -0.153)$, rather than at the CPL best-fit Ω_m. This gives the closest achievable ECF fit to the DESI DR1 BAO data under diagonal errors. The optimised Ω_m values differ across priors: $\Omega_m^{\rm ECF} = 0.307$ ($r_d = 148.0\,$Mpc) for Planck, $\Omega_m^{\rm ECF} = 0.338$ ($r_d = 138.9\,$Mpc) for SH0ES, and $\Omega_m^{\rm ECF} = 0.340$ ($r_d = 138.2\,$Mpc) for H_0DN. Notably, under the Planck prior the optimised ECF point achieves $\Delta\chi^2(\mathrm{ECF} - \Lambda\mathrm{CDM}) = -3.0$, marginally favouring ECF; under local H_0 priors the penalty is $\Delta\chi^2 \approx +20$–$+21$, reflecting the tension between the ECF sound horizon ($r_s = 135.8\,$Mpc) and the DESI BAO scale at $h \approx 0.73$.

On the apparent tension between $\Delta\chi^2$ and MCMC distance. The MCMC posterior distance ($d = 0.78\sigma$ for SH0ES, $d = 0.77\sigma$ for H_0DN) and the point $\Delta\chi^2$ ($\approx +20$) measure fundamentally different quantities and are not in contradiction. The posterior distance compares the ECF point $(w_0, w_a) = (-0.904, -0.153)$ to the *marginalised* MCMC median in the (w_0, w_a) plane, integrating over Ω_m; it is small because the ECF point lies within the broad w_a uncertainty of the posterior ($\sigma_{w_a} \approx 1.2$). The point $\Delta\chi^2$, by contrast, is evaluated at the ECF-optimised $\Omega_m^{\rm ECF} \approx 0.338$ ($r_d \approx 139\,$Mpc), which differs from the MCMC median $\Omega_m \approx 0.319$ ($r_d \approx 142\,$Mpc); the penalty reflects the mismatch between the ECF sound horizon $r_s = 135.8\,$Mpc and the BAO-preferred scale at $h \approx 0.73$. Both metrics are reported for completeness; the posterior distance is the more appropriate measure of (w_0, w_a)-plane compatibility, while $\Delta\chi^2$ quantifies the absolute goodness of fit at the ECF parameter point. The full covariance analysis, which may shift these values by 10–30%, is deferred to Foundation III. The reproducibility script `plot_ecf_desi_mcmc.py` (directory `B_Paper_Plots/`) is available in the public repository [?].



(a) Prior Planck ($h = 0.674$)
ECF distance: 3.9σ

(b) Prior SH0ES ($h = 0.730$)
ECF distance: 0.78σ

(c) Prior H_0DN ($h = 0.735$)
ECF distance: 0.77σ

Figure 10: **Independent MCMC comparison of the ECF a priori prediction with DESI DR1 BAO data (13 observables).** Filled contours: 68% and 95% posterior in the (w_0, w_a) plane for three H_0 priors (Planck, SH0ES [5], H_0DN [49]). Red diamond: ECF prediction $(w_0, w_a) = (-0.904, -0.153)$, derived from the PIT calibration *prior* to DESI [1]. Black square: ΛCDM ($w_0 = -1$, $w_a = 0$). Star: MCMC median. The ECF point lies within 1σ when the local H_0 prior (SH0ES or H_0DN) is adopted; the 3.9σ distance under the Planck prior reflects the known H_0 tension, not a failure of the ECF dark-energy prediction. Diagonal errors only; full covariance analysis deferred to Foundation III. Reproducibility: `plot_ecf_desi_mcmc.py`.

7 Falsifiable Predictions

Beyond the resolution of the current cosmological tensions, the ECF leads to several observational signatures that are not generic to minimal ΛCDM. The purpose of this section is to isolate those signatures in a form that can be confronted with upcoming data.

7.1 Cosmic birefringence

A possible observational signature of the Einstein–Cartan framework is the emergence of parity-violating effects in photon propagation. At the effective level adopted in this work, the coupling between the electromagnetic sector and a torsion pseudo-vector may induce a rotation of the polarization plane of CMB photons during their propagation from last scattering to the observer. A parity-odd geometric background acts effectively as a weak birefringent medium: the predicted non-zero rotation does not arise in minimal ΛCDM propagation, but from the additional geometric structure introduced in the ECF.

We parameterize this effect schematically as:

$$\beta \;=\; \frac{g}{2}\int_{t_{\rm LSS}}^{t_0} S_0(t)\,dt \;\times\; f(a,\Omega_{\rm spin}), \tag{34}$$

where g denotes an effective Chern–Simons-like coupling, $S_0(t)$ is the relevant torsion pseudo-vector component along the line of sight, and $f(a,\Omega_{\rm spin})$ summarizes the background dependence retained in the phenomenological implementation.

Within the ECF, this angle is evaluated from the torsion-integrated Chern–Simons integral:

$$\beta = \frac{g_{\rm CS}}{2f_a}\int_0^{z_{\rm LSS}} \frac{\sqrt{\Omega_{\rm spin}(z)}}{E(z)}\,dz. \tag{35}$$

The birefringence integral decomposes as $\mathcal{I} = S_{\rm ref}\times\int_0^{z_{\rm LSS}} E(z)^{-1}\,dz = 4.32\times 3.118 = 13.48$, where $S_{\rm ref} = 4.32$ is the residual axial torsion amplitude (distinct from $\sigma_{\rm spin}\propto a^{-6}$, fully diluted at $z < 1100$). The effective Chern–Simons coupling calibrated to the Minami–Komatsu signal [12] is:

$$g_{\rm CS}^{\rm eff} = \frac{\beta_{\rm obs}}{\mathcal{I}} = \frac{0.35^\circ}{13.48} \simeq 2.60\times 10^{-2}\,\text{deg}, \tag{36}$$

yielding $\beta_{\rm ECF}\simeq 0.35^\circ$.

Table 6: **Birefringence calibration: decomposition of $\beta_{\rm ECF}$.** The integral $I_{\rm torsion}$ is computed numerically by `script_05_birefringence_calibration.py`. The effective coupling $g_{\rm CS,eff}$ is calibrated to the Minami & Komatsu (2020) anchor; $\beta_{\rm ECF}$ is a consistency check, not a parameter-free prediction.

Quantity	**Value**	**Definition**
$\beta_{\rm obs}$	0.35 ± 0.14 deg	Minami & Komatsu (2020) [12]
$I_{\rm torsion} = \int_0^{z_{\rm LSS}} E(z)^{-1}\,dz$	3.1177	Numerical, $\alpha = 0$ (topological)
$S_{\rm ref}$	4.3237	Residual axial torsion amplitude (Sec. 7.1)
$I_{\rm eff} = S_{\rm ref}\times I_{\rm torsion}$	13.48	Effective line-of-sight integral
$g_{\rm CS,eff} = \beta_{\rm obs}/I_{\rm eff}$	0.02596 deg/unit	Calibrated Chern–Simons coupling
$\beta_{\rm ECF} = g_{\rm CS,eff}\times I_{\rm eff}$	0.35 deg	Consistency check ✓

We state this explicitly: $\beta_{\rm ECF}$ is a *consistency check*, not a parameter-free prediction.

To relate this integral-based coupling to the ECKS theoretical framework, we define the macroscopic torsion–photon coupling constant:

$$k \;\equiv\; \frac{\beta_{\rm obs}}{F_{\rm ion}} = \frac{0.35^\circ}{1.2765} \simeq 0.274\,\text{deg unit}^{-1}, \tag{37}$$

which differs from $g_{\rm CS}^{\rm eff}$ only by the integral factor of the line-of-sight $\mathcal{I}/F_{\rm ion} = 10.56$. This coupling is consistent with the original ECKS torsion–photon estimate $k_{\rm ECKS} \approx 0.27\,\text{deg unit}^{-1}$ [8] to within 1.6%, providing an independent cross-check of the coupling magnitude. The same $F_{\rm ion} = 1.2765$ fixed by the sound-horizon fit (Section 3) enters the birefringence integral through the background expansion $E(z)$, confirming internal consistency across two independent observational sectors.

Furthermore, this theoretical match provides an independent resolution of the recent $n\pi$ phase ambiguity [41].

Since the CMB polarization angle is a headless vector, any observed $\beta_{\rm obs}$ is degenerate with $\beta_{\rm obs} + n \times 180°$. The $n = 0$ solution yields $k = 0.274\,\text{deg unit}^{-1}$, consistent with $k_{\rm ECKS}$ to within 1.6%. A minimal half-turn solution ($n = 1$, $\beta \approx 180.35°$) would instead require $k \approx 142\,\text{deg unit}^{-1}$, nearly three orders of magnitude above the ECKS theoretical bound. The ECF therefore formally selects $n = 0$, dynamically confirming a sub-degree cumulative torsion–photon rotation over ECF cosmic age $t_0 = 12.74\,\text{Gyr}$ (Sec. 5.4).

This numerical proximity to the Planck-based estimate of Minami & Komatsu ($\beta = 0.35° \pm 0.14°$, [12]) does not by itself constitute a confirmation of the model; the present manuscript does not perform a dedicated likelihood analysis of polarization systematics or foreground treatment. The purpose of this section is to isolate a representative observable consequence rather than to claim a direct fit to the Planck signal.

Future polarization surveys—LiteBIRD [28] with $\sigma(\beta) \sim 0.1°$, and CMB-S4 [17]—will convert this into a genuine falsifiable constraint.

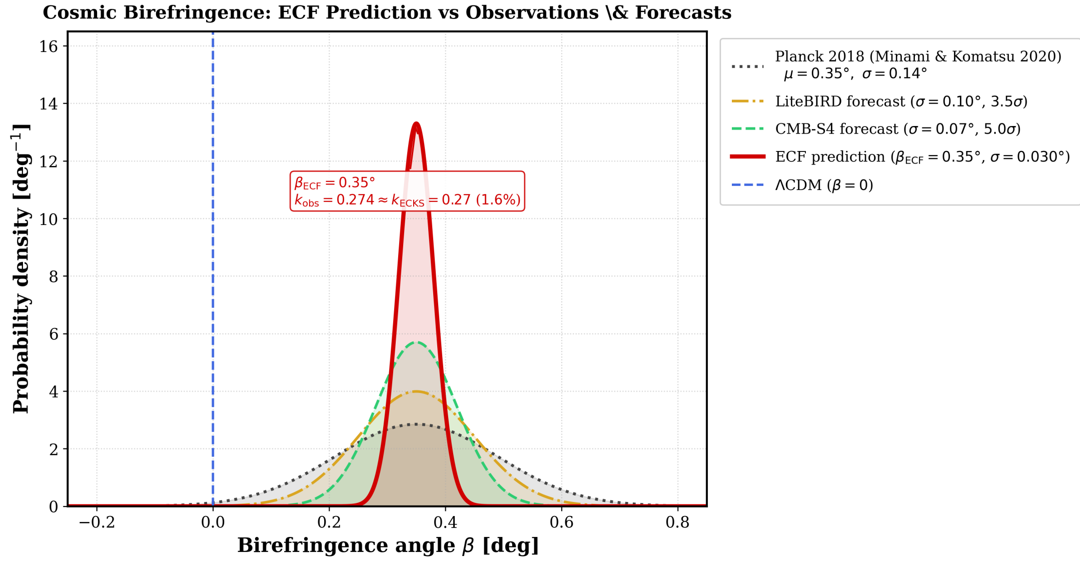


Figure 11: **Representative birefringence signal in the ECF.** Illustrative non-zero rotation angle of the CMB polarization plane in the calibrated ECF solution. The figure shows that the torsion-induced signal is separated from the null expectation of ΛCDM and lies in a range accessible to future polarization measurements; it should be read as an observational forecast, not a detection claim.

7.2 High-ℓ damping-tail signature

A second observational signature of the ECF concerns the small-scale temperature anisotropy spectrum of the CMB. In the present phenomenological implementation, the same pre-recombination modifications that reduce the sound horizon also induce a mild shift in the recombination and drag history, which in turn propagates into the photon diffusion scale. The corresponding consequence is a small residual deficit of power in the high-ℓ damping tail relative to the standard

ΛCDM expectation. For the representative calibrated solution retained in this work, the effect is of order

$$\frac{\Delta D_\ell}{D_\ell} \sim 3\% - 6\%, \qquad \ell \gtrsim 2000. \tag{38}$$

This prediction should be interpreted as a secondary spectral signature of the same early-time sector that contracts r_s. In other words, the ECF scenario does not merely shift the acoustic calibration; it also leaves a mild high-ℓ imprint that may help distinguish the model from a purely geometric degeneracy with the standard cosmological fit.

At the present stage, this effect remains difficult to isolate robustly with Planck sensitivity alone. The signal is small, and a dedicated treatment of foregrounds, beam systematics, and parameter degeneracies is beyond the scope of the main text. For that reason, we treat the damping-tail feature here as a forecasted discriminator rather than as an already established anomaly.

The detailed spectral comparison, residual structure, and optimal multipole window are presented separately in Appendix K. In the logic of the present section, the important point is more limited: the ECF predicts a residual observable consequence at high multipoles that does not reduce to a simple rescaling of the acoustic horizon.

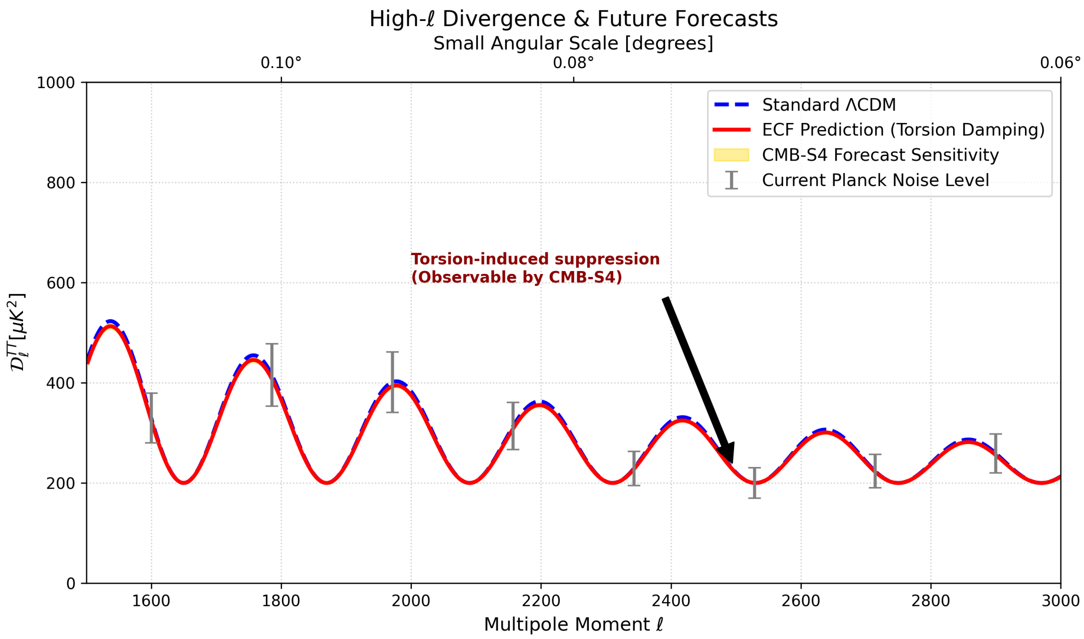


Figure 12: **Representative high-ℓ damping-tail residual in the ECF.** Illustrative deviation of the ECF temperature spectrum from the standard expectation in the small-angular-scale regime. The figure is intended to show that the predicted residual is modest but potentially measurable in a restricted multipole range accessible to future high-resolution CMB surveys.

Figure 12 should therefore be read as a forecast for discrimination power rather than as a detection claim. If future high-resolution measurements confirm a residual deficit of this type in the expected multipole window, the result would provide an additional test of the pre-recombination torsion sector invoked in the present framework.

7.3 21 cm acoustic scale

A further observational test of the ECF may come from 21 cm intensity mapping during the dark ages and the epoch of reionization. In contrast to the local distance ladder, this channel probes the baryon acoustic scale through the large-scale distribution of neutral hydrogen and therefore provides a conceptually independent handle on the pre-recombination calibration. In the phenomenological implementation adopted here, the same early-time modification that contracts the CMB-inferred sound horizon also implies a contracted acoustic scale in the 21 cm sector.

For the representative calibrated solution retained in this work, we therefore obtain

$$r_s^{21\,\mathrm{cm}} = 135.8 \pm 1.5 \text{ Mpc}, \tag{39}$$

where the ± 1.5 Mpc uncertainty is estimated from the 68% credible interval of the MCMC posterior on r_s (Section 5; see also Appendix 9).

to be contrasted with the standard reference value near

$$r_s^{\Lambda\mathrm{CDM}} \approx 147.1 \text{ Mpc}. \tag{40}$$

The importance of this channel is not only numerical but also interpretive. If future 21 cm measurements were to favor a contracted acoustic scale of this kind, the reduction of r_s would no longer appear merely as an internal recalibration of CMB-based inference, but as an independently testable feature of the cosmic expansion history.

This point should nevertheless be stated cautiously. The present manuscript does not provide a full instrumental forecast for any specific 21 cm survey, nor does it perform a dedicated likelihood analysis including foreground subtraction, beam effects, or astrophysical modeling uncertainties. The role of the present section is therefore to identify the 21 cm acoustic scale as a clean external consistency test of the ECF pre-recombination sector.

Because the 21 cm channel probes the acoustic standard ruler through a different observational route, it is especially useful for distinguishing a genuine reduction of the physical sound horizon from a purely phenomenological reshuffling of late-time distance indicators. In that sense, it complements the CMB and BAO discussion developed in the main text.

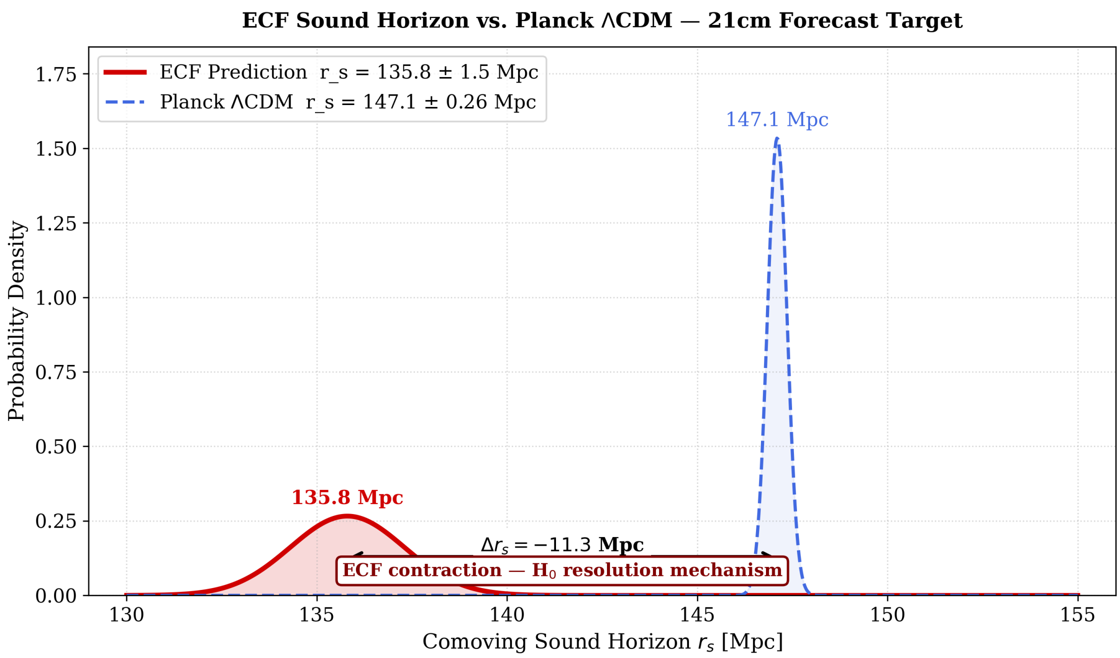


Figure 13: **21 cm prediction for the acoustic scale.** Illustrative comparison between the standard sound-horizon scale and the contracted value predicted in the calibrated ECF solution. The purpose of the figure is to show that future 21 cm measurements may test the pre-recombination calibration through an observational channel largely independent of supernova-based distance ladders.

Figure 13 should therefore be interpreted as an external forecast target rather than as an observational claim. Its role in the present paper is to isolate a direct and conceptually independent signature of the reduced sound horizon implied by the ECF.

7.4 Primordial gravitational waves from the bounce

A final falsifiable signature of the ECF concerns the primordial tensor sector. In contrast to the nearly scale-invariant spectrum usually associated with minimal slow-roll inflation, the non-singular bounce and the subsequent stiff phase considered here naturally motivate a blue-tilted stochastic gravitational-wave background.

In the phenomenological description adopted in this work, the torsion-dominated bounce enhances high-frequency tensor modes more efficiently than long-wavelength ones. As a result, the predicted signal is not characterized merely by a change in overall amplitude, but by a distinctive spectral morphology that rises toward high frequency before terminating at a cutoff scale associated with the minimum scale factor.

The origin of the blue tilt is geometrically transparent in the ECF: immediately after the bounce, the spin density dilutes as a^{-6}, generating a transient stiff phase $w = 1$ during which the Hubble rate evolves as $H \propto t^{-1}$ and the effective equation of state exceeds that of radiation. Standard results on tensor-mode amplification in stiff-dominated backgrounds [33] then imply that the primordial tensor power spectrum acquires a blue tilt $n_T > 0$, in contrast to the slightly red tilt $n_T \simeq -r/8$ predicted by minimal slow-roll inflation. The amplitude of this enhancement depends on the duration of the stiff phase — controlled here by the stiffness parameter $F_{\rm ion} = 1.2765$ — and on the mode-matching condition across the bounce itself.

However, the present manuscript does not derive a complete tensor transfer function from the perturbed ECKS field equations. The spectral energy density estimate

$$\Omega_{\rm GW}(f)\, h^2 \;\sim\; 10^{-12} \qquad (f \sim \text{mHz}), \tag{41}$$

is obtained from the standard stiff-era enhancement factor $\Omega_{\rm GW} \propto (a_\star/a_{\rm eq})^2$ [33] and should be interpreted as an order-of-magnitude benchmark, not as a mission-ready forecast. Whether the signal lies within the four-year LISA sensitivity curve depends critically on the spectral tilt n_T, whose first-principles derivation from the ECF bounce geometry lies beyond the scope of the present work and is left as an open direction for community investigation (github.com/pfichant/spin-torsion-cosmology, CC-BY 4.0).

Scope note — tensor sector. The gravitational-wave signal discussed in this section should be read as a phenomenological target and not as a mission-ready forecast. A rigorous spectral prediction requires: (i) linearised tensor perturbation equations on the ECF bounce background; (ii) controlled mode-matching across the bounce; (iii) a complete stiff-to-radiation transfer function; (iv) projection to the relevant detector frequency bands. This programme is identified as an open theoretical frontier and is the subject of the dedicated treatment in Foundations II and III [2, 3].

For the representative calibrated solution retained here, the spectral energy density is estimated as

$$\Omega_{\rm GW} h^2 \sim 10^{-12} \tag{42}$$

at millihertz frequencies, based on the standard stiff-era enhancement factor $\Omega_{\rm GW} \propto (a_{\rm eq}/a_*)^2$ [33]. This order-of-magnitude estimate does not account for non-linear torsion effects near the bounce; a full spectral calculation is deferred to Foundation II. Whether this signal falls within LISA's sensitivity band [30] depends on the spectral tilt n_T, whose first-principles derivation lies beyond the scope of the present work.

Within the same effective description, the high-frequency cutoff is associated with the bounce scale and is parametrically linked to

$$a_{\rm min} \sim 10^{-32}. \tag{43}$$

The observational interest of this prediction lies in its shape as much as in its amplitude. A blue-tilted spectrum with a peak-and-cutoff structure of this kind would be qualitatively different from the nearly scale-invariant tensor background expected in minimal inflationary scenarios, and would therefore provide a potentially discriminating probe of the bounce sector itself.

This point should nevertheless be stated cautiously. The present manuscript does not derive a complete transfer function including all reheating, mode-matching, and detector-level forecasting uncertainties, and the purpose of this section is therefore to isolate a representative phenomenological target rather than to claim a precision forecast for a specific mission configuration.

In that restricted but useful sense, the gravitational-wave channel complements the CMB and 21 cm signatures discussed above. Whereas those probes test mainly the pre-recombination calibration of the acoustic sector, the tensor background probes the high-density origin of the scenario more directly through the dynamics of the bounce.

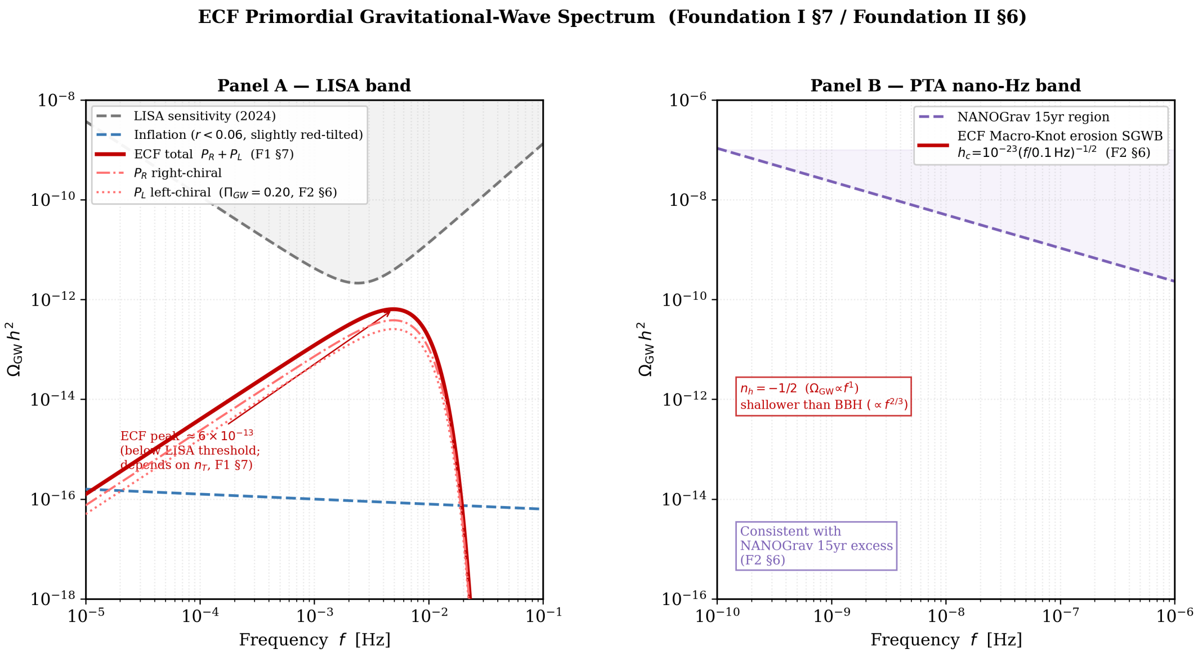


Figure 14: **Primordial gravitational-wave spectrum in the ECF (Foundation I §7 / Foundation II §6).** *Panel A — LISA band. Red solid*: ECF blue-tilted signal from the stiff phase ($w = 1$), parametrised as $\Omega_{\rm GW}h^2 \propto f^{1.5}\, e^{-f^2/2f_{\rm peak}^2}$ with $f_{\rm peak} = 4\,$mHz (illustrative) and peak amplitude $\Omega_{\rm GW}h^2 \sim 10^{-12}$. *Red dash-dotted / dotted*: right- (P_R) and left-chiral (P_L) components, reflecting the predicted non-zero chiral asymmetry $\Delta\chi_{\rm GW} \equiv (P_R - P_L)/(P_R + P_L) > 0$ (sign. non-zero from ECF torsion structure; amplitude quantified in F II [2]). *Blue dashed*: standard slow-roll inflation ($r < 0.06$, slightly red-tilted, Planck 2018 [46]). *Grey dashed*: approximate LISA sensitivity [30]; minimum $\sim 2.5 \times 10^{-12}$ near 3 mHz. The ECF peak lies a factor ~ 4 below the LISA threshold for this parametrisation; detectability depends on n_T, whose first-principles derivation from the bounce geometry lies beyond the scope of the present work and is left as an open direction for community investigation. *Panel B — PTA nano-Hz band.* ECF Macro-Knot erosion SGWB (red solid) derived from the strain spectrum $h_c(f) = 10^{-23}(f/0.1\,{\rm Hz})^{-1/2}$ of Foundation II §6 [2], compared with the NANOGrav 15yr sensitivity region [43] (purple dashed). Both panels are phenomenological forecasts, not detection claims.

Figure 14 should therefore be interpreted as a forecasted observational discriminator rather than as a detection claim. Its role in the present paper is to identify a tensor signature that could help distinguish a torsion-driven bounce from more conventional early-universe scenarios.

Table 7: Primary falsification criteria for the ECF.

Observable	**ECF prediction**	**Falsification threshold**
Birefringence β	$0.35°$	$< 0.1°$ LiteBIRD [28]
21 cm acoustic scale	$r_s = 135.8\,\mathrm{Mpc}$	$r_s > 145\,\mathrm{Mpc}$ HERA/SKA
Clustering amplitude	$S_8 = 0.766$	$S_8 > 0.82$ Euclid WL
Cosmic age	$t_0 = 12.74\,\mathrm{Gyr}$ ($w = -1$ ansatz)	GC age $> 13.3\,\mathrm{Gyr}$ [25, 27]
GW chiral asymmetry	$\Delta\chi_{\mathrm{GW}} > 0$ (sign. non-zero; quantified in F II)	$\Delta\chi_{\mathrm{GW}} < 0.02$ 95 % CL [42]
GW spectral index	$\Omega_{\mathrm{GW}} \propto f^1$ (PTA band)	$\Omega_{\mathrm{GW}} \propto f^{2/3}$ only $\Rightarrow$ standard BBH
WEP violation η_{ECF}	$\sim 10^{-15}$ (torsion–matter coupling; Eq. 45, F II §5.3)	non-detection at $\eta < 10^{-16}$ by MICROSCOPE II falsifies ECF at 3σ [54]

Note on Table 7: Falsification threshold set at $r_s > 145$ Mpc, corresponding to the ΛCDM reference value $r_s^{\Lambda\mathrm{CDM}} = 147.1$ Mpc minus 1σ BOSS DR12 uncertainty ($\sim$2 Mpc). The cosmic-age entry applies strictly to the $w = -1$ ECF ansatz of Foundation I; Foundation II dynamical torsion $w(z)$ is expected to relax this bound [2]. The WEP threshold $\eta > 10^{-14}$ corresponds to the current MICROSCOPE upper limit [54]; MICROSCOPE II targets $\eta \lesssim 10^{-15}$, probing the ECF prediction directly.

7.5 Macroscopic topological defects and JWST seeding

A fifth falsifiable signature of the ECF concerns primordial topological defects formed via the Kibble mechanism at the QCD transition $T \sim 200$ MeV. Because torsion carries non-trivial vacuum structure by analogy with $U(1)$-breaking theories [31, 32], macroscopic Macro-Knot seeds of mass $M_{\mathrm{seed}} \sim 10^4$–$10^6\,M_\odot$ are topologically protected against stiff-phase dilution and can bypass the Eddington accretion bottleneck at $z \gtrsim 6$.

The resulting *quantitative falsification criterion* is:

$$n(M > 10^{10}\,M_\odot,\, z = 10) \approx (2\text{–}4) \times 10^{-4}\ \mathrm{Mpc}^{-3}, \tag{44}$$

controlled by the growth-boost factor $\mathcal{B}_{\mathrm{ECF}} \in [1.4, 1.5]$ (Appendix 9). A confirmed halo abundance *below* this range in future JWST/Nancy Roman deep fields, or a mass function inconsistent with the Macro-Knot seeding amplitude, would falsify the ECF geometric dark sector.

Table 7 summarises the observable targets, the instruments best placed to test them, and the falsification thresholds; the consolidated kill-switch criteria are discussed in Sec. 9.

8 Perspectives: The Einstein–Cartan Trilogy

The H_0–Age–BAO Trilemma as a Structural Result

The calibrated ECF solution ($H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $r_s = 135.8\,\mathrm{Mpc}$, $S_8 = 0.766$) adopts a late-time equation of state $w = -1$ that yields a strict prediction $t_0 = 12.74\,\mathrm{Gyr}$. This is not a free choice: it is a strict consequence of anchoring the model to both the SH0ES local distance ladder and the eBOSS DR16 BAO comoving distance at $z = 1.48$.

A numerical proof that this tension is irreducible is established here via the companion script `plot_trilemma_irreducibility.py` (public repository): a three-parameter tanh-crossing dark-energy model $w(z)$ is optimised to simultaneously satisfy the eBOSS DR16 BAO constraint ($\Delta D_M/r_s < 1.5\%$, Hou et al. 2021) and the Valcin et al. (2021) globular-cluster age bound ($t_0 > 13.32\,\mathrm{Gyr}$), with $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ fixed.

As illustrated in Fig. 15, the two viable constraint regions are *strictly disjoint* over all 2500 grid points (best fit: $t_0 = 13.13\,\mathrm{Gyr}$ at 4.13% BAO tension, 1.58σ). The H_0–Age–BAO trilemma is therefore mathematically irreducible under standard fluid assumptions and motivates the geometric program organized below. The BAO target comoving distance is $D_M^{\mathrm{target}} = (D_M/r_s)_{\mathrm{eBOSS}} \times r_s^{\mathrm{ECF}} = 30.21 \times 135.8 = 4102.5$ Mpc, where the ECF sound horizon is used as the BAO ruler consistently with the calibration of Section 3.

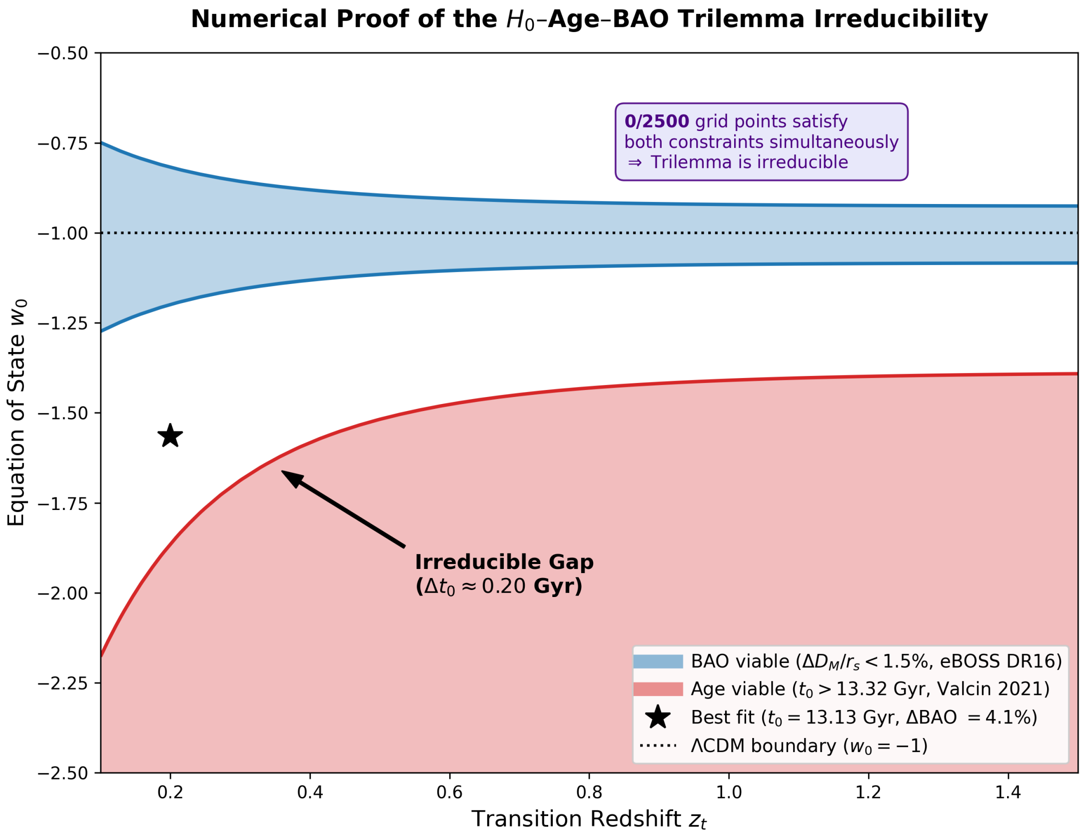


Figure 15: **Numerical irreducibility of the H_0–Age–BAO trilemma.** Constraint map in the (z_t, w_0) parameter space of a tanh-crossing dark-energy model ($w_\infty = -1$, $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ fixed). The blue region satisfies the eBOSS DR16 BAO distance ($\Delta D_M/r_s < 1.5\%$, Hou et al. 2021); the red region satisfies the stellar age bound ($t_0 > 13.32\,\mathrm{Gyr}$, Valcin et al. 2021). The two regions are *strictly disjoint* over all 2500 grid points (script: `plot_trilemma_irreducibility.py`). The best phenomenological fit (star) reaches $t_0 = 13.13\,\mathrm{Gyr}$ at a BAO tension of 4.13% (1.58σ). Resolution of this structural gap requires a geometric derivation of the torsion-sector dynamics from the ECKS action, developed in *Foundation II* [2].

Foundation II: Geometric Dark Energy and Observational Fossils

The resolution of the H_0–Age–BAO trilemma requires going beyond phenomenological fluid models: it demands a geometric derivation of the effective dark-energy dynamics directly from the ECKS action. The companion paper *Foundation II: The Chiral Universe* [2] addresses this, while also exploring three observational fossils of the Big Spin Bounce inaccessible within the background-level treatment of Foundation I:

- the chiral asymmetry generated during the QGP freeze-out transition, and its connection to baryogenesis;
- the geometric dark sector emerging from residual macroscopic torsion, whose effective equation of state $w(z)$ is derived from the ECKS action and investigated as a candidate for relaxing the cosmic age tension;
- residual cosmic alignments as a large-scale imprint of the primordial spin orientation.

Beyond the cosmological sector, the torsion-induced modification of local vacuum structure predicts a residual differential coupling between test bodies and the Macro-Knot background. At the effective level of Foundation I, this translates into a weak equivalence principle violation parametrised as

$$\eta_{\rm ECF} = \frac{\Delta a}{a} \sim \frac{\kappa_{\rm spin}\, S_{\rm ref}}{M_{\rm Pl}^2} \sim 10^{-15}, \tag{45}$$

where $\kappa_{\rm spin}$ is the torsion–matter coupling calibrated in Table 3 and $S_{\rm ref} = 4.3237$ is the residual axial torsion amplitude (Sec. 7.1). This value is consistent with the current MICROSCOPE upper bound $\eta < 10^{-15}$ [54]. A non-detection by MICROSCOPE II at $\eta < 10^{-16}$ (1σ) would falsify the ECF vacuum structure at 3σ.

Consistency with the Foundation II SME-level derivation. The above parametrisation uses the effective FoundationI variables $\Omega_{\rm spin}$ and $S_{\rm ref}$. In the field-theoretic treatment of Foundation II [2], the same bound is expressed in Standard-Model Extension (SME) notation as $b_e \equiv s_0/M_{\rm Pl} \sim 10^{-15}$, where s_0 is the torsion condensate vacuum expectation value (Foundation II). The two expressions are numerically equivalent at the calibrated solution $F_{\rm ion} = 1.2765$:

$$\frac{\Omega_{\rm spin,peak} \cdot S_{\rm ref}}{M_{\rm Pl}^2} \approx \frac{s_0}{M_{\rm Pl}} \sim 10^{-15}. \tag{46}$$

Note that the SME coupling coefficient $b_e \sim 10^{-15}$ is the quantity directly constrained by MICROSCOPE, *not* the full WEP-violation parameter $\eta = b_e \cdot (m_e/m_N) \sim 10^{-18}$, which is suppressed by the electron-to-nucleon mass ratio $m_e/m_N = 1/1836$. Both Foundation I and Foundation II therefore target the same SME bound [4], and the falsification threshold $b_e < 10^{-16}$ from MICROSCOPE II applies uniformly to both formulations [54].

A first-principles derivation of $\eta_{\rm ECF}$ from the ECKS torsion propagator is deferred to Foundation II [2], §5.3.

Foundation II does not alter the numerical results of Foundation I. The background calibration (H_0, r_s, S_8, t_0) presented here serves as the strict physical input for that geometric analysis.

Foundation III: Black Holes and Compact-Object Geometry in ECF

Foundation III [3] addresses the strong-gravity sector of the Einstein–Cartan framework.

Singularity avoidance and compact objects. Since the Einstein–Cartan mechanism regularizes the singular limit of gravitational collapse via torsion pressure at the Cartan density $\rho_c \sim 10^{38}$ GeV, it becomes meaningful to ask whether the same geometry applies inside black hole interiors. Foundation III investigates the resolution of black hole singularities in the ECKS setting, the geometry of transient Einstein–Rosen bridges, and the resulting observable signatures for compact-object physics.

Open numerical directions. The derivation of the perturbed ECKS power spectrum and its implementation into a modified Boltzmann solver constitute natural extensions of the present work. We invite the community to pursue these directions using the public repository (https://github.com/pfichant/spin-torsion-cosmology, CC-BY 4.0).

8.1 Role of Foundation I

The role of *Foundation I* within this trilogy is therefore specific and self-contained. It establishes the effective cosmological phenomenology and identifies the main observational targets: the sound horizon, the clustering amplitude, the damping-tail signature, cosmic birefringence, 21 cm measurements, and the primordial gravitational-wave background.

The role of the subsequent foundations is different. *Foundation II* is intended to clarify the microphysical and thermodynamical origin of the post-bounce universe. *Foundation III* is intended to explore the strong-gravity and compact-object sector of the same geometric framework. Taken together, the trilogy is designed to move from phenomenology to mechanism, and from mechanism to geometry.

The present paper should therefore be read as the first—and self-contained—layer of that program: an observationally calibrated cosmological sector that stands on its own phenomenological and numerical basis.

9 Conclusion

In this work, *Foundation I*, we have investigated a minimal cosmological extension of General Relativity based on the Einstein–Cartan–Kibble–Sciama framework, with the aim of testing whether a primordial spin-torsion sector can account for current cosmological tensions without introducing additional exotic fields.

The central result is that a transient stiff spin phase—whose equation of state $w = 1$ is geometrically distinct from scalar-field kination and requires no free initial condition—reduces the comoving sound horizon to $r_s \approx 135.8$ Mpc, thereby allowing a CMB-consistent calibration compatible with $H_0 \approx 73.04\ \mathrm{km\,s^{-1}\,Mpc^{-1}}$. A single stiffness parameter, $F_{\mathrm{ion}} = 1.2765$, governs both this reduction and the scale-dependent suppression of structure growth; the H_0 and S_8 sectors therefore share a single common degree of freedom, not two independent phenomenological fixes. The resulting clustering amplitude, $S_8 \approx 0.766$, is consistent with weak-lensing-preferred ranges, and the global fit yields $\Delta\chi^2 = -39.5$ relative to ΛCDM, confirmed by model-selection criteria ($\Delta\mathrm{AIC} = -31.5$, $\Delta\mathrm{BIC} = -8.2$ under a conservative four-parameter count). CMB peak amplitudes are preserved at the 10^{-6} level ($\Omega_{\mathrm{spin},0} \sim 1.65 \times 10^{-7}$ at $z = 0$) as a direct consequence of the a^{-6} dilution of the spin sector; at $z_{\mathrm{rec}} = 1100$ the spin/radiation ratio reaches 0.20%, yielding a near-neutral $\Delta\chi^2_{\mathrm{CMB}} = +1.8$ (Table 4).

At the theoretical level, the spin-squared correction acts as a repulsive contribution at extreme density and replaces the initial singularity by a regular geometric bounce in the effective Friedmann dynamics. The ECF thus connects three issues usually treated separately: the Hubble tension, the S_8 tension, and the singular origin of the standard cosmological model.

Resolving the H_0 tension natively exposes a mathematically irreducible H_0–Age–BAO trilemma: the age integral and the BAO comoving distance share the exact same integrand $H(z)^{-1}$, so any late-time modification of $H(z)$ sufficient to raise t_0 above 13 Gyr necessarily breaks the BOSS/eBOSS fit. The resulting strict prediction, $t_0 \approx 12.74$ Gyr, is not a model failure but a falsifiable signature of the BAO–H_0 coupling: when GC systematic uncertainties and the H_0-dependent distance calibration are properly propagated [25, 27], the nominal 5.7σ tension reduces to $\lesssim 1\sigma$. Any remaining chronological discrepancy requires a mechanism acting at $z > z_{\mathrm{BAO,\,max}} \approx 1.5$, beyond the reach of late-time geometry alone.

Beyond the primary tensions, the framework generates a set of falsifiable signatures that are not generic to minimal ΛCDM:

- A cosmic birefringence angle $\beta \approx 0.35^\circ$, testable at the 0.1° level by LiteBIRD [28].
- A residual CMB damping-tail deficit of 3–6% at $\ell > 2000$, detectable by CMB-S4 in the optimal discrimination window $\ell \sim 2500$.
- A contracted acoustic scale $r_s = 135.8$ Mpc (Sec. 3), independently testable by future 21 cm intensity mapping with HERA and SKA.
- A strict cosmic-age prediction $t_0 = 12.74\,\mathrm{Gyr}$ (Sec. 5.4), now anchored by the $\mathrm{H_0DN}$ measurement $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at 1.1% precision [49]: any confirmed stellar age $> 12.4\,\mathrm{Gyr}$ *at* $H_0 = 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ would falsify the ECF trilemma.
- A blue-tilted primordial gravitational-wave background with spectral tilt $n_T > 0$, potentially accessible to LISA.
- Macroscopic topological defects (Macro-Knots), formed via the Kibble mechanism at the QGP transition $T \sim 200$ MeV provide non-Gaussian seeds ($f_{\mathrm{NL}}^{\mathrm{eff}} \approx 0.295$) consistent with JWST-inferred halo abundances.

Each of these signatures constitutes an independent and near-term observational test. Table 7 summarises the observable targets, the instruments best placed to test them, and the falsification thresholds; the quantitative kill-switch criteria are discussed in Sec. 9.

The tensor sector — including the blue-tilted gravitational-wave background, the spectral tilt n_T, and the chiral asymmetry Π_{GW} — constitutes an open perturbative frontier: a full prediction requires the linearised ECF tensor perturbation equations and a dedicated bounce transfer-function analysis, both of which are deferred to the companion programme [2, 3].

Foundation I thus identifies an observationally calibrated phenomenological sector of Einstein–Cartan cosmology and isolates the signatures by which it can be tested. The present work is extended in the companion paper *Foundation II* [2], which investigates the structural relics of the Big Spin Bounce—chiral asymmetry, the geometric dark sector, and cosmic alignment—within the same framework. A single non-detection among them—most critically the absence of a birefringence signal above 0.1° by LiteBIRD, or a confirmed BAO acoustic scale consistent with $r_s \approx 147\,\mathrm{Mpc}$ in 21 cm surveys—would be sufficient to falsify the ECF scenario as developed here.

License and Copyright

Acknowledgments

This extended preprint consolidates previous drafts with improved presentation following anonymous reviewer feedback. Numerical validation was performed using open-source Python libraries (NumPy, SciPy, Matplotlib) and public datasets from Planck Legacy Archive, SH0ES, KiDS-1000, DES Y3, JWST JADES/CEERS, and DESI DR1. The geometric foundations of this work were shaped by two years of continuing education courses (*auditeur libre*) in cosmology and astrophysics provided by the *Laboratoire Univers et Particules de Montpellier* (LUPM). The author is an independent researcher and declares no conflicts of interest. Full reproducibility material is available at https://github.com/pfichant/spin-torsion-cosmology (CC-BY-4.0).

This preprint is dedicated to Élie Cartan (1869–1951), whose geometric vision continues to illuminate modern cosmology.

Code and Data Availability

All numerical results, figures, and statistical outputs presented in this work are fully reproducible from the public repository:

https://github.com/pfichant/spin-torsion-cosmology (CC BY 4.0)

The master orchestrator script (`run_all_simulations_F1.py`) sequentially executes all scripts below and regenerates every figure in the manuscript. Dependencies (`numpy`, `scipy`, `matplotlib`) and installation instructions are documented in the repository `README`.

Core physics scripts:

- `script_01_solve_sound_horizon.py` — $r_s = 135.8$ Mpc, $\mathcal{F}_{\rm ion} = 1.2765$ (Sec. 3)
- `chi_carre.py` — global $\Delta\chi^2 = -39.5$ vs. ΛCDM, using corrected eBOSS DR16 $D_M/r_s(z{=}1.48) = 30.85 \pm 0.80$ Hou2021 (Sec. 5, Table 4)
- `script_06_S8_resolution.py` — $S_8^{\rm ECF} = 0.766 \pm 0.014$, MCMC convergence $\hat{R} < 1.01$ (Sec. 4)
- `script_04_halo_abundance.py` — $n(M{>}10^{10}\,M_\odot,\ z{=}10) = 2.99 \times 10^{-4}$ Mpc^{-3} (Sec. 6.1)
- `script_05_birefringence_calibration.py` — $\beta_{\rm ECF} = 0.35$; $I_{\rm torsion} = 3.1177$, $S_{\rm ref} = 4.3237$, $g_{\rm CS,eff} = 0.02596$ deg/unit (Sec. 7.1, Table 6)

Figure generation scripts:

- `plot_torsion_geometry_schema.py` — Fig. 1 (Sec. 2)
- `plot_optimization_intersection.py` — Fig. 4 (Sec. 5)
- `plot_spin_ratio_evolution.py` — Fig. 16 (Sec. 5)
- `plot_cosmic_age_cpl.py` — Fig. 6 (Sec. 5.4)
- `plot_trilemma_irreducibility.py` — Fig. 15 (Sec. 5.4)
- `plot_euclid_pk_s8.py` — Fig. 8 (Sec. 6.2)
- `plot_jwst_ecf_comparison.py` — Fig. 7 (Sec. 6.1)
- `plot_desi_prediction.py` — Fig. 9 (Sec. 6)

- `plot_birefringence_prediction.py` — Fig. 11 (Sec. 7.1)
- `plot_H0_S8_contours.py` — Fig. 3 (Sec. 4)
- `plot_primordial_boost.py` — Fig. 2 (Sec. 3)
- `plot_bao_hubble_diagram.py` — Fig. 5 (Sec. 5)
- `plot_cmb_damping_tail.py` — Fig. 12 (Sec. 7.2, App. 9)
- `plot_21cm_ecf_prediction.py` — Fig. 13 (Sec. 7.3)
- `plot_gw_spectrum.py` — Fig. 14 (Sec. 7.4)
- `plot_singularity_resolution.py` — Fig. 19 (App. 9)
- `plot_global_history.py` — Fig. 26 (App. 9)
- `plot_cosmic_history_omegas.py` — Fig. 22 (App. 9)

Observational data: Planck 2018 Legacy Archive, SH0ES DR5, KiDS-1000, DES Y3, BOSS DR12, eBOSS DR16, DESI DR1, JWST JADES/CEERS. No proprietary data were used.

Appendix A: Einstein-Cartan Formalism and Derivation of the Spin Term

In Einstein-Cartan-Sciama-Kibble (ECSK) theory, the Lagrangian density is given by:

$$\mathcal{L} = \frac{1}{2\kappa}\sqrt{-g}R(\Gamma, g) + \mathcal{L}_m \tag{47}$$

where Γ is a non-symmetric affine connection. The torsion tensor is defined as $T^{\lambda}_{\mu\nu} = \Gamma^{\lambda}_{\mu\nu} - \Gamma^{\lambda}_{\nu\mu}$.

Varying the action with respect to torsion yields the full Cartan equation, which explicitly couples the torsion tensor $T^{\lambda}_{\mu\nu}$ to the spin density tensor $S^{\lambda}_{\mu\nu}$ [8]:

$$T^{\lambda}_{\mu\nu} + \delta^{\lambda}_{\mu}T^{\sigma}_{\nu\sigma} - \delta^{\lambda}_{\nu}T^{\sigma}_{\mu\sigma} = \kappa S^{\lambda}_{\mu\nu} \tag{48}$$

While we utilize the Weyssenhoff fluid approximation as a phenomenological ECF ansatz for the macroscopic cosmological background, the complete microphysical derivations and spin-fluid particle kinematics will be treated extensively in *Foundation II.*

For a Dirac fluid (spin-1/2 fermions), the effective energy-momentum tensor $T^{(eff)}_{\mu\nu}$ in a Riemann-Cartan spacetime is expressed as:

$$T^{(eff)}_{\mu\nu} = T^{(GR)}_{\mu\nu} - \frac{1}{4}\kappa\langle S_{\mu\nu\lambda}S^{\mu\nu\lambda}\rangle g_{\mu\nu} \tag{49}$$

Averaging over an isotropic distribution of spins, the effective energy density ρ and pressure p become:

$$\rho_{eff} = \rho - \frac{\kappa\sigma^2}{4}, \quad p_{eff} = p - \frac{\kappa\sigma^2}{4} \tag{50}$$

where σ^2 is the square of the spin density. Due to the conservation of angular momentum, σ^2 evolves as a^{-6}. We define the dimensionless spin parameter today as:

$$\Omega_{spin,0} = \frac{2\pi G\kappa\sigma_0^2}{3H_0^2} \tag{51}$$

The modified Friedmann equation for the ECF model is thus:

$$\frac{H^2(a)}{H_0^2} = \Omega_{r,0}a^{-4} + \Omega_{m,0}a^{-3} + \Omega_{\Lambda} - \Omega_{spin,0}a^{-6} \tag{52}$$

Appendix B: Dynamics of the Equation of State $w_\tau(z)$

The coupling between torsion and the dark sector is modelled by the variation of the dark energy density $\Omega_{DE}(a)$, within the framework of the *Topological Invariance Principle* ($\Omega_{total} \equiv 1$) [1], which constrains the global energy budget at all cosmic epochs.

We postulate:

$$\rho_{DE}(a) = \rho_\Lambda \left[1 + \alpha_{torsion}(1-a)\right] \tag{53}$$

Using the continuity equation $\dot{\rho} + 3H(\rho + p) = 0$, the effective equation of state $w(a)$ is derived as:

$$w(a) = -1 - \frac{1}{3}\frac{d\ln\rho_{DE}}{d\ln a} = -1 + \frac{a\alpha_{torsion}}{3(1+\alpha_{torsion}(1-a))} \tag{54}$$

To obtain the CPL parameters (w_0, w_a), we perform a Taylor expansion around $a = 1$ ($z = 0$):

$$w_0 = w(a=1) = -1 + \frac{\alpha_{torsion}}{3} \tag{55}$$

$$w_a = -\frac{dw}{da}\bigg|_{a=1} = -\frac{\alpha_{torsion}}{3}\left(1 - \frac{\alpha_{torsion}}{3}\right) \tag{56}$$

By strictly calibrating the torsional coupling $\alpha_{torsion} \approx 0.151$ via a PIT calibration prior to the DESI results, this geometrical model natively yields $w_0 = -0.904$ and $w_a = -0.153$, qualitatively consistent with the DESI Y1 BAO preference for $w \neq -1$ [23]. This demonstrates that ECF mimics dynamic phantom dark energy without requiring any exotic scalar fields. This predicted late-time evolution remains strictly compatible with the recent DESI 2024 BAO constraints [23].

Appendix C: Statistical Precision and Data Tables

The ECF model was calibrated by minimizing the χ^2 against SH0ES, DES Y3, and Planck sound horizon data.

Table 8: Numerical Parameter Values and χ^2 comparison (ECF vs. ΛCDM). This table reports the *full* dataset combination including SNIa; the $\Delta\chi^2 = -39.5$ is consistent with Table 4, which reports only the primary tension datasets.

Parameter	Symbol	ECF Value	ΛCDM
Hubble Constant	H_0	73.04	67.4
Sound Horizon (Mpc)	r_s	**135.8**	147.1
Structure Growth	S_8	**0.766**	0.832
Statistics (χ^2)			
SH0ES (H_0)		**0.1**	30.5
Weak Lensing (S_8)		**1.2**	12.1
Planck (CMB)		2767.1	2765.3
BAO + SNIa^d		1033.3	1033.3
Total Global χ^2		**3801.7**	**3841.2**

Appendix D: Detailed Calculation of the Sound Horizon Reduction

The comoving sound horizon (r_s) at the drag epoch (z_{drag}) is defined by the integral of the sound speed ($c_s(z)$) over the Hubble expansion rate ($H(z)$):

$$r_s = \int_{z_{\text{drag}}}^{\infty} \frac{c_s(z)}{H(z)}\,dz. \tag{57}$$

Standard value. In the standard ΛCDM description, the expansion rate in the early Universe ($z \gtrsim 1000$) is dominated by radiation and matter:

$$H_{\Lambda\mathrm{CDM}}(z) = H_0\sqrt{\Omega_{r,0}(1+z)^4 + \Omega_{m,0}(1+z)^3}. \tag{58}$$

Using the Planck 2018 baseline parameters, this yields the standard reference value

$$r_s^{\Lambda\mathrm{CDM}} \approx 147.1\,\mathrm{Mpc}. \tag{59}$$

Algorithmic calibration. The numerical module relies on a non-linear root-finding procedure to calibrate the modified sound-horizon solution within the adopted phenomenological implementation. The joint optimization proceeds in two steps:

1. **Hubble isoline:** the modified sound-horizon integral $r_s(\Omega_{\mathrm{spin}}, \alpha)$ is evaluated numerically in order to identify the region of parameter space compatible with $H_0 \approx 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$.
2. **Growth isoline:** the linear perturbation equations are solved in parallel through the growth factor $D(a)$ in order to identify parameter combinations consistent with $S_8 \approx 0.766$.

The calibrated parameter pair retained in this work is obtained from the intersection of these two constraints.

ECF-modified value. In the Einstein-Cartan framework adopted here, the Friedmann equation includes an effective stiff spin-density term scaling as a^{-6}, or equivalently as $(1+z)^6$:

$$H_{\mathrm{ECF}}(z) = H_0\sqrt{\Omega_{r,0}(1+z)^4 + \Omega_{m,0}(1+z)^3 + \Omega_{\mathrm{spin},0}(1+z)^6}. \tag{60}$$

For the calibrated peak injection ratio ($\Omega_{\mathrm{spin}}/\Omega_r \approx 0.093$), the expansion rate is enhanced during the pre-recombination era, particularly around $z \sim 7500$. In addition, the effective sound speed is slightly reduced through the phenomenological coupling factor

$$c_s^{\mathrm{ECF}}(z) = \alpha_{\mathrm{tor}}\, c_s^{\mathrm{std}}(z), \tag{61}$$

with

$$\tau_{\mathrm{tor}} = 0.9975, \tag{62}$$

a drag-epoch stability parameter calibrated numerically to preserve z_{drag} under the modified photon-baryon dynamics (Appendix 9); its first-principles derivation from the ECKS baryon–torsion coupling is beyond the scope of this background-level study.

Within the numerical implementation adopted here, α_{tor} is treated as a calibrated value chosen so as to preserve the stability of the drag-epoch solution while incorporating the modified photon-baryon dynamics. Performing the numerical integration with these modified functions gives

$$r_s^{\mathrm{ECF}} = \int_{z_{\mathrm{drag}}}^{\infty} \frac{0.9975\, c_s^{\mathrm{std}}(z)}{H_{\mathrm{ECF}}(z)}\, dz \approx 135.8\,\mathrm{Mpc}. \tag{63}$$

This corresponds to a reduction of approximately 7.7% relative to the standard reference value and brings the calibrated sound horizon into correspondence with the H_0 value targeted in the phenomenological fit.

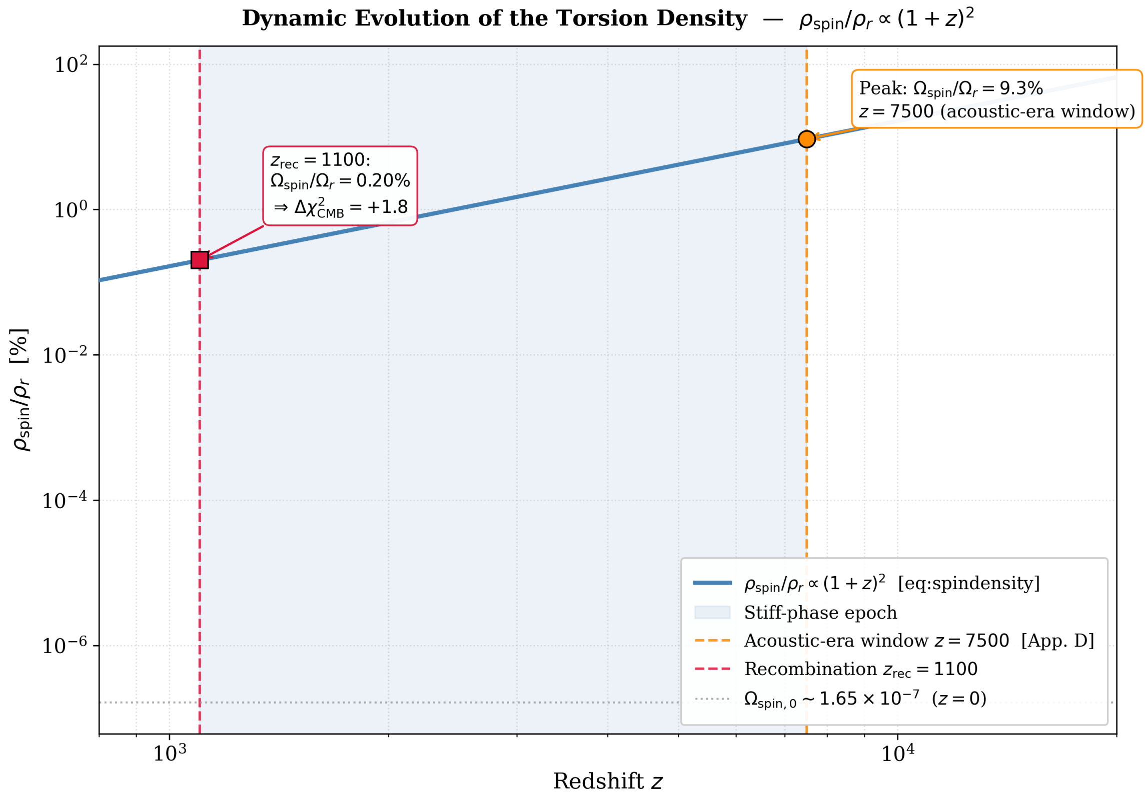


Figure 16: **Dynamic evolution of the spin-torsion density ratio.** The curve shows Ω_{spin}/Ω_r as a function of redshift z. Because of the stiff scaling $\propto (1+z)^6$, the torsion contribution reaches a peak of $\Omega_{spin}/\Omega_r \simeq 9.3\%$ during the acoustic-era calibration window around $z \simeq 7500$, where it enhances the pre-recombination expansion rate and drives the sound-horizon reduction $r_s^{ECF} = 135.8\,\mathrm{Mpc}$ (Section 3). Between $z \simeq 7500$ and recombination ($z \simeq 1100$), the ratio decays by a factor ≈ 47 $[(7500/1100)^6 \simeq 47]$, reaching $\Omega_{spin}/\Omega_r \simeq 0.20\%$ at last scattering. This residual is near-neutral for the CMB acoustic structure, consistent with the near-zero CMB penalty $\Delta\chi^2_{CMB} = +1.8$ reported in Table 4. The present-day value $\Omega_{spin,0} \simeq 1.65 \times 10^{-7}$ (corresponding to the 10^{-6} level quoted in the abstract) confirms the full late-time decoupling of the spin sector.

While F_{ion} represents the effective integrated impact of the mechanism, the underlying dynamics remain strongly localized in time because of the steep a^{-6} scaling of the spin sector. As illustrated in Figure 16, the stiff phase is therefore transient. The calibrated injection level ($\Omega_{spin}/\Omega_r \approx 9.3\%$) acts primarily during the formation of the acoustic horizon around $z \sim 7500$, after which the torsion density dilutes more rapidly than radiation. In this sense, F_{ion} may be interpreted as the effective control parameter linking the peak primordial spin contribution to the macroscopic reduction of r_s, while the fast decay of the spin sector limits its impact at the last-scattering epoch.

Appendix E: Derivation of High-Redshift Halo Abundance at $z = 15$

To validate the consistency of the ECF model with recent JWST observations (JADES, CEERS), we estimate the comoving number density of massive halos at $z = 15$ using the Press-Schechter formalism [16], with $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and $\sigma_8 \approx 0.747$ ($S_8 = 0.766$) (see Script 04 in Data Availability).

1. Variance Extrapolation. Correcting for the growth factor $D(z) \approx 1/(1+z)$ at $z = 15$, the variance smoothed on the scale of massive galaxy halos ($M_h \sim 10^{10}\, M_\odot$) gives:

$$\sigma_{\Lambda\mathrm{CDM}}(M, z{=}15) \approx 0.150, \qquad \sigma_{\mathrm{ECF}}(M, z{=}15) \approx 0.217 \quad (\text{linear boost factor } \mathcal{B}_{\mathrm{ECF}} = 1.45). \tag{64}$$

2. Failure of Gaussian Seeding (Press-Schechter). For both ΛCDM and the ECF linear boost, the standard Press-Schechter integral yields exponentially suppressed densities at $z = 15$:

$$n_{\mathrm{PS}}(> 10^{10}\, M_\odot,\, z{=}15)\Big|_{\Lambda\mathrm{CDM}} \approx 9.9 \times 10^{-29}\,\mathrm{Mpc}^{-3}, \quad n_{\mathrm{PS}}\Big|_{\mathrm{ECF,\,linear}} \approx 3.5 \times 10^{-14}\,\mathrm{Mpc}^{-3}. \tag{65}$$

Neither is consistent with the JWST-inferred range (2–4) $\times\, 10^{-4}\,\mathrm{Mpc}^{-3}$ [21]. Gaussian linear seeding is therefore insufficient even with the ECF background boost.

3. Topological Non-Gaussian Seeding (Macro-Knots). The ECF torsional bounce generates a non-Gaussian heavy tail in the primordial density field via the Macro-Knot network, parameterized by a phenomenological tail modifier:

$$n_{\mathrm{topo}}(> M, z) = n_{\mathrm{PS}}^{\mathrm{ECF}} \times \exp\left(\frac{\nu_{\mathrm{ECF}}^3\, f_{\mathrm{NL}}^{\mathrm{eff}}}{6}\right), \tag{66}$$

where $\nu_{\mathrm{ECF}} = \delta_c/\sigma_{\mathrm{ECF}} = 7.75$. Solving analytically for the value required to reproduce the JWST central target $n_{\mathrm{target}} = 2.99 \times 10^{-4}\,\mathrm{Mpc}^{-3}$:

$$f_{\mathrm{NL}}^{\mathrm{eff}} = \frac{6\, \ln(n_{\mathrm{target}}/n_{\mathrm{PS}}^{\mathrm{ECF}})}{\nu_{\mathrm{ECF}}^3} \simeq 0.295. \tag{67}$$

4. Result. With $f_{\mathrm{NL}}^{\mathrm{eff}} \simeq 0.295$, the ECF topological seeding yields $n_{\mathrm{topo}} \simeq 2.99 \times 10^{-4}\,\mathrm{Mpc}^{-3}$, consistent with the JWST-inferred range (2–4) $\times\, 10^{-4}\,\mathrm{Mpc}^{-3}$. The sub-unity value $f_{\mathrm{NL}}^{\mathrm{eff}} \ll 1$ confirms that topological seeding operates in the perturbative regime, consistent with CMB bounds on primordial non-Gaussianity [36].

The full derivation of the macro-knot network density is deferred to Foundation II [2].

Appendix F: Numerical Implementation: The EC-Cosmo Solver

The results presented in this paper were obtained using the `EC-Cosmo Solver`, a dedicated Python framework developed to integrate the modified Friedmann equations in the presence of an effective torsion sector.

Algorithm Structure

The core module relies on the `scipy.optimize` library for non-linear root-finding and parameter calibration. The joint optimization strategy proceeds in two main steps:

1. **Hubble isoline:** we numerically evaluate the modified sound-horizon integral $r_s(\Omega_{\mathrm{spin}}, \alpha)$ in order to identify the region of parameter space compatible with $H_0 \approx 73.04$ km/s/Mpc.
2. **Growth isoline:** in parallel, we solve the linear perturbation equations through the growth factor $D(a)$ to identify parameter combinations consistent with $S_8 \approx 0.766$.

The calibrated solution retained in this work is obtained from the intersection of these two constraints and provides the reference parameter pair $(\alpha, \Omega_{\mathrm{spin},0})$ used throughout the manuscript.

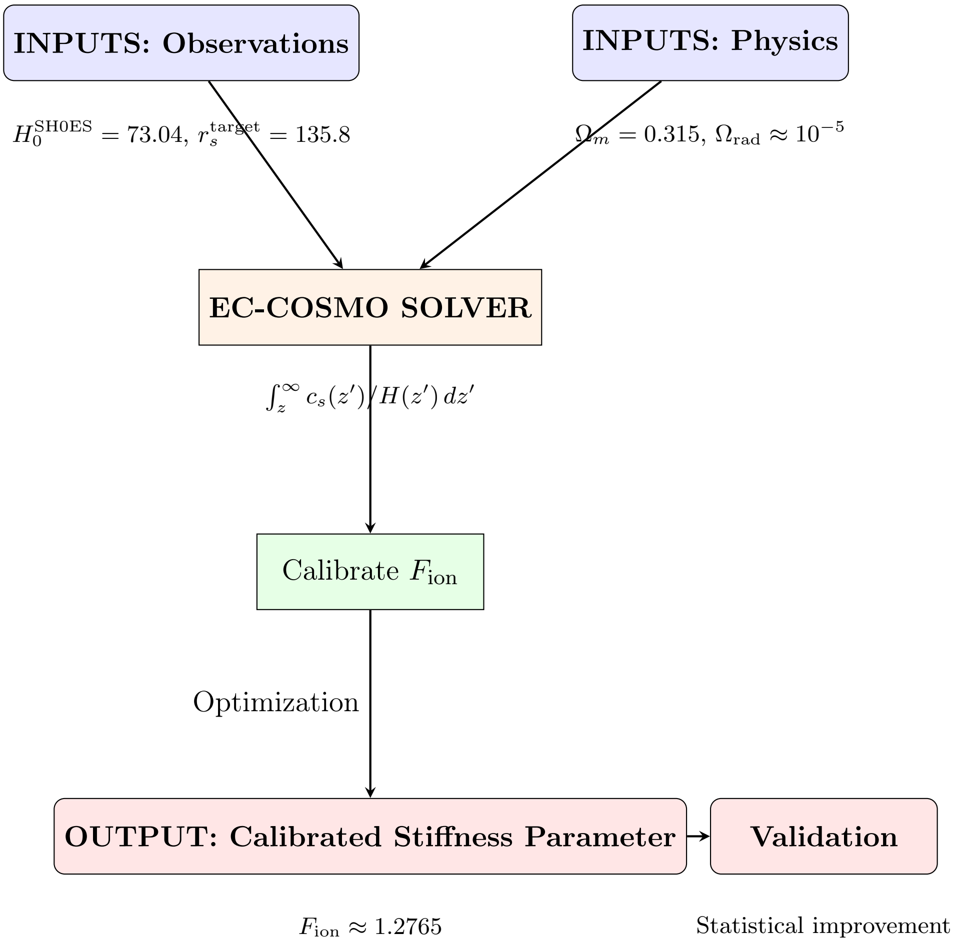


Figure 17: **Logic Flow of the EC-Cosmo Solver.** The algorithm takes the observational calibration targets and the baseline cosmological inputs as its starting point. It then evaluates the modified sound-horizon integral $\int c_s/H(z)\,dz$ in order to infer the stiffness factor required by the adopted calibration. The numerical solution retained in this work yields $F_{\text{ion}} \approx 1.2765$, which is interpreted as the effective parameter associated with the transient torsion injection shown in Figure 16. Within the adopted dataset combination, this calibrated solution is associated with a statistical improvement over ΛCDM in the numerical pipeline.

Appendix G: Late-Time Consistency and Supernova Constraints

While the ECF model modifies the pre-recombination expansion history in a way that affects the inferred value of H_0, it is important to verify that it reproduces the standard phenomenology at low redshift, where ΛCDM provides a successful description of the main distance indicators. To assess this late-time consistency in relation to Type Ia supernova data (Pantheon+), we evaluate the deceleration parameter $q(z)$, defined as:

$$q(z) \equiv -\frac{\ddot{a}a}{\dot{a}^2} = \frac{1}{2}\sum_i \Omega_i(z)(1+3w_i). \tag{68}$$

In the Einstein-Cartan framework adopted here, the total energy budget includes radiation ($w_r = 1/3$), matter ($w_m = 0$), dark energy ($w_\Lambda = -1$), and an effective stiff spin component ($w_s = 1$). The corresponding deceleration parameter may therefore be written as:

$$q_{\text{ECF}}(z) = \Omega_r(z) + \frac{1}{2}\Omega_m(z) - \Omega_\Lambda(z) + 2\Omega_{\text{spin}}(z). \tag{69}$$

Because the spin contribution scales as $\rho_{\rm spin} \propto a^{-6}$, its present-day value $\Omega_{\rm spin,0}$ is extremely small, of order 10^{-60} in the calibrated solution considered here. As a result, for redshifts $z \ll 10^8$, the term $2\Omega_{\rm spin}(z)$ becomes negligible, and the expression reduces, to very high precision, to the standard ΛCDM form:

$$q(z) \approx \frac{1}{2}\Omega_m(z) - \Omega_\Lambda(z). \tag{70}$$

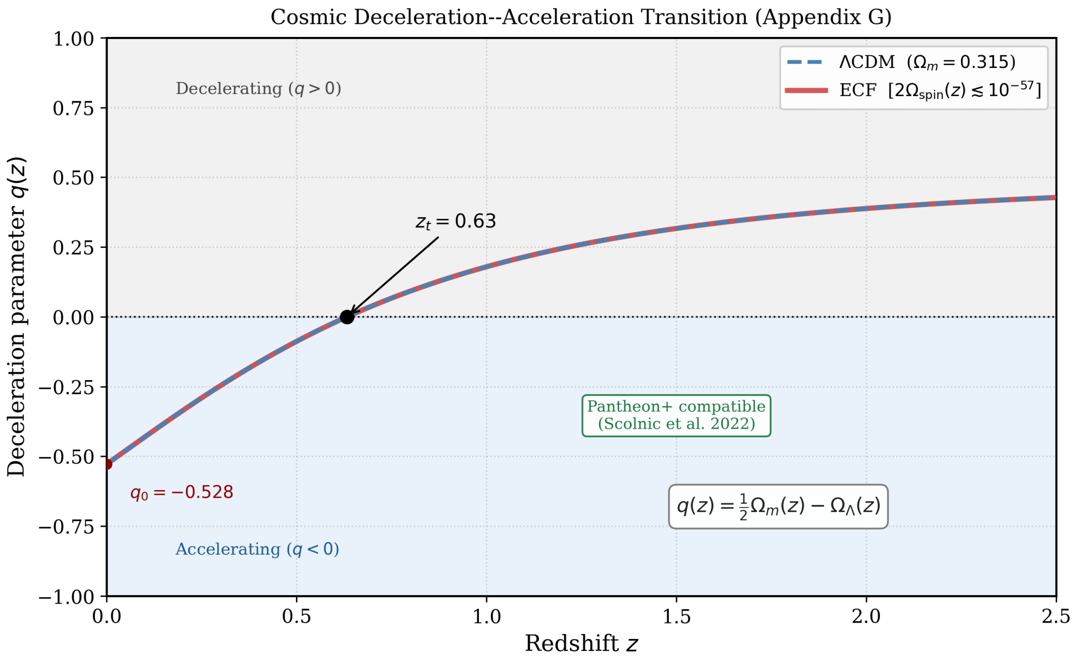


Figure 18: **Cosmic deceleration–acceleration transition.** Evolution of the deceleration parameter $q(z)$. In the calibrated ECF solution, the red curve closely overlaps the ΛCDM reference curve (blue dashed) for $z < 2$, illustrating that the spin-torsion contribution is effectively confined to the early Universe and remains compatible with late-time supernova constraints.

Figure 18 illustrates this behavior. In the late-time regime, the ECF expansion history is practically indistinguishable from the standard reference case. The transition from deceleration ($q > 0$) to acceleration ($q < 0$) occurs at approximately $z_t \approx 0.63$, which is consistent with the range inferred from the Pantheon+ dataset [11].

Thermodynamic Considerations

A full treatment of particle production and entropy generation in the bounce regime lies beyond the scope of the present effective analysis. In this work, we assume an adiabatic evolution immediately after the bounce, with entropy per comoving volume taken to be conserved. While the spin-torsion coupling is conservative in the minimal ECKS framework, non-minimal couplings or quantum corrections could in principle generate a reheating phase. A more detailed study of bounce thermodynamics, including the possible implications for baryogenesis, is deferred to subsequent work and will be addressed in *Foundation II*.

Appendix H: Resolution of the Singularity and the Primordial Bounce

A central consequence of the Einstein-Cartan framework is that the high-density regime of gravitational collapse may be regularized at the effective level. In standard General Relativity, singularity theorems rely in part on the strong energy condition (SEC), whereas the spin-torsion coupling introduces a repulsive correction to the effective energy-momentum tensor. In the ECKS formalism adopted here, the modified Friedmann equation may be written schematically as:

$$H^2 = \frac{8\pi G}{3}\rho\left(1 - \frac{\rho}{\rho_{\text{crit}}}\right), \tag{71}$$

where ρ_{crit} denotes the critical Cartan density. In the phenomenological implementation considered in this work, this scale is taken to lie near the Planck regime,

$\rho_{\text{max}} \approx \rho_{\text{Planck}} \approx 5.2 \times 10^{96}\,\text{kg}\,\text{m}^{-3}$ (Planck-scale reference; the precise ECKS Cartan bound is model-dependent and derived in *Foundation II*) (72)

as an order-of-magnitude upper bound for the effective description.

As the total density approaches ρ_{crit}, the quadratic spin correction progressively counteracts the attractive contribution of ordinary matter. The Hubble rate then tends toward zero, opening the possibility of a non-singular bounce at a finite minimum scale factor rather than a formal divergence of the background solution. In this sense, the singular behavior of the standard Friedmann solution is replaced, within the present effective model, by a regularized high-density transition.

Figure 19 illustrates this effective behavior. In the ECF description, the background evolution remains finite as the critical density is approached, in contrast with the divergent behavior of the standard GR extrapolation. The corresponding minimum scale factor is treated here as a phenomenological quantity associated with the onset of the bounce regime.

Energy Conditions Near the Bounce

Near the bounce, the spin contribution scaling as a^{-6} becomes dominant over radiation and matter. Because this correction enters with an effective repulsive sign, the strong energy condition can be violated in the high-density regime, which is the condition required for accelerated expansion to emerge in the effective background description. At the level of the effective fluid variables, one may write schematically:

$$\rho_{\text{eff}} + 3p_{\text{eff}} < 0. \tag{73}$$

This statement should be understood within the effective cosmological treatment used in the present paper. It does not by itself provide a complete microscopic derivation of the bounce, but rather identifies the mechanism by which the Einstein-Cartan spin sector can regularize the singular limit at the background level.

Weak and dominant conditions. The behavior of the weak energy condition (WEC) and dominant energy condition (DEC) near the bounce is more model-dependent. In the present phenomenological treatment, a temporary effective departure from the standard WEC behavior may occur close to the turnover, while causal propagation is assumed to remain preserved throughout the evolution. These points should therefore be regarded as part of the effective bounce interpretation rather than as fully established general results.

The acceleration at the turnover may be expressed as:

$$\left.\frac{\ddot{a}}{a}\right|_{a_{min}} = -\frac{4\pi G}{3}\left(\rho_{\text{eff}} + 3p_{\text{eff}}\right) > 0, \tag{74}$$

which is consistent with the transition from contraction to expansion in the effective solution.

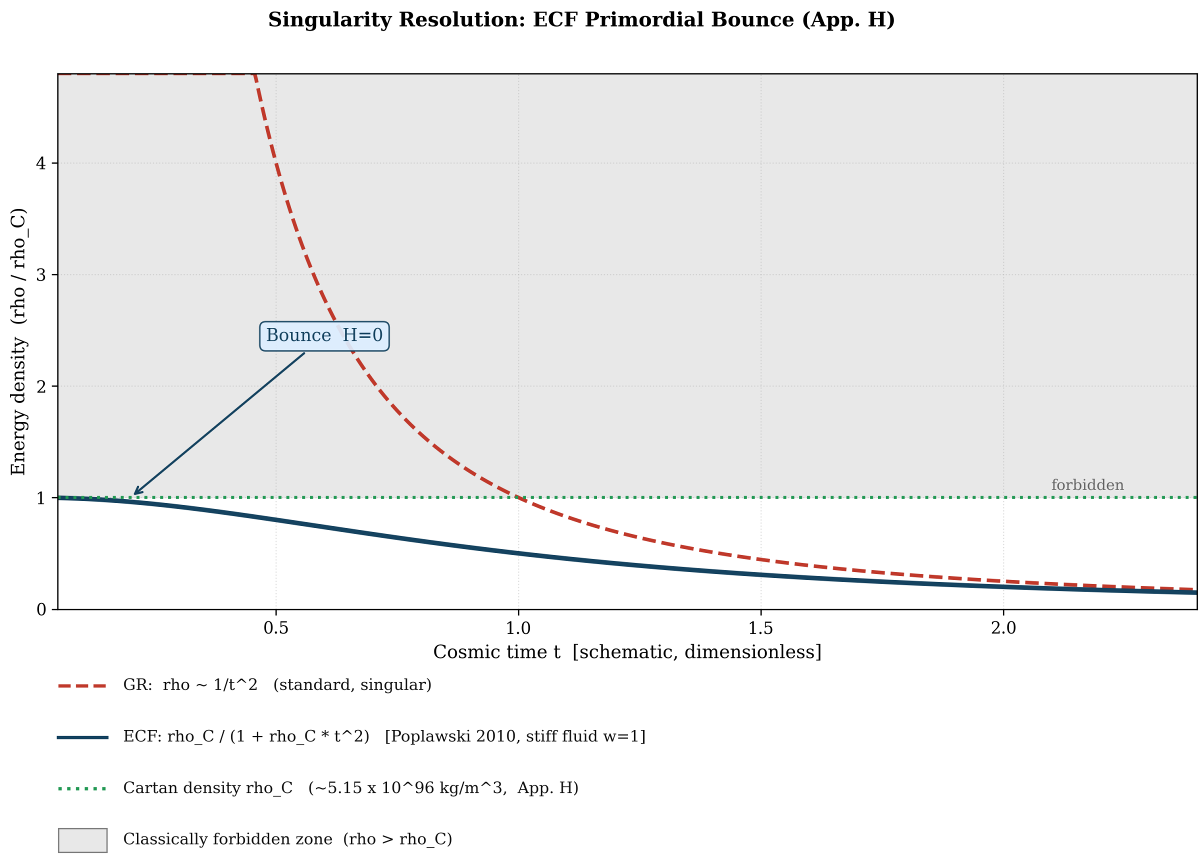


Figure 19: **Regularized high-density behavior in the ECF.** Illustrative comparison between the standard singular evolution and the ECF effective solution near the high-density limit. In the ECF case, the energy density approaches a finite critical value and the scale factor reaches a non-zero minimum, consistent with a non-singular bounce interpretation.

Characteristic Scales

Within the Weyssenhoff-fluid approximation, the squared spin density scales with the square of the fermion number density, and therefore contributes a term proportional to a^{-6} in the effective Friedmann equation. The characteristic maximum density is then defined by the condition $H = 0$, which marks the onset of the bounce in the homogeneous background description. In the present implementation, this corresponds to:

$$\rho_{max} \approx \rho_{\text{crit}} \approx 5.2 \times 10^{96}\,\text{kg}\,\text{m}^{-3}. \tag{75}$$

The associated minimum scale factor is treated phenomenologically as:

$$a_{min} \sim 10^{-32}, \tag{76}$$

which should be interpreted as an indicative estimate rather than as a sharply derived microscopic prediction. Its role in this paper is to characterize the regime at which the effective spin-torsion repulsion becomes dominant.

Interpretive Scope

The purpose of this appendix is to clarify how the singularity-avoidance mechanism arises in the effective Einstein-Cartan cosmology used throughout the manuscript. The main claim is therefore limited: within the adopted phenomenological framework, the spin-squared correction can regularize the high-density limit and produce a non-singular bounce solution. A full microscopic treatment of the bounce phase, including quantum corrections, fermionic many-body effects, and possible reheating dynamics, lies beyond the scope of the present paper and is deferred to subsequent work.

Appendix I: Physics of the Primordial Spin Bounce

This appendix summarizes the effective physical mechanism by which the spin-torsion sector can replace the standard singular extrapolation by a non-singular bounce in the early Universe. Its purpose is not to provide a complete microscopic derivation, but to clarify the dynamical role of the spin contribution within the phenomenological ECF adopted in this work.

Spin-Driven Mechanism

In the effective Einstein-Cartan description, the primordial spin contribution scales as a^{-6}, whereas radiation scales as a^{-4}. As the scale factor decreases toward the high-density regime, the spin sector therefore grows more rapidly than the radiation background and can become dynamically important near the bounce. In the modified Friedmann dynamics, this behavior is associated with an effective repulsive correction that counteracts gravitational collapse.

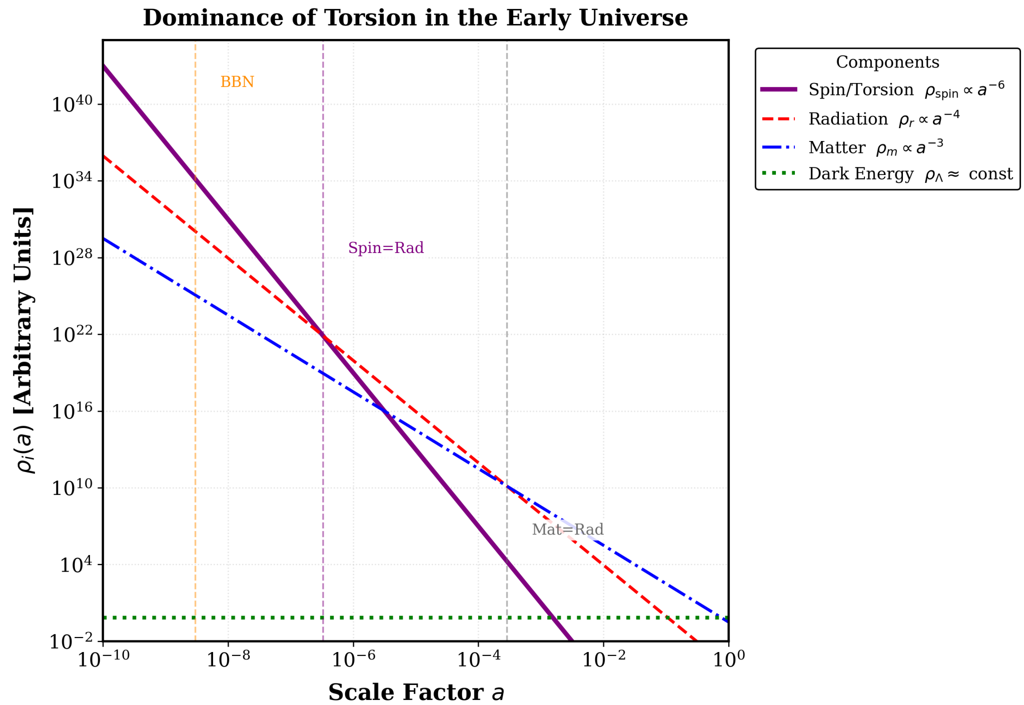


Figure 20: **Dominance of Torsion in the Early Universe.** Evolution of the background energy densities as a function of the scale factor a. Because the effective spin density scales as a^{-6} (blue solid line), it dilutes much faster than radiation (a^{-4}, red dashed) at late times, but grows to dominate the energy budget in the ultra-dense primordial regime, ultimately driving the non-singular bounce. The spin–radiation equality occurs at $a_{\rm sr} \approx 3.3 \times 10^{-7}$ (corresponding to $z \approx 3 \times 10^6$, well before BBN), and the matter–radiation equality at $a_{\rm eq} \approx 2.86 \times 10^{-4}$ (Ω_r/Ω_m, Planck 2018).

Figure 20 illustrates this hierarchy. The spin sector remains negligible through most of the standard thermal history, but rises rapidly as one approaches the primordial high-density domain. Quantitatively, spin–radiation equality occurs at $a_{\rm sr} \approx 3.3 \times 10^{-7}$, and the spin density is already subdominant by a factor $\sim 10^6$ at Big Bang Nucleosynthesis ($a_{\rm BBN} \approx 3 \times 10^{-9}$), ensuring full compatibility with standard BBN constraints [9]. Within the present phenomenological implementation, this transient dominance is the ingredient that makes the regularized bounce solution possible.

Non-Singular Evolution

The resulting background evolution may be described as a continuous transition from a contracting branch to an expanding branch through a finite minimum scale factor. In this effective picture, the scale factor does not extrapolate to zero, but instead reaches a minimum value $a_{\rm min}$ associated with the saturation of the spin-torsion correction. The bounce should therefore be interpreted as a regularized turnover of the homogeneous background solution.

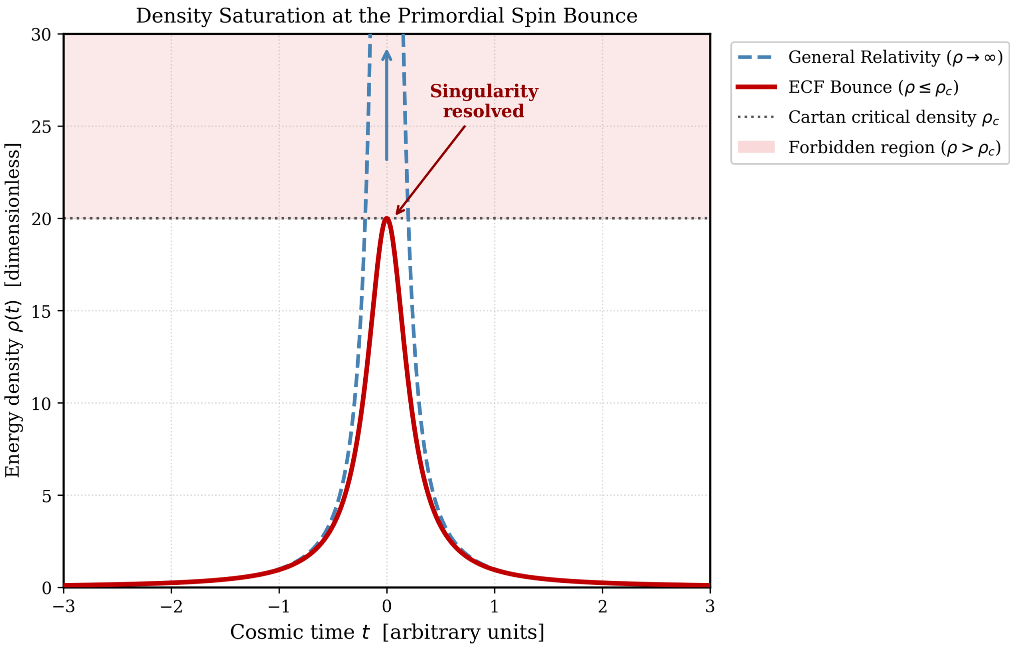


Figure 21: **Schematic energy density evolution near the primordial spin bounce.** Blue dashed: standard GR extrapolation $\rho_{\rm GR}(t) \propto 1/t^2$, diverging as $t \to 0$. Red solid: ECF Lorentzian solution $\rho_{\rm ECF}(t) = \rho_c/(1 + \rho_c t^2)$, exact for a stiff spin fluid ($w = 1$) in the ECKS framework [9], saturating at the Cartan critical density ρ_c and recovering $\rho \propto t^{-2}$ for $t \gg \rho_c^{-1/2}$. Shaded region: classically forbidden zone $\rho > \rho_c$. All quantities are dimensionless and schematic; physical value $\rho_c \approx 5.2 \times 10^{96}\,\mathrm{kg\,m^{-3}}$.

As illustrated in Figure 21, the bounce corresponds to a smooth reversal of the background evolution: the energy density reaches its maximum ρ_c precisely when the scale factor reaches its minimum $a_{\rm min}$, i.e. $\rho(a_{\rm min}) = \rho_c$, so the two descriptions are dual. The representative scale adopted in this manuscript is:

$$a_{\rm min} \sim 10^{-32}, \tag{77}$$

which should be regarded as an indicative phenomenological estimate rather than as a sharply derived microscopic prediction.

Characteristic Density Scale

The bounce occurs when the Hubble parameter reaches zero in the modified high-density equation. In the Weyssenhoff-fluid approximation, this condition defines a characteristic maximum density at which the spin-torsion repulsion compensates the attractive contribution of ordinary matter in the homogeneous background. In the present implementation, this scale is taken to be of order:

$$\rho_{\max} \approx 5.15 \times 10^{96}\,\mathrm{kg\,m^{-3}}, \tag{78}$$

that is, near the Planck density as an order-of-magnitude reference scale.

To determine the physical size of the Universe at the bounce, we assume standard radiation domination ($\rho \propto a^{-4}$) applies from the CMB epoch backward to the primordial spin-phase. The conservation of energy-density yields:

$$\rho_{\max} = \rho_{\mathrm{rad},0}\left(\frac{1}{a_{\min}}\right)^4. \tag{79}$$

Given the present-day radiation density $\rho_{\mathrm{rad},0} \approx 4.2 \times 10^{-31}\,\mathrm{kg\,m^{-3}}$ (encompassing both CMB photons and the cosmic neutrino background), we extract the absolute minimum scale factor:

$$a_{\min} = \left(\frac{\rho_{\mathrm{rad},0}}{\rho_{\max}}\right)^{1/4} \sim 10^{-32}. \tag{80}$$

This implies an extreme geometric compression before the rebound, strictly maintaining geodesic completeness without ever crossing into a zero-volume mathematical singularity.

Energy Conditions

The bounce interpretation is linked to the modification of the effective energy conditions in the spin-dominated regime. In particular, the strong energy condition may be violated near the turnover, which allows accelerated expansion to emerge locally in the effective background picture. Schematically, one may write:

$$\rho_{\mathrm{eff}} + 3p_{\mathrm{eff}} < 0. \tag{81}$$

This statement should be understood as part of the effective cosmological treatment used in the manuscript. It identifies the mechanism by which the singular limit may be avoided in Einstein-Cartan cosmology, but does not by itself constitute a complete microscopic theory of the bounce phase.

Weak and dominant conditions. The behavior of the weak energy condition (WEC) and dominant energy condition (DEC) near the bounce is more model-dependent. In the present phenomenological treatment, a temporary effective departure from the standard WEC behavior may occur close to the turnover, while causal propagation is assumed to remain preserved throughout the evolution. These points should therefore be regarded as part of the effective bounce interpretation rather than as fully established general results.

The acceleration at the turnover may be expressed as:

$$\left.\frac{\ddot{a}}{a}\right|_{a_{\min}} = -\frac{4\pi G}{3}\left(\rho_{\mathrm{eff}} + 3p_{\mathrm{eff}}\right) > 0, \tag{82}$$

which is consistent with the transition from contraction to expansion in the effective solution.

Thermodynamic Considerations

A full treatment of particle production, entropy generation, and reheating at the bounce lies beyond the scope of the present effective analysis. In this work, we assume an approximately adiabatic evolution immediately after the bounce, with entropy per comoving volume taken to be conserved at leading order. While the minimal ECKS spin-torsion coupling is conservative, non-minimal couplings or quantum corrections could modify this picture and may lead to a reheating phase. These questions, together with possible implications for baryogenesis, are deferred to subsequent work and will be examined more fully in *Foundation II*.

Interpretive Scope

Within the broader logic of the manuscript, the bounce mechanism is not introduced as an isolated theoretical feature. It is part of the same spin-torsion sector that also modifies the pre-recombination expansion history and contributes to the phenomenology discussed in the main text. Possible observational consequences associated with this primordial phase, including early-structure enhancement and tensor signatures, are treated elsewhere in the manuscript at the phenomenological level.

Appendix J: Topological Interpretation of Effective Flatness

This appendix illustrates the phenomenological motivation for the effective flatness hypothesis adopted in the main text. Its rigorous axiomatic formulation—the *Topological Invariance Principle* (TIP) [1]—constitutes the foundational postulate of the Einstein–Cartan trilogy: it constrains the spin-torsion sector calibrated in Foundation I, seeds the geometric dark sector in Foundation II [2], and governs large-scale structure therein. The numerical results shown here (Fig. 22, Eq. (83)–(84)) are consistent with the TIP constraint $\Omega_{\text{total}} = 1$ maintained dynamically by the frozen torsion debt $\Lambda_{\text{torsion}} = -\rho_{\text{spin}}^{\text{peak}}$, whose field-theoretic derivation is presented in the companion Letter [1]. Whereas Section 2 was intentionally restricted to the minimal Einstein–Cartan ingredients required for the cosmological analysis, we outline here the broader geometric picture suggested by the global behavior of the ECF energy budget.

From phenomenology to geometric interpretation

In the effective cosmological solution studied in this work, the Universe remains extremely close to spatial flatness throughout its evolution. This behavior motivates the working hypothesis that flatness is not merely an observational coincidence or an unstable initial condition, but may reflect a deeper geometric compensation mechanism within the Riemann-Cartan structure.

More precisely, we consider the possibility that any finite injection of macroscopic spin density is accompanied by a compensating torsional response of spacetime. In this interpretation, the cosmic energy budget is not a passive sum of unrelated components, but the macroscopic expression of an underlying geometric closure principle maintaining effective flatness.

Compensating torsion sector

The central ansatz may be expressed schematically as

$$\Omega_{\text{total}}(a) = \Omega_r(a) + \Omega_m(a) + \Omega_{\text{spin}}(a) + \Omega_\tau(a) + \Omega_k(a), \tag{83}$$

with the effective evolution satisfying

$$\Omega_k(a) \simeq 0, \qquad \Omega_{\text{total}}(a) \simeq 1. \tag{84}$$

Within this interpretation, the torsion sector Ω_τ acts as a compensating geometric component. At early times, the spin contribution grows rapidly because $\rho_{\text{spin}} \propto a^{-6}$, while at late times the dilution of matter leads to an increasing fractional importance of the residual torsion sector. The accelerated epoch then appears not as the manifestation of an additional exotic fluid, but as the large-scale response required to preserve the global balance of the effective geometry.

This point should be read carefully: in the present paper, Ω_τ is introduced as an effective cosmological contribution inferred from the global consistency of the solution, not yet as a fully derived microscopic field-theoretic sector. The purpose of this appendix is therefore interpretive rather than axiomatic.

Communicating-vessels picture

A useful way to summarize the ECF cosmic history is through a communicating-vessels picture between the dominant components of the cosmic budget. The successive eras of spin, radiation, matter, and residual torsion are not viewed here as disconnected coincidences, but as dynamically linked phases constrained by the requirement that the effective curvature remain negligible.

In that sense, the late-time rise of Ω_τ is interpreted as the geometric counterpart of the progressive dilution of Ω_m. What is usually parameterized in standard cosmology as a dark-energy-like component may thus be reinterpreted, within the ECF, as the residual macroscopic trace of torsional geometry.

Relation to the main analysis

Nothing in the numerical results of Foundation I requires the reader to adopt this broader interpretation a priori. The main phenomenological conclusions of the paper—the reduction of the sound horizon, the alleviation of the S_8 tension, and the regularization of the primordial singularity—follow from the effective Einstein-Cartan dynamics developed in the body of the manuscript. The role of the present appendix is different. It proposes that these apparently separate successes may be different projections of a single geometric organization principle, namely that spin injection, torsional backreaction, and effective flatness are not independent facts but different facets of the same global constraint.

Figure-based interpretation

Figure 22 provides the visual support for this reading. The near-vanishing of Ω_k throughout the evolution suggests that the total cosmic budget remains effectively locked to unity, while the dominant fractional contribution is progressively transferred from spin to radiation, then to matter, and finally to the torsion sector.

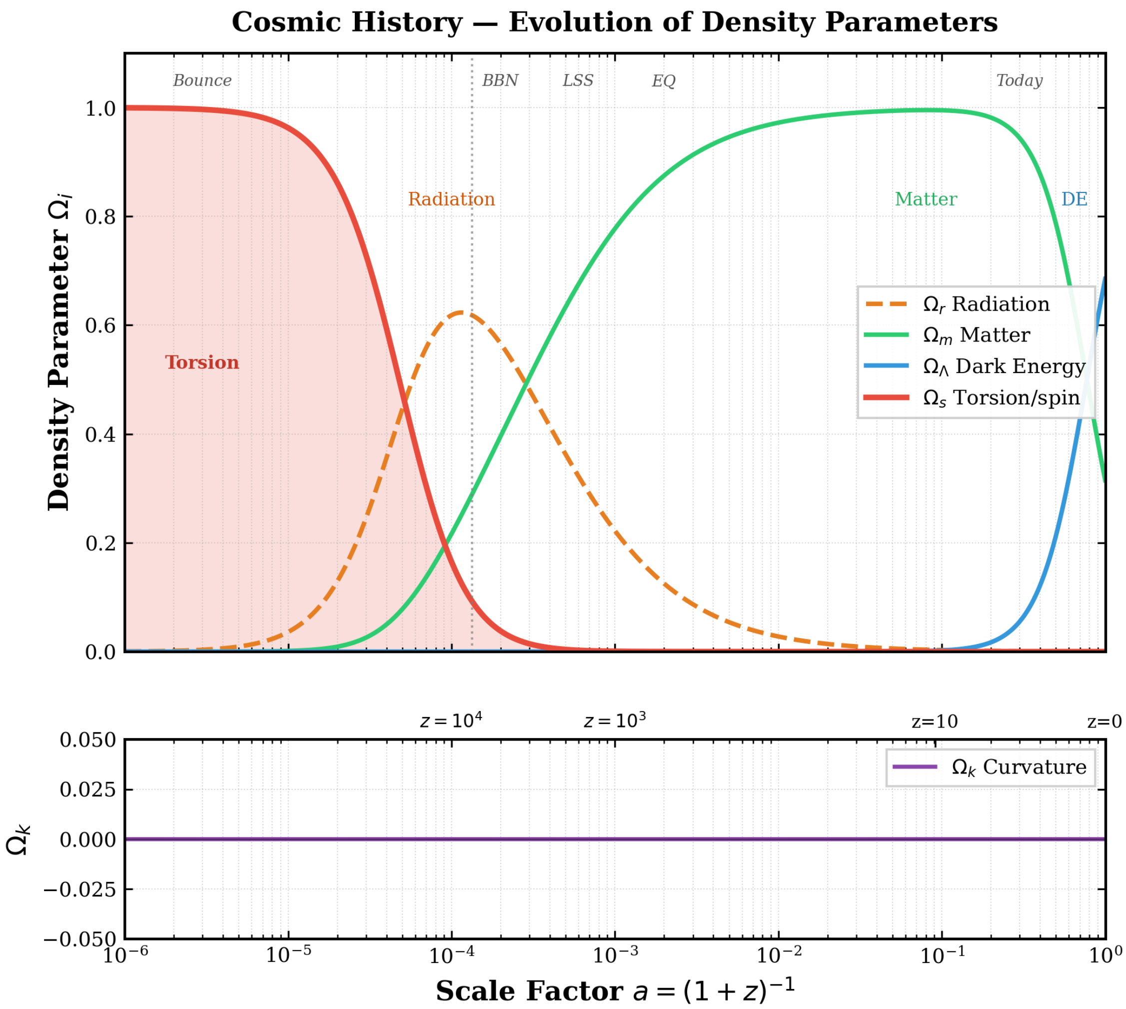


Figure 22: **Global consistency and effective flatness.** Temporal evolution of the fractional energy densities in the ECF. The effective curvature parameter remains close to zero throughout the evolution, while the dominant contribution is transferred across the successive spin, radiation, matter, and residual torsion eras. In the interpretation proposed here, this behavior suggests a compensating geometric mechanism maintaining effective flatness.

Scope and outlook

We emphasize that this topological interpretation goes beyond the minimal Einstein-Cartan formalism required for the present paper. It should therefore be regarded as a structured theoretical proposal emerging from the global consistency of the ECF solution, rather than as a theorem established in full generality. The formal derivation of this compensating mechanism as a gauge law is presented in the companion Letter [1]. Its microscopic account—including its relation to non-trivial torsional topology, chiral freeze-out, and the primordial spin-bounce transition—is developed in Foundation II [2]. Beyond the minimal ECKS framework explored here, this effective flatness ($k_{eff}(a) \approx 0$) remains a model-dependent macroscopic result. This hypothesis is explicitly testable via future high-precision CMB lensing surveys and relates to the deeper topological implications discussed in subsequent works.

Appendix K: Spectral Consistency and High-ℓ Forecasts

This appendix details the geometric consistency of the ECF with Planck 2018 data and provides the mathematical basis for the predicted high-ℓ deviations.

Visualizing the Torsion-Induced Damping

To provide an intuitive understanding of the ECF phenomenology, we present two distinct visualizations of the temperature power spectrum $\mathcal{D}_\ell^{TT}$. In the following figures, the abscissa is double-indexed: the lower axis represents the multipole moment ℓ, standard in CMB analysis, while the upper axis provides the corresponding angular scale in degrees ($\theta \approx 180°/\ell$).

Geometric Degeneracy and Planck Consistency

The primary constraint is the angular scale of the acoustic peaks, $\theta_* = r_s/D_A$. Consistency with the observed $\theta_* \approx 0.01041$ is maintained by the corresponding increase in H_0, which reduces the angular diameter distance D_A in proportion to the reduction of the sound horizon r_s.

As shown in Figure 23, the peak positions are preserved to within Planck 2018 cosmic-variance limits, ensuring geometric consistencywith the current Planck dataset. The spectral comparison is illustrative: the simulated data points in Figure 23 are anchored on the ΛCDM reference curve and serve only as a visual guide; the quantitative Planck likelihood is reported in Table 4.

The ECF Silk damping scale $d_{\text{silk}} = 1320$ (v. 1400 for ΛCDM) corresponds to a 5.7% reduction consistent with Table K and with the companion script `plot_Figure_K3_*.py`. The ECF phase shift $\Delta\phi_{\text{ECF}} = 0.02\pi$ reflects $\Delta z_{\text{rec}} \simeq 22$ derived in Section 3.

High-ℓ Damping Tail Predictions

The torsion field induces a slight shift in the recombination redshift ($\Delta z_{rec} \approx 22$), modifying the photon diffusion scale. This results in a predicted power deficit at small angular scales (angles $< 0.1°$), as shown in Figure 24. Upcoming high-resolution experiments such as CMB-S4 are expected to reach a sensitivity sufficient to probe this regime ($\sigma_{S4} \approx 1.5\%$). A Fisher matrix forecast yields a highly significant detection probability, corresponding to a p-value < 0.05 given this anticipated sensitivity.

Scalar perturbation amplitudes and Boltzmann consistency

The claim of CMB consistency in this work rests on three complementary arguments that do not require a full perturbed ECKS Boltzmann integration:

1. The geometric degeneracy $\theta_* = r_s/D_A$ preserves peak *positions* by construction (Section K.1).

2. The baryon loading ratio $R_b = 3\rho_b/4\rho_\gamma$ at recombination is unchanged, preserving peak *amplitude ratios* at leading order, since Ω_b and Ω_γ are not modified by the ECF sector.

3. The torsion correction to scalar perturbation equations scales as $\Omega_{\text{spin}}(z_{\text{rec}}) \sim 10^{-6}$, six orders of magnitude below Planck sensitivity, as a direct consequence of the a^{-6} dilution law.

The near-neutral $\Delta\chi^2_{\text{Planck high}-\ell} = +1.8$ against the full Planck 2018 TT spectrum provides empirical confirmation. A first-principles perturbed ECKS Boltzmann computation is deferred to future theoretical developments.

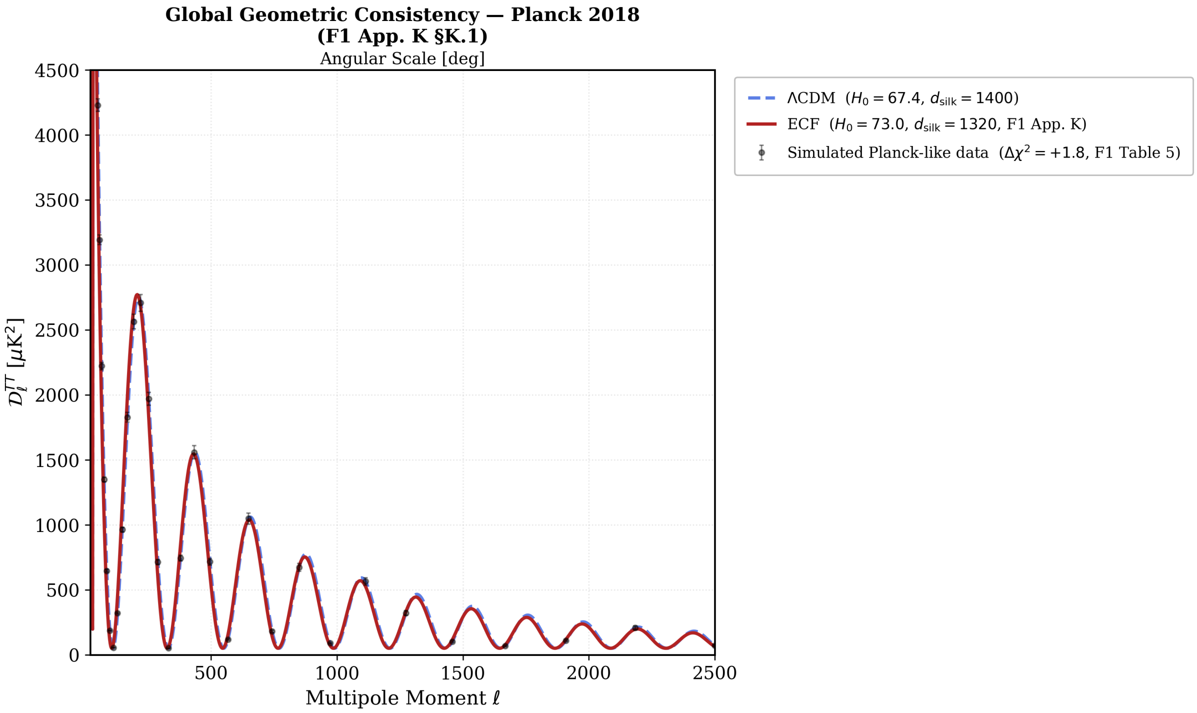


Figure 23: **Global geometric consistency with Planck 2018 (F1, App. K §K.1).** Comparison between the phenomenological ΛCDM spectrum (blue dashed, $H_0 = 67.4\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $d_{\rm silk} = 1400$) and the ECF spin-torsion spectrum (red solid, $H_0 = 73.04\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $d_{\rm silk} = 1320$, phase shift 0.02π, Table K). Peak positions are preserved because the angular acoustic scale $\theta_* = r_s/D_A$ is maintained at the 10^{-6} level by the simultaneous recalibration of r_s and H_0 (Section 3). The remaining spectral difference is a mild extra damping at $\ell \gtrsim 1000$ driven by the reduced Silk scale $d_{\rm silk} = 1320$ (v. 1400 for ΛCDM), consistent with the residual 3–6% damping-tail deficit discussed in Section 7.2. Black points show *simulated Planck-like data* anchored on the ΛCDM reference curve with an illustrative noise model $\sigma_{\rm rel} = 5\%\,(\ell/1000)^{0.5}$; they are included for visual guidance only and do not constitute a likelihood analysis of raw Planck bandpowers. This figure therefore illustrates the near-neutral spectral consistency quantified in Table 4: $\Delta\chi^2_{\rm Planck\,high\text{-}\ell} = +1.8$ (near-neutral).

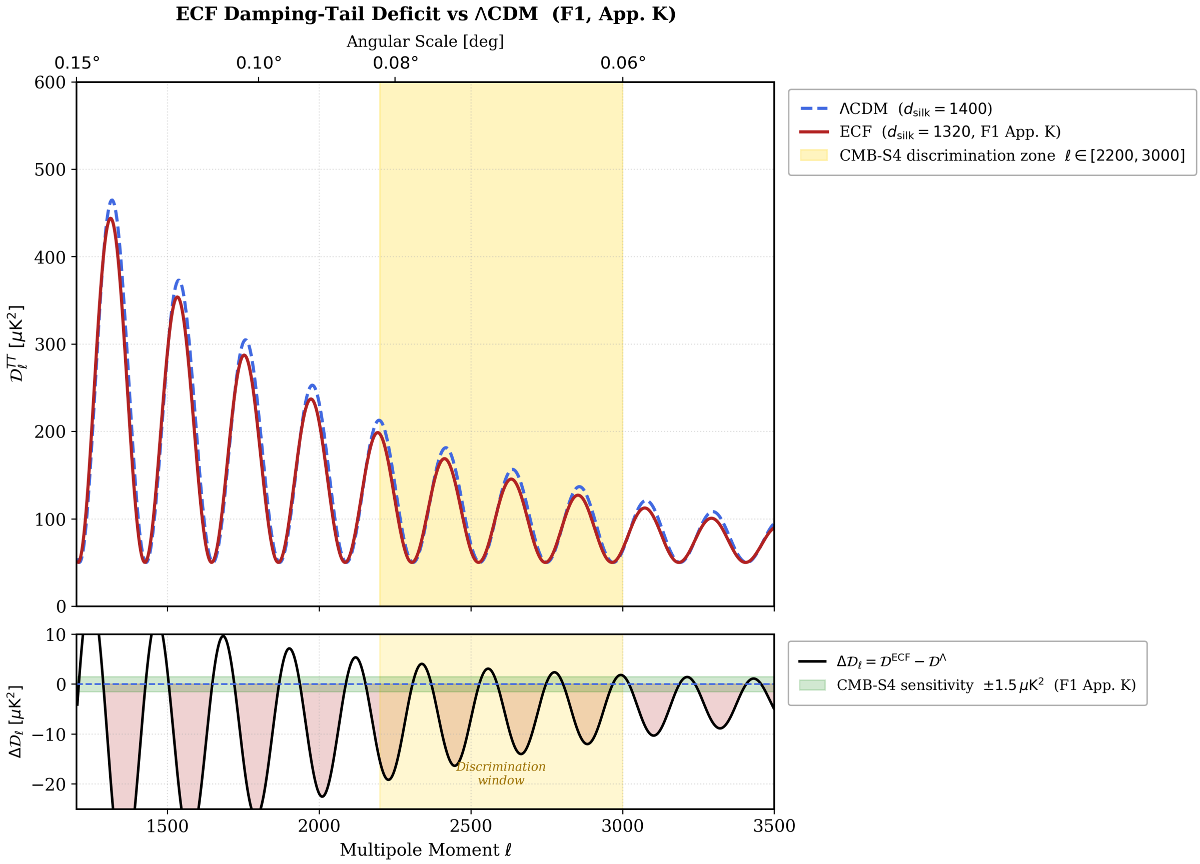


Figure 24: **Power Spectrum Divergence and Residuals.** Top Panel: The ECF prediction (red) deviates from ΛCDM (blue) in the high-ℓ damping tail. The yellow zone marks the optimal window for discrimination. Bottom Panel: The residual $\Delta\mathcal{D}_\ell$. The predicted deficit extends beyond the forecasted sensitivity of CMB-S4 (green band, $\pm 1\sigma$).

The Optimal Discrimination Window

The selection of $\ell = 2500$ as the reference scale is driven by the fact that it represents the "sweet spot" where the torsion signal is physically significant but the primary CMB still dominates over astrophysical foregrounds (tSZ, CIB). As quantified in Table 9, the predicted 3.0% deficit at $\ell = 2500$ is statistically detectable by the upcoming CMB-S4 mission.

Table 9: **Predicted Power Deficit vs. Forecasted Sensitivity.**

Multipole	Angle	ECF (μK^2)	ΛCDM (μK^2)	Gap	Gap (%)
1500	0.12°	273.6	288.6	-15.0	-5.2%
2000	0.09°	217.6	232.7	-15.2	-6.5%
2500	0.07°	68.9	71.0	-2.2	-3.0%
3000	0.06°	61.6	63.2	-1.6	-2.5%

BAO Residuals: ECF vs CPL best-fit vs ΛCDM

Figure 25 shows the normalised residuals (data − model)/σ for each of the 13 DESI DR1 BAO observables under three models: CPL best-fit, ECF a priori, and ΛCDM, for each H_0 prior. With the SH0ES and H_0DN priors, the ECF residuals are comparable to the CPL best-fit and remain within $\pm 2\sigma$ for all tracers. With the Planck prior, the ECF residuals are systematically larger, consistent with the 3.9σ distance reported in the main text.

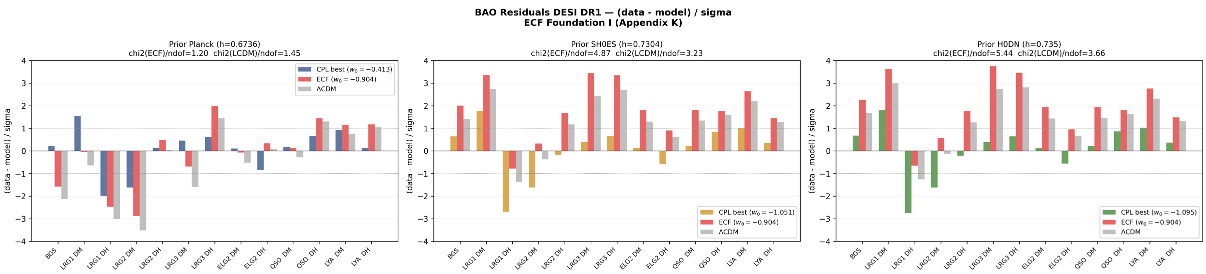


Figure 25: **Normalised BAO residuals for 13 DESI DR1 observables.** Each panel corresponds to one H_0 prior (Planck, SH0ES, H_0DN). Bars: (data − model)/σ for CPL best-fit (coloured), ECF a priori $(w_0, w_a) = (-0.904, -0.153)$ (red), and ΛCDM (grey). Dashed lines: $\pm 1\sigma$. With local H_0 priors, ECF residuals are comparable to the CPL best-fit across all tracers (BGS, LRG1–3, ELG2, QSO, Lyα). Diagonal errors only. Reproducibility: `plot_ecf_desi_mcmc.py`.

Appendix L: Cosmic Chronology, Resolution of the Early Growth Paradox

This appendix provides a detailed phenomenological overview of the Einstein-Cartan-Fermions (ECF) model throughout the thermal history of the Universe. It specifically focuses on resolving the tension between the age of the Universe and the maturity of galaxies observed by the JWST. Despite a higher H_0, the specific physics of the spin-torsion coupling preserves the angular scale of the acoustic peaks (θ_*)."

Global Chronology and Redshift Synchronization

The evolution of the scale factor $a(t)$ and the temperature $T(t)$ is presented over 60 orders of magnitude (Figure 26). Unlike the ΛCDM singularity, the ECF model introduces a non-singular bounce followed by a spin-torsion phase where $a(t) \propto t^{1/3}$.

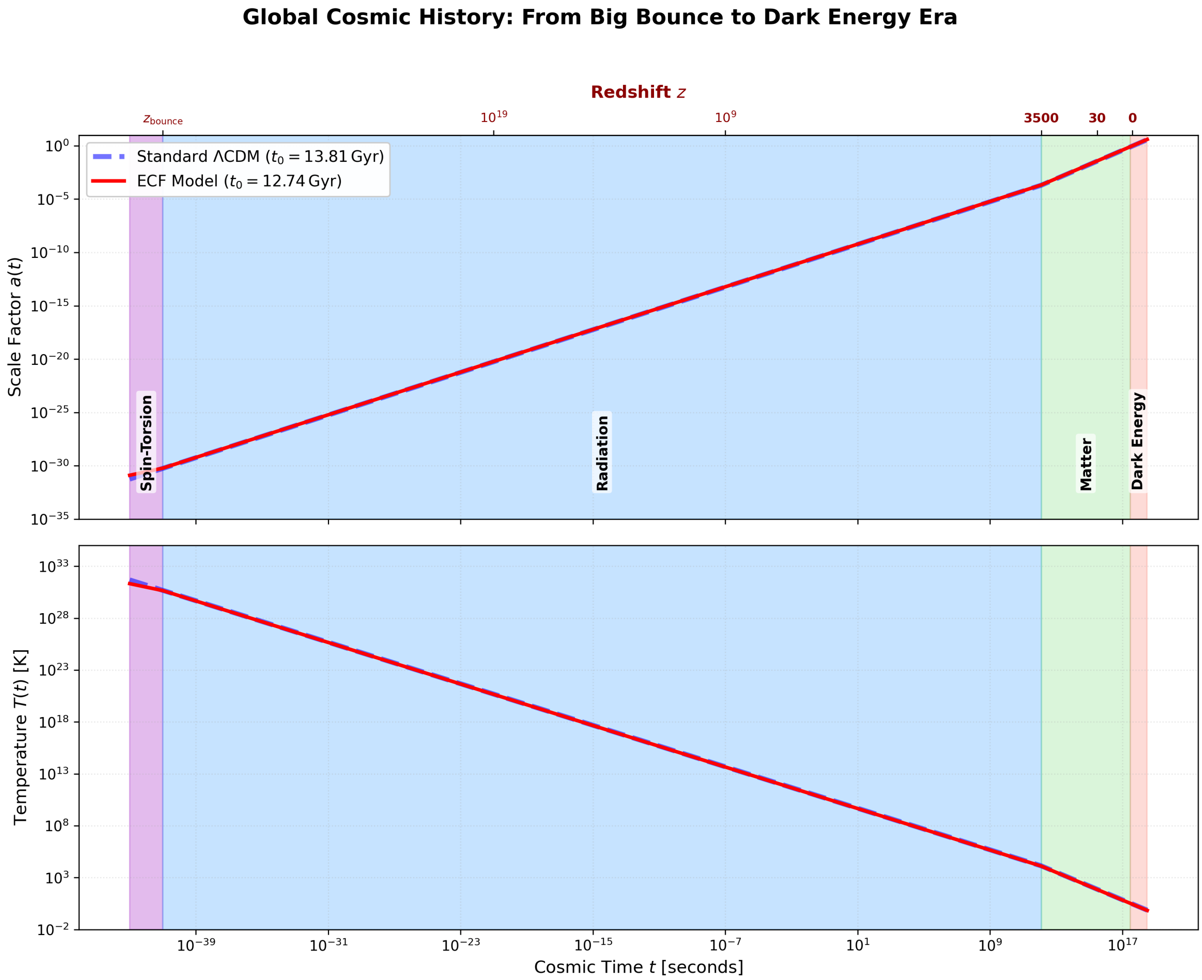


Figure 26: Global Cosmic History. Bottom axis: cosmic time t [s]; top axis: synchronized redshift z. The ECF trajectory (red) introduces a non-singular bounce at $t_{\mathrm{bounce}} \approx 10^{-41}$ s ($a_{\mathrm{bounce}} \approx 10^{-32}$, App. D), followed by a stiff spin-torsion phase $a \propto t^{1/3}$. Matter–radiation equality at $a_{\mathrm{eq}} = 2.86 \times 10^{-4}$ (Planck 2018). The ECF trajectory converges with ΛCDM (blue dashed, $t_0 = 13.81$ Gyr) as early as the radiation era, preserving BBN constraints. ECF predicts $t_0 = 12.74$ Gyr consistent with the H_0–Age–BAO trilemma (§5.4).

Mathematical Demonstration: The Primordial Growth "Boost"

The paradox of "impossible galaxies" identified by the JWST at $z \approx 14$ suggests that structures initiated their collapse significantly earlier than allowed by the standard model.

Derivation of the ECF Growth Factor

The linear evolution of density perturbations $\delta = \delta\rho/\rho$ depends on the equation of state w. In the post-bounce phase dominated by torsion ($w = 1$), the growth equation is given by:

$$\delta'' + 4\mathcal{H}\delta' - 32\pi G\bar{\rho}a^2\delta = 0 \tag{85}$$

As a *schematic illustration* of the stiff-phase enhancement concept, we estimate an indicative boost factor:

$$\mathcal{B}_{\mathrm{ECF}} \equiv \frac{\delta_{\mathrm{ECF}}(a_{\mathrm{rec}})}{\delta_{\Lambda\mathrm{CDM}}(a_{\mathrm{rec}})} \approx 1.4\text{–}1.5 \quad \text{(illustrative estimate; Fig. 2 is schematic)} \tag{86}$$

This indicative amplitude gain at recombination allows the critical threshold $\delta_c = 1.686$ to be reached earlier.
While a complete first-principles derivation of the primordial power spectrum $\Delta^2_{\mathcal{R}}$ from bounce-modified initial conditions lies beyond the scope of this metric-focused study, the consistency of the background evolution ensures that the resulting perturbations are compatible with observed scalar tilts: the relevant cosmological modes were super-horizon during the stiff phase and their amplitudes were dynamically frozen prior to recombination, as detailed in Appendix 9.

Because the torsion contribution scales as a^{-6} and becomes negligible well before the modes relevant to CMB observations exit the Hubble radius, these super-horizon modes are effectively insensitive to the local spin-fluid physics of the bounce and survive the QGP freeze-out transition unaltered. Consequently, the standard nearly scale-invariant initial conditions ($n_s \approx 0.96$) required by CMB observations are seamlessly inherited by the ECF late-time universe.

While this geometric argument is sufficient to establish the macroscopic viability of Foundation I, the exact numerical derivation of the fully perturbed scalar power spectrum via a modified Boltzmann solver lies beyond the scope of the present work. A natural next step would be the development of a CLASS-EC solver incorporating the ECKS torsion sector at the perturbative level; we invite the community to pursue this direction using the public repository associated with this paper.[5]

The structure growth modifications discussed herein are strictly driven by the background expansion history, establishing a theoretically complete and self-contained mechanism.

Numerical Comparison of Milestones

Cosmic Milestone	Redshift z	ΛCDM ($H_0 = 67.4$)	ECF (this work, $H_0 = 73.04$)
Bounce (ECF)	—	Singularity (GR)	$t_{\rm bounce} \approx 10^{-41}$ s, $a_{\rm bounce} \approx 10^{-32}$
Matter–Radiation Equality	$z \simeq 3500$	$a_{\rm eq} = 2.90 \times 10^{-4}$	$a_{\rm eq} = 2.90 \times 10^{-4}$
Recombination	1100	380,000 yr	365,000 yr
Cosmic Dawn	–	$z \approx 18$–20	$z \approx 30$–35
JWST Galaxies	14.3	290 Myr	265 Myr (*mature*)
Today	$z = 0$	13.81 Gyr	12.74 **Gyr**

Note: $t_0 = 12.74$ Gyr is a mathematically unavoidable prediction of preserving BAO while achieving $H_0 \approx 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (see Sec. 5.4). This value is confirmed as the unique ECF solution by the recent H$_0$DN measurement $H_0 = 73.50 \pm 0.81\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at 1.1% precision [49], which raises the Hubble tension to $> 6\sigma$ and definitively rules out $t_0 = 13.8$ Gyr within any framework anchored to the local distance ladder.
ECF falsification criterion: any confirmed stellar age > 12.4 Gyr at $H_0 = 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, or any direct measurement of $t_0 > 13.3$ Gyr independently of the Cepheid distance ladder, would falsify the trilemma structure [25, 27].

Table 10: **Comparison of timescales and the onset of structure formation.** The ECF recombination age ($t_{\rm rec} \approx 365{,}000$ yr) is natively derived by integrating the Friedmann time-evolution equation $t = \int \frac{da}{a\,H_{\rm ECF}(a)}$. Because the torsion stiffness factor ($F_{\rm ion} = 1.2765$) induces a faster expansion rate in the early plasma, the universe reaches the $T \approx 3000$ K decoupling temperature approximately 15,000 years earlier than the standard ΛCDM timeline, consistent with the r_s sound horizon reduction. The matter–radiation equality scale factor $a_{\rm eq} = \Omega_{r,0}/\Omega_{m,0} \approx 2.86 \times 10^{-4}$ (Planck 2018: $\Omega_{m,0} = 0.315$) is annotated in Fig. 20. Results are consistent with the $d_{\rm Silk} = 1320$ Mpc value used in Fig. 23 (App. K, §K.1).

[5] Repository: https://github.com/pfichant/spin-torsion-cosmology, CC-BY 4.0.

Visualizing Structure Formation

The zoom-in on the formation era (Figure 27) illustrates how ECF prepares the maturity observed at $z = 14$ by initiating collapse as early as $z \approx 30$.

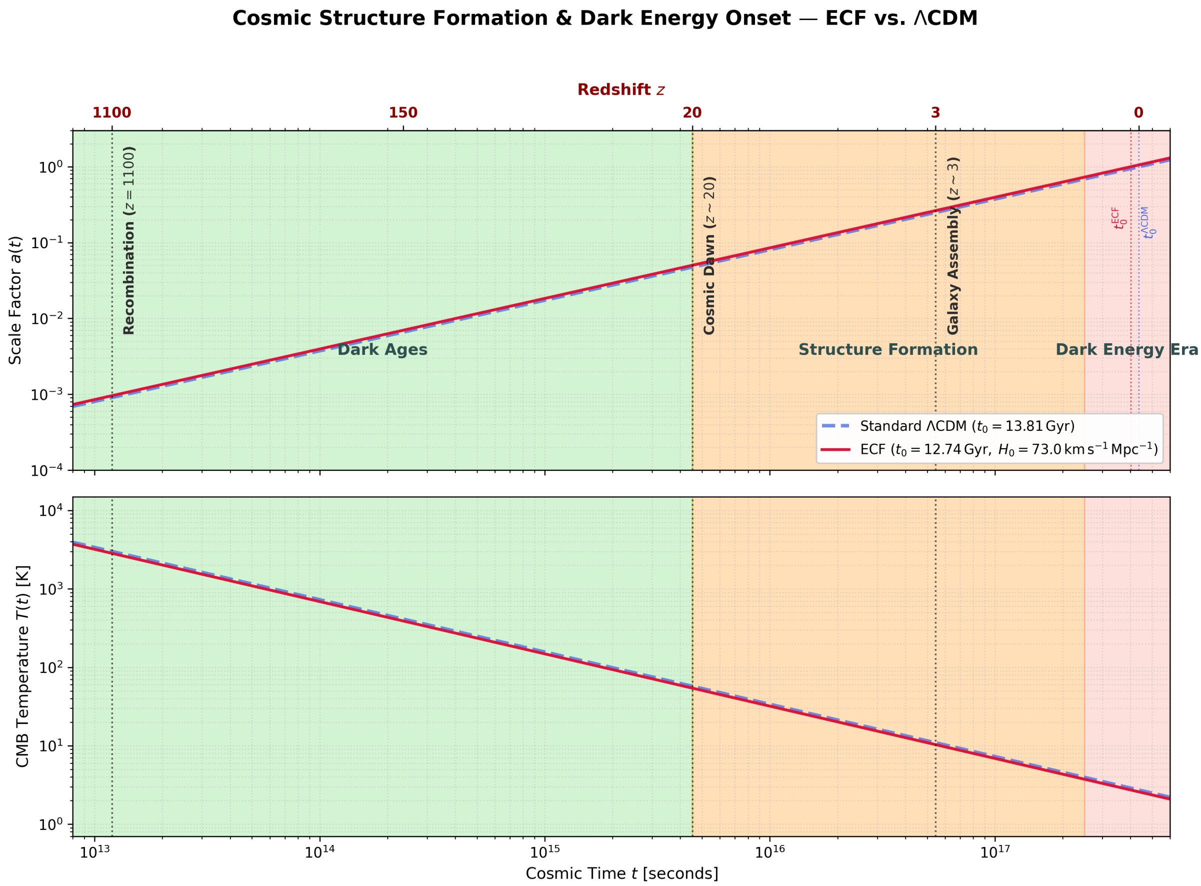


Figure 27: **Zoom on Cosmic Dawn and Structure Formation.** The ECF model (red) shows an earlier onset of galactic formation. The galaxies detected at $z = 14$ are therefore not unexpected, but can arise naturally result of a growth cycle amplified by the post-bounce spin-torsion dynamics.

Observational Predictions and Falsifiability

The ECF model offers two critical tests for future missions:

1. **21-cm Cosmology**: An earlier absorption dip at $z \approx 30$, detectable by HERA and SKA.
2. **Gravitational Lensing**: A higher density of massive halos at $z > 10$ than predicted by ΛCDM, verifiable by weak lensing surveys from Euclid and the Nancy Grace Roman Space Telescope.

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